



Baseline Survey for 'Operationalizing Green Economy Transition in Africa: Status of Green Economy initiatives in Nakuru and Mombasa Counties



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Operationalizing Green Economy Transition
in Africa: Status of Green Economy
Initiatives in Nakuru and Mombasa Counties

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Executive Summary

Devolution is perhaps the most transformative aspect of the Constitution of Kenya (2010). The Constitution established a National Government and 47 County Governments, which are distinct but inter-dependent entities. These, the national and county Governments, are to conduct their mutual relations mainly on the basis of consultation and cooperation.

It is in this regard that the Institute for Climate Change Adaptation of the University of Nairobi, the Consultant, undertook a study on behalf of: the International Climate Initiative (IKI); the German Federal Ministry for the Environment, Nature Conservation, Building and Nuclear Safety (BMUB); UNEP; GIZ; and Ministry of Environment, Natural Resources and Regional Development, Kenya; and County Governments of Nakuru and Mombasa.

The study sought to establish the status of Green Economy initiatives in Nakuru and Mombasa Counties. The specific issues the survey sought to examine are; (1) Six priority sectors for Mombasa (cities and built environment, fisheries, tourism, transport and infrastructure, energy and waste) and Five priority sectors for Nakuru (agriculture, energy, forestry, transport and water) counties, selected out of the classification of eleven sectors (agriculture, buildings, cities, energy, fisheries, forestry, manufacturing, tourism, transport, waste, water) (2) the predefined indicators for Green Economy in the sectors, the aim is to acquire and update information for each of the selected sectors and (3) analyse and state the main findings of Green Economy related assessments for the study counties

The data analysis was informed by (i) Consultants experiences as lead Institute on Climate Change adaption and Green Economy concepts (ii) the Constitution of Kenya, relevant laws and policies focusing on devolution and relevant articles (iii) personal stories and experiences by respondents. This report documents the findings and challenges as analyzed from the field visits, discusses the implications of the findings on the Operationalizing Green Economy Transition in Africa: with specific focus on Kenyan Counties and concludes with some recommendations on the way forward.

The measure of performance in Green Economy in the study counties was done using some indicators, categorised in four, namely the (1) General economic sector indicators; (2) Environmental issues and related indicators, (3) Green Economy policy interventions and related indicators, (4) Well-being and equity indicators. The issues which emerged from the analyses are summarized below:

- a) *General economic sector indicators:* It was found that there is generally no data available neither on the GDP of most assessed sectors such as waste nor on the contribution of the growth of the sectors to the GDP at national or county level. However, there is general feeling that the sectors' contribution to Counties' GDP and employment was, at the very least, important.
- b) *Environmental issues and related indicators:* The study established that waste management is major environmental issue for both counties. Poor management of waste has the highest impact on all sectors. Solid and liquid effluent discharge from industry, municipality, hotels, residential areas and others all have major negative impacts on both land and water bodies. Addressing this issue and land degradation will help to greatly reduce the factors that negatively impact on the growth of the tourism sector.
- c) *Wellbeing and quality indicators:* The study found no reliable figures on the percentage of people employed in the assessed sectors at County level relative to the total workforce within the Counties. For some, the work income is the sole livelihood, but others have small scale income

generating activities to supplement their earnings. The sectors contribution to household budgets range from very low to medium and low to very high.

- d) *Green economy policy interventions*: It was established that many sectoral policies and laws with significant relevance to Green Economy are not harmonized with each other and with the Constitution. This is enhanced by Counties' use of sectoral rather than integrated and ecosystem-based approach to development. In addition, weak enforcement of laws and weak implementation of policies remain major issues of concern.
- e) *County planning cycle*: It was found that both Counties incorporate a number of key documents, tools, methods and public participatory platforms which guide development planning at the County level. Some of these documents are: the County Integrated Development Plans (CIDP), County Annual Development Plans, and Zoning Plans.
- f) *Planning tools*: Both Counties are using such tools as Acts e.g. Public Finance Management Act, IFMIS, Monitoring Committees, Master Plans, Strategic Plans, Zoning Plans, and Land Use Plans.
- g) *Planning methods*: In regards to planning methods, the study found that the Counties employ such methods for planning as: Information Communications Technology (ICT) forum with toll free numbers, policy briefs, SWOT, Cost Benefit Analysis.
- h) *Engagement platforms*: Generally, it was established that both Counties are members of intergovernmental forums and inter-county forums.
- i) *Players and actors*: There are various governmental and non-governmental, communities based and private sector organizations involved in Green Economy with significant initiatives.

Conclusion and Recommendations

Operationalization of Green Economy at the County level is largely on course. The required legislations, regulations, systems and structures are at various stages. The progress made so far sets the foundation for effective mainstreaming and capacity development at local level and pilot applications of the toolbox. That notwithstanding, it was also evident from the study that there was need to expedite the establishment as well as the review of existing policies, legislation and some institutions in order to ensure that they facilitate mainstreaming of Greening concepts in development process and service delivery.

Some of the primary threats to the effective implementation of the Green Economy Strategies identified are: (1) High population and competition for diminishing resources; (2) Corruption and high costs of investment (3) climate change and high costs of rehabilitation of damaged ecosystems (4) lack of legal framework for enforcement (5) perennial water and power shortages (6) poor infrastructure due to budget constraints (7) weak land tenure system and widespread poverty (8) politicization and conflict of interest (9) limited funding and expensive alternative energy sources (10) lack of sufficient policy frameworks to support enforcement of greening initiatives.

In order to address these challenges and mitigate other related risks, the Consultant recommends the following:

- *Political commitment*- enhance fiscal and monetary tools to steer economies away from business-as-usual- to the green growth path
- *Legal reforms*: Legal reforms require revisiting existing laws and tuning their jurisdictions through amendments and directives. This will anchor decisions and policies in a binding and enforceable framework with systemic assessment of legal standards.
- *Financial Instruments*: Government bodies re-evaluating subsidy provisions and redirecting funds towards ventures that provide permanent solutions to energy security issues.

-
- *Technology Viability:* Even with financial instruments in place, the choice of renewable technologies may prove to be a significant challenge. This calls for the use of public funds to support working groups to debate the type of renewable technology most suited to current geopolitical and climatic conditions.
 - *Human Capital Formation:* Give due attention to the Green Economy human capital investments-enhance the know-what, know-how and the know-why.
 - *Institutional Role:* Initiate institutional adjustments or realignment at the county level to enable the adoption and adaptation to innovative solutions.
 - *Develop a common language:* It entails the standardizations of targets, benchmarks, indicators and measurement units and methods.
 - *Green Economy Communications strategy:* Establish a well thought out sensitization and communications programme. This will help bridge the clearly immanent knowledge gap around issues pertaining to or even the necessity for a Green economy campaign.

Table Contents

ACKNOWLEDGEMENTS	II
EXECUTIVE SUMMARY	III
TABLE CONTENTS.....	VI
LIST OF TABLES.....	X
LIST OF FIGURES	XIII
ACRONYMS	XIV
1. INTRODUCTION	1
1.1 General	1
1.2 Objectives.....	1
1.3 Scope of the Study	2
1.4 Purpose of the Study	2
1.5 National Green Economy Analysis	2
1.6 The Policy and Institutional Framework for a Green Economy	3
2. METHODOLOGY	6
2.1 Overall Approach and Performance Indicators.....	6
2.2 Research Methodology	6
2.3 Sampling Methods	6
2.4 Data Collection Tools.....	7
2.5 Data Analysis Methods.....	7
2.6 Limitations/Constraints	7
3. NAKURU COUNTY FINDINGS.....	9
3.1 Overview	9
3.2 Socioeconomic Setting	10
3.3 Sector Analysis	12
3.3.1 Agriculture.....	12
I. General economic sector indicators.....	12
II. Environmental issues and related indicators.....	14
III. Well-being and equity indicators.....	15
IV. Green economy policy interventions	18
3.3.2 Water Sector	18

I.	General economic sector indicators	18
II.	Environmental issues and related indicators.....	19
III.	Wellbeing and equity indicators	20
IV.	Green economy policy interventions	21
3.3.3	Forestry	22
I.	General economic sector indicators	22
II.	Environmental issues and related indicators.....	23
III.	Wellbeing and equity indicators	24
IV.	Green economy policy interventions	24
3.3.4	Energy Sector	25
I.	General economic sector indicators	25
II.	Environmental issues and related indicators.....	26
III.	Wellbeing and equity indicators	27
IV.	Green economy policy interventions	28
3.3.5	Transport and Infrastructure.....	30
I.	General economic sector indicators	30
II.	Environmental issues and related indicators.....	31
III.	Wellbeing and equity indicators	31
IV.	Green economy policy interventions	31
3.3.6	Tourism.....	32
I.	General economic sector indicators	32
II.	Environmental issues and related indicators.....	35
III.	Wellbeing and equity indicators	35
IV.	Green economy and policy interventions	36
3.3.7	Waste Management.....	36
I.	General economic sector indicators	36
II.	Environmental issues and related indicators.....	37
III.	Wellbeing and equity indicators	38
IV.	Green economy and policy interventions	38
3.3.8	Prioritisation of Analysed Sectors	39
3.4	County Planning Cycle	45
3.5	Institutional arrangements for annual development planning.....	46
I.	Governor.....	46

II. Council of County Governors.....	46
III. Intergovernmental Relations Technical Committee.....	46
IV. Senate	46
V. County Assembly	47
VI. County Executives.....	47
VII. County Public Service Board	47
VIII. City and Municipality Boards.....	47
3.6 Strategic Documents guiding development and decision making	47
3.7 Relevant actors.....	49
4. MOMBASA COUNTY FINDINGS.....	50
4.1 Overview	50
4.2 Biophysical Setting	51
4.3 Socioeconomic Setting	52
4.4 Sectoral Analysis.....	60
4.4.1 Waste Sector	61
I. General economic sector indicators.....	61
II. Environmental issues and related indicators.....	65
III. Wellbeing and equity indicators	68
IV. Green economy policy interventions	69
4.4.2 Tourism.....	73
I. General economic sector indicators.....	73
II. Environmental issues and related indicators.....	76
III. Wellbeing and equity indicators	78
IV. Green economy policy interventions	80
4.4.3 Cities and Built Environment.....	83
I. General economic sector indicators.....	83
II. Environmental issues and related indicators.....	85
III. Wellbeing and equity indicators	87
IV. Green economy policy interventions	87
4.4.4 Transport and Infrastructure.....	90
I. General economic sector indicators.....	90
II. Environmental issues and related indicators.....	92
III. Wellbeing and equity indicators	93

IV. Green economy policy interventions	94
4.4.5 Fisheries.....	97
I. General economic sector indicators	97
II. Environmental issues and related indicators.....	98
III. Wellbeing and equity indicators	100
IV. Green economy policy interventions	101
4.4.6 Energy.....	103
I. General economic sector indicators	103
II. Environmental issues and related indicators.....	106
III. Wellbeing and equity indicators	108
IV. Green economy policy interventions	109
4.4.7 Prioritisation of Analysed Sectors	114
4.5 County Planning Cycle	121
4.5.1 Institutional arrangements for annual development planning.....	121
4.5.2 Relevant actors.....	126
4.5.3 Best entry point for the project in Mombasa County	127
CHAPTER 5 - CONCLUSIONS	129
5.1. Priority sectors for each study county	129
5.2. The predefined indicators	129
5.3. Analysis and main findings of Green Economy	129
5.4. Areas that need to be addressed for an effective Green Economy implementation	129
BIBLIOGRAPHY	131
ANNEXES	139
ANNEX 1 - THE PERFORMANCE (SECTOR) INDICATORS	139
ANNEX 2 - STAKEHOLDER MEETING MINUTES AND REPORTS	157
ANNEX 3 - SAMPLE QUESTIONNAIRES	162
ANNEX 4 - FILLED INDICATOR DATA SHEETS FOR NAKURU AND MOMBASA	177
NOTES TO THE COMPILATION OF COUNTY DATA	177
ANNEX 5 - THE DEVELOPMENT PLANNING CYCLE UNDERTAKEN BY MOMBASA COUNTY PLANNERS: RESPONDENT ANALYSIS BY SECTOR	217

List of Tables

Table 3-1: Administrative Units and Area of Nakuru County	10
Table 3-2: County Population Projections	10
Table 3-3: Agriculture Sector Statistics	12
Table 3-4: Flower Production.....	14
Table 3-5: Value of Pesticides Imported from Financial Year 2003/04 – 2009/10.....	15
Table 3-6: Water Sector Statistics	19
Table 3-7: Green Economy Initiatives in the Water Sector.....	22
Table 3-8: Forestry Sector Statistics	24
Table 3-9: Green Economy Initiatives in the Forestry Sector.	25
Table 3-10: Sources of Lighting Fuel in Percentage	25
Table 3-11: Sources of Cooking Fuel in Percentage.....	26
Table 3-12: Renewable Energy Installed Capacity 2004 – 2012 (MW)	27
Table 3-13: Energy sector commitment for Green Economy in Nakuru County	29
Table 3-14: Green Economy interventions in the energy sector.....	29
Table 3-15: Transport Sector Statistics	30
Table 3-16: Green Economy Interventions	31
Table 3-17: Tourism Sector Statistics.....	32
Table 3-18: Green Economy interventions in the Tourism Sector	36
Table 3-19: Sector Statistics for Waste Management	37
Table 3-20: Green Economy intervention in waste sector.....	38
Table 3-21: Assessment of the agriculture sector.....	39
Table 3-23: Assessment of the energy sector against the 10 point criteria for sector prioritisation (1- low; 2- medium; 3 - High).....	40
Table 3-24: Assessment of the water sector against the 10 point criteria for sector prioritisation (1- low; 2- medium; 3 - High).....	40
Table 3-25: Assessment of the tourism sector against the 10 point criteria for sector prioritisation (1- low; 2- medium; 3 - High).....	41
Table 3-26: Assessment of the waste sector against the 10 point criteria for sector prioritisation (1- low; 2- medium; 3 - High).....	42
Table 3-27: Assessment of the forestry sector against the 10 point criteria for sector prioritisation (1- low; 2- medium; 3 - High).....	43
Table 4-1: Administrative/political jurisdictions in Mombasa County.	51
Table 4-2: Population distribution and density by Sub-County, Mombasa County.	52
Table 4-3: Mombasa County household statistics based on KNBS.....	53
Table 4-4: Total recorded employment ¹ at national level in thousands, 2011 – 2015	55
Table 4-5: County labour force, required total employment (in numbers) by year.....	55
Table 4-6: Overall employment by education levels in Mombasa County	55
Table 4-7: Estimated Mombasa wage employment (numbers in thousands) by industry: 2011 – 2014.	56
Table 4-8: Estimated Mombasa wage employment (%) by industry: 2011 – 2014.	58
Table 4-9: Contribution of "water supply; sewerage, waste management" sector to GDP at national level.....	61
Table 4-10: City planners' views on the waste sector contribution to the County GDP and employment.	61
Table 4-11: Future Sewage Generation in Mombasa, computed based on population, water supply and sewage generation rates (2015)	63
Table 4-12: Proposed sewerage development plan for Mombasa - National Water Master Plan, 2030	64

Table 4-13 Distribution of Transfer Stations within the County.....	64
Table 4-14: County planners' views on issues relating to the energy, wastewater treatment, and ecolabelling indicators in the waste sector.	66
Table 4-15: County planners' views on the environmental issues and impacts, mitigation measures in place and challenges faced in the County - waste sector.....	67
Table 4-16: Estimated waste generation rate by socio-economic group for Mombasa County	68
Table 4-17: Contribution of "water supply; sewerage, waste management" sector to wage employment at national level.....	68
Table 4-18: Existing Green Economy initiatives, challenges, and opportunities in the waste sector...	70
Table 4-19: Existing Green Economy initiatives, challenges, and opportunities in the waste sector according to the County planners - baseline survey.	71
Table 4-20: City planners' views on the tourism sector contribution to the County GDP and employment.	74
Table 4-21 Hotel rooms and beds available and occupied by Coastal Zone, 2014.....	74
Table 4-22: Mombasa/Coastal Hotel Rooms – Count of Beds Available and Occupied	75
Table 4-23: County planners' views on issues relating to the energy, wastewater treatment, and ecolabelling indicators in the tourism sector.	77
Table 4-24: County planners' views on the environmental issues and impacts, mitigation measures in place and challenges faced in the County tourism sector.	78
Table 4-25: National wage employment in the tourism sector (number in thousands).....	79
Table 4-26: Existing Green Economy initiatives, challenges, and opportunities.....	80
Table 4-27: Existing Green Economy initiatives, challenges, and opportunities in the tourism sector according to the County planners.	81
Table 4-28: Proposed Land use for Mombasa County	84
Table 4-29: City planners' views on the cities and built environment sector contribution to the County GDP and employment.	84
Table 4-30: County planners' views on issues relating to the energy, wastewater treatment, and ecolabelling indicators in the cities and built environment sector.....	86
Table 4-31: County planners' views on the environmental issues and impacts, mitigation measures in place and challenges faced in the County cities and built environment sector.....	86
Table 4-32: : National wage employment in the cities and built environment sector (number in thousands).....	87
Table 4-33: Existing Green Economy initiatives, challenges, and opportunities in the cities and built environment sector.....	88
Table 4-34: Existing Green Economy initiatives, challenges, and opportunities in the cities and built environment sector.....	89
Table 4-35: National GDP by Transport and Infrastructure Sector Activity: 2011-2014.....	90
Table 4-36: Mombasa port activity indicators.....	91
Table 4-37: City planners' views on the transport and infrastructure sector contribution to the County GDP and employment.	92
Table 4-38: County planners' views on issues relating to the energy, wastewater treatment, and ecolabelling indicators in the transport and infrastructure sector.....	92
Table 4-39: County planners' views on the environmental issues and impacts, mitigation measures in place and challenges faced in the County transport and infrastructure sector.	93
Table 4-40: National wage employment in transportation and storage (number in thousands) by gender: 2012 -2014.	94
Table 4-41: Existing Green Economy initiatives, challenges, and opportunities in the cities and built environment sector according to county documents.....	95
Table 4-42: Existing Green Economy initiatives, challenges, and opportunities in the transport and infrastructure sector according to the County planners.	96

Table 4-43: Quantity of Fish Landed, 2010 - 2014	98
Table 4-44: Value of Fish Landed, 2010 - 2014.....	98
Table 4-45: City planners' views on the fisheries sector contribution to the County GDP and employment.	98
Table 4-46: Fisheries production systems and their impacts to the environment (Source, Draft ISUDP)	99
Table 4-47: County planners' views on issues relating to the energy, wastewater treatment, and ecolabelling indicators in the fisheries sector.	99
Table 4-48: County planners' views on the environmental issues and impacts, mitigation measures in place and challenges faced in the County fisheries sector.	100
Table 4-49: National wage employment in agriculture, fishing and forestry (number in thousands)	101
Table 4-50: Existing Green Economy initiatives, challenges, and opportunities according to County documents	102
Table 4-51: Existing Green Economy initiatives, challenges, and opportunities in the fisheries sector according to the County planners	102
Table 4-52 City planners' views on the energy sector contribution to the County GDP and employment.	106
Table 4-53: County planners' views on issues relating to the energy, wastewater treatment, and ecolabelling indicators in the energy sector.....	108
Table 4-54: County planners' views on the environmental issues and impacts, mitigation measures in place and challenges faced in the County energy sector	108
Table 4-55: Mombasa Wage Employment (in numbers of thousands) in Energy Sector and Gender.	109
Table 4-56: Existing Green Economy initiatives, challenges, and opportunities in the energy sector according to the County planners.	109
Table 4-57: Existing Green Economy initiatives, challenges, and opportunities.....	111
Table 4-58 Existing Green Economy initiatives, challenges, and opportunities in the energy sector according to the County planners.	111
Table 4-59: Assessment of the waste sector	115
Table 4-60 Assessment of the waste sector	116
Table 4-61: Assessment of the Cities and Built Environment sector	117
Table 4-62: Assessment of the Transport and Infrastructure sector	118
Table 4-63: Assessment of the Fishery sector	119

List of Figures

Figure 1-1: Sectors and Indicators	2
Figure 3-1: Location Map Showing Nakuru County and its Neighbours	9
Figure 3-2: Nakuru County Budgets Estimates 2015/16	11
Figure 3-3: Volume of Pesticides Imported from Financial Year 2003– 2009 (PCPB, 2010).	15
Figure 3-4: County Total food Crop Production 2012 - 2014	16
Figure 3-5: County Horticulture Production 2012 - 2014	16
Figure 3-6: Nakuru County Beef and Wool Production; Milk and Eggs Production 2013 to 2015	17
Figure 3-7: Nakuru County Livestock Production 2013 to 2015	17
Figure 3-8: Land use change in upper Gilgil area	20
Figure 3-9: Smallholder Agroforestry Plot	23
Figure 3-10: Reliability as an important factor in energy source	27
Figure 3-11: Cost as an important factor in energy source	27
Figure 3-12: Eco-friendly (low carbon footprint)	28
Figure 3-13: Availability as an important factor in energy source	28
Figure 3-14: Beds Occupied, 2010 - 2014	33
Figure 3-15: Number of Visitors to Nakuru National Park, 2010 - 2014	34
Figure 4-1: Mombasa County (Source: NK/PADECO/EJEC, 2016).	50
Figure 4-2: Population growth of Mombasa County	53
Figure 4-3: Population growth in each zone, 1979-2009	54
Figure 4-4: Percentage of solid waste generated by type in Mombasa County	62
Figure 4-5: Solid Waste Generation Projection Mombasa County Government	63
Figure 4-6: Solid waste streams and scavengers role in Mombasa (Draft ISUDP, 2015)	65
Figure 4-7: National tourism arrivals, 1995-2014.	73
Figure 4-8: National Tourism Earnings	74
Figure 4-9: Number of visitors to Fort Jesus, Haller's Park, and Mombasa Marine National Park	76
Figure 4-10: Mombasa wage employment (in thousands) in accommodation and food service activities	79
Figure 4-11: Growth in employment in construction and real estate activities (Cities and Built Environment Sector), Mombasa County, 2011 - 2015	85
Figure 4-12: Mombasa wage employment (numbers in thousands), transportation and storage activities	94
Figure 4-13: Mombasa wage employment (numbers in thousands) in agriculture, forestry and fishing, 2011 - 2014	100
Figure 4-14: Nationally installed capacity by type of power (MW)	104
Figure 4-15: Domestic sale of petroleum fuels ('000 tonnes) by consumer category	105
Figure 4-16: Electricity Sales By Area in Millions of KWh (June 2006 -2014)	106
Figure 4-17: Mombasa County percentage distribution of households by cooking fuel source	107
Figure 4-18: Current and Projected Power Demand (source - ISUDP).	107
Figure 4-19: Estimated Mombasa County wage employment (numbers in thousands)	109

Acronyms

ADC	Agricultural Development Corporation
CBOs	Community Based Organizations
CEC	County Executive Committee Member
CIDP	County Integrated Development Plan
CORDIO	Coastal Organization on Research
CSOs	Civil Society Organizations
EU	European Union
FY	Financial Year
GDC	Geothermal Development Company
GIZ	German International Organization
HOD	Head of Department
HSE	Health, Safety and Environment
ICCA	Institute for Climate Change and Adaptation
ILO	International Labour Organization
IWASP	Integrated Waste and Sanitation Plan
KAA	Kenya Airports Authority
KALRO	Kenya Agriculture and Livestock Research Organization
KAPAP	Kenya Agricultural Productivity and Agribusiness Project
KEMFRI	Kenya Maritime and Fisheries Research Institute
KenGen	Kenya Electricity Generating Company
KEPSA	Kenya Private Sector Alliance
KIPPRA	Kenya Public Policy Research
KMA	Kenya Maritime Authority
KPA	Kenya Ports Authority
KPC	Kenya Power Company
KPHC	Population and Housing Census
KRA	Kenya Revenue Authority
KWS	Kenya Wildlife Service
MCIDP	Mombasa County Integrated Development Plan
MP	Master Plan
MTEF	Medium Term Economic Framework
NAWASCO	Nakuru Water and Sewerage Company
NBA	National Building Authority
NMK	National Museums of Kenya
PCPB	Pests Control Products Board
SCODE	Sustainable Community Development Services
TOR	Terms of Reference
UNECE	United Nations Economic Commission for Europe
WARMA	Water Resources Management Authority
WWF	World Wide Fund for Nature

1. INTRODUCTION

1.1 General

UNEP's work on green economy raised the visibility of this concept in 2008, particularly through a call for a Global Green New Deal (GGND). The GGND recommended a package of public investments and complementary policy and pricing reforms aimed at kick-starting a transition to a green economy, while reinvigorating economies and jobs and addressing persistent poverty (Barbier 2010). Designed as a timely and appropriate policy response to the economic crisis, the GGND proposal was an early output from the United Nations' Green Economy Initiative (UNEP 2011).

UNEP defines a green economy as one that results in "improved human well-being and social equity, while significantly reducing environmental risks and ecological scarcities" (UNEP 2010). In its simplest expression, a green economy is low-carbon, resource efficient, and socially inclusive. In a green economy, growth in income and employment are driven by public and private investments that reduce carbon emissions and pollution, enhance energy and resource efficiency, and prevent the loss of biodiversity and ecosystem services (UNEP 2011).

1.2 Objectives

Noting that a number of African countries have recognised the potential of the Green Economy (GE) approach and are in the process of developing appropriate Green Economy strategies, GIZ and UNEP launched a joint project titled "Operationalizing Green Economy Transition in Africa: Status of green economy initiatives in Ethiopia, Ghana, Kenya and Rwanda" in order to complement national efforts in selected countries in Africa in transforming their economies towards efficient and competitive engines that reduce poverty and do not threaten the environment. They observed that what is lacking is the practical implementation of these strategies at the sub-national or local level, and therefore, through this project, have the overall objective of supporting the selected countries by: developing key planning and management tools for Green Economy at the sub-national and local level; building the capacity of local governments and other relevant stakeholders from the public and private sector; supporting pilot demonstrations; and developing an appropriate national framework for replication.

The main objective of this assignment, which is one component of the overall project, was to conduct a baseline survey in Mombasa and Nakuru counties to document the status of the Green Economy in pilot communities (sub-national units) which in the case of Kenya is the County level.

The baseline survey was to specifically do the following;

1. Carry out an assessment of six priority sectors for Mombasa (cities and built environment, fisheries, tourism, transport and infrastructure, energy and waste) and five priority areas for Nakuru (agriculture, energy, forestry, transport and water) counties, that were selected out of the classification of eleven sectors (agriculture, buildings, cities, energy, fisheries, forestry, manufacturing, tourism, transport, waste, water) through stakeholder consultations;
2. Adopting the GIZ-predefined indicators for Green Economy in the sectors, acquire and update information for each of the selected sectors; and
3. Analyse and state the main findings of Green Economy related assessments for the study counties.

1.3 Scope of the Study

The baseline survey provides a snapshot of green economy in the Nakuru and Mombasa pilot Counties, based on the priority sectors selected by the County stakeholders as indicated in section 1.2 above. This was done within the context of a review of the state of four indicator categories that collectively could be used to characterise each of the selected sectors as illustrated below. The sector indicators that were used in this study to assess the green economy of the pilot areas were developed by GIZ (see Annex I).

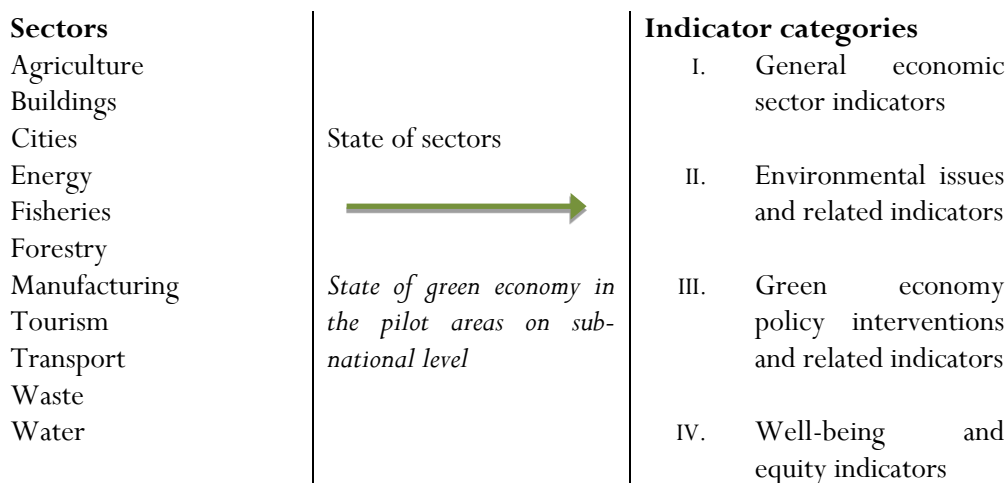


Figure 1-1: Sectors and Indicators

1.4 Purpose of the Study

The rationale behind the project has already been addressed above. The project is expected to lead to an enhanced integrated planning as well as changes in local policies that lead to pro-poor and financially viable, resource efficient, ecosystem restoring, and low carbon or climate resilient projects implemented within the pilot Counties. These actions, policies and practices are then to be up scaled to the rest of the country, and more broadly, to build national capacities and provides the framework for further replication of the pilot lessons and experiences.

1.5 National Green Economy Analysis

The Government of Kenya (GoK) has begun the effort of introducing and implementing policies and initiatives that support the Green Economy agenda. In that context, it has set in motion a series of steps that are meant to advance the economic transition from business-as-usual to a Green Economy. The actions are aligned to the Kenya Constitution (2010), the Vision 2030 tenets, the Green Economy Assessment Report on Kenya by UNEP (2014), the Kenyan National Climate Change Action Plan, the Strategic Plans completed (Kenya Response to Climate Change Strategy (2010) and (Kenya Green Economy Strategy and Implementation Plan 2015), and Kenya Climate Smart Agriculture (2015). The country is also party to a number of Multilateral Environmental Agreements (MEAs) which precede the notion of a "Green Economy", but whose objectives support, in part, the Green Economy agenda. Implementation of some of the completed strategic plans is lagging behind due to either lack of funds or technical experts.

In the Kenyan context, Green Economy should be understood as economic activities and processes that are equitable and inclusive, but do not compromise environmental sustainability. The Constitution of Kenya 2010 clearly brings forward the environmental objectives of the Country by calling for "sustainable exploitation,

utilization, management and conservation of the environment and natural resources” and Article 42 of the Bill of Rights states that “every person has the right to a clean and healthy environment, which includes the right to have the environment protected for the benefit of present and future generations”. Further, the country’s Vision 2030 envisions a country whose citizens can enjoy a high quality of life in a clean and secure environment.

It is important to note that the draft Medium Term Plan (MTP II) of the Government of Kenya acknowledges that the growth strategy of the country embraces measures to ensure a greener economy and that the economy is put on a low carbon trajectory. Furthermore, the African Development Bank (AfDB) in cooperation with UNEP, UNDP, WWF and ILO are supporting the Government in preparing a Green Economy Strategy and Implementation Plan 2015.

The Green Economy in Kenya is seen as having great potential for ensuring economic growth in the long run and, as stated in a scoping study carried out by the Kenya Institute for Public Research and Analysis (KIPRA, 2012), Kenya should take advantage of the opportunities that arise from green growth and effectively address challenges and barriers. A Green Economy approach would allow for ensuring the sustainable consumption of natural resources, energy efficiency, whilst sustaining economic growth and development. The key sectors of the Green Economy in which Kenya has begun and continues its efforts are renewable energy; promotion of sustainable production and consumption; pollution control and waste management; environmental planning and governance; and restoration of forest ecosystems.

1.6 The Policy and Institutional Framework for a Green Economy

Many sectoral policies and laws regarding environment are not harmonized with each other and with the Constitution. These include policies and laws relating to agriculture, land, water, forests, trade and industry, which have significant links with, and implications for, the environment. The government’s sectoral rather than integrated and ecosystem-based approach to management of natural resources has proved insufficient in addressing environmental challenges. In addition, weak enforcement of laws and weak implementation of policies remain major issues of concern. However, with the promulgation of the Constitution of Kenya 2010, an opportunity arose to align anew our policies and regulations, some of which were decades old, with the new realities underpinning the need for heightened environmental stewardship and for embarking on sustainable low carbon development pathways. Below are outlined some of the many policies pertaining to environmental sustainability (with a focus on climate change, which arguably is largest strategic environmental challenge facing Kenya), key actors in the field, and lessons learned.

Kenya’s current institutional framework on environment is characterised by fragmentation. Various aspects of environmental policy cut across different institutions. Although the Environmental Management and Coordination Act of 1999 was a major landmark, with the primary objective of improving coordination and management of the environment, legislation of relevant laws and regulations have not yet been completed. The National Climate Policy (NCCP) is the de facto policy document for the country on matters climate change. Other climate change response policies in Kenya include Vision 2030, the National Climate Change Response Strategy (NCCRS - 2010), and The Second Medium Term Plan 2013-2017. Climate change is further addressed in the agriculture sector strategies such as ASDS (2010-2020). The National Climate Change Response Strategy (NCCRS) provides the framework for integrating climate concerns into development priorities, government planning and budgeting. Agriculture is a major focus of the NCCRS, with substantial needs for adaptation and mitigation identified. It captures Kenya’s early experiences, within the context of climate change and agriculture, as instructive for the country’s future development. These lessons inform the efforts of Kenya in the shift towards landscape planning with positive results such as the development of a National Catchment Management Policy, in which agricultural landscapes would be managed not only for agriculture and watershed

management objectives, but also related goals such as food security, climate adaptation and biodiversity conservation. The NCCRS includes indicative budgets and plans for line ministries. The implementation of the full strategy is estimated to cost US\$3 billion annually over the next 20 years, and roughly US\$100 million per year would be needed for adaptation and mitigation activities in agriculture (GoK 2010b).

Kenya's Agricultural Sector Development Strategy makes climate adaptation a priority, but the agricultural components of the NCCRS provide more details on prioritized activities. On adaptation, the NCCRS calls for accelerated investment in weather information systems, research on drought tolerant crop varieties, soil and water conservation, water harvesting, and strengthening integrated pest management systems, among others. On the mitigation front, emphasis will be on agricultural mitigation by hosting a variety of innovative land-based carbon projects, including agroforestry's Agricultural Carbon Project which turns sustainable agricultural practices into carbon credits, as well as biogas development programs. Kenya plans to build on this experience and others in mitigation by prioritizing activities such as proper management of agricultural waste, organic farming, mulching, agroforestry, and selected application of biotechnology.

To operationalize and guide the NCCRS, the government in March 2013 finalized the development of the National Climate Change Action Plan (NCCAP). This was also developed through a consultative process that engaged actors across government, the private sector and over 200 civil society groups. NCCAP is meant to operationalize NCCRS by providing the analysis and enabling mechanisms to make implementation successful. It will also support efforts towards the implementation of the Kenya Constitution 2010 and the attainment of Vision 2030; as well as encourage people-centred (inclusive) development, ensuring that climate change actions help the country move toward its long-term development goals. In particular, the NCCAP sets out a vision for a low carbon climate resilient development pathway; summarises analysis of mitigation and adaptation options and recommended actions; recommends an enabling policy and regulatory framework; and sets out next steps for knowledge management and capacity development, technology requirements, a financial mechanism, and a national performance and benefit measurement system (NPBM). Vision 2030 captures many environmental issues.

The Kenya Vision 2030 Flagship project which covers the period 2008 to 2030 is anchored on four pillars: the Economic, Social, Political, and Foundations pillars. The economic pillar aims to improve the prosperity of all Kenyans through an economic development programme that aims for an average GDP growth rate of 10% per annum and has targeted six priority sectors (Tourism, Agriculture and Livestock, Wholesale and Retail Trade, Manufacturing, Business Process Outsourcing and Financial Services) to achieve this and create opportunities. The social pillar seeks to build a just and cohesive society with social equity in a clean and secure environment, and with elements such as health, education and training, environment, water and sanitation, human resource development, and urbanisation and housing, provides a strong entry point for Green Economy practices. The political pillar seeks to expand the democratic space and espouses a people-centred, results-oriented, and accountable system that will guarantee Kenya's attainment of its specific goals, strategies and flagship projects outlined under Vision 2030's economic and social pillars. The foundations pillar is cross-cutting, and encompasses factors that are referred to as "enablers" that underpin the country's envisaged transformation. It includes an improved and expanded national physical infrastructure including the roads, railways, and communications network, land utilisation and management, and also embraces the overarching role of Science, Technology and Innovation (STI). These all provide scope for the embedding of Green Economy principles and practices in development.

The Vision 2030 flagship projects form the basis of the governments Medium Term Plans. The government recently launched the Second Medium Term Plan 2013-2017 (MTP) which identifies key policy actions,

reforms, programmes and projects to be implemented in the 2013-2017 period in line with government priorities, the Kenya 2010 constitution and the long-term objective of Vision 2030. Accordingly, the theme of this MTP is “Transforming Kenya: Pathway to Devolution, Socio-Economic Development, Equity and National Unity”. The MTP gives priority to devolution as spelt out in the constitution and to more rapid socio-economic development with equity as a tool for building national unity. The flagship projects under the social pillar of the MTP that will help build resilience against climate change and embrace sustainable development include strengthening environmental governance; waste management and pollution control; rehabilitation of urban rivers; land reclamation; irrigation and drainage infrastructure installation; water harvesting and storage programme; rehabilitation and protection of water towers among others. Disaster risk management (with the purpose of ending or reducing the negative effects of drought emergencies) is stated in the MTP as a key pillar for national transformation. The MTP highlights that the most critical foundations for drought resilience are security (reduce conflicts over natural resources), improved infrastructure (increase access to markets and basic services, hence decrease vulnerability) and formation of human capital (improve health and education). These areas are all essential for strengthened resilience.

The successful transition to a Green Economy which will result in Sustainable Development requires reforms that support economic transformation through capacity building, reform of regulatory frameworks and improved financing mechanisms. Given this background and the potential for the Green Economy to boost economic growth and create employment and sustainable livelihoods in Kenya, it is essential that all relevant stakeholders understand the PESTEL (Politics, Economic, Social, Technological, Environmental, Legal) framework and consider how to integrate or factor in Green Economy principles and practices.

2. METHODOLOGY

2.1 Overall Approach and Performance Indicators

The overall approach adopted was as follows:

1. Identification of Pilot Counties in consultation with GIZ;
2. Literature Review focused on Green Economy in Kenya;
3. Web- and Literature-based data mining for quantitative indicator data;
4. Development, Testing and Refining of Questionnaires
5. Key Stakeholder Meetings for Sector Prioritisation at County Level
6. Field Surveys in Mombasa and Nakuru Counties: Included Key Informant Interviews, Focused Group Discussions, Questionnaire Surveys
7. Data Entry, Analysis and Reporting

The GIZ adopted the UNEP categorisation of 10 industry sectors, but split the Agriculture and Fishery sector into two, ending up with 11 sectors. Within each of these sectors, indicators were identified/developed for sub-national planning and Green Economy mainstreaming for monitoring and evaluation. For themes that cut across sectors and that had some level of complexity, a "General" category was created. Thus, in total, there are 12 categories of indicators. These indicators (See illustration in Section 1.3 above and Annex 1) are comprehensive but not exclusive and can be reduced or enhanced in the further process of the project. For the "General" category, indicators were grouped into five classes, as follows: (1) Economic indicators, (2) Environmental issues and related indicators, (3) Green economy policy interventions and related indicators, (4) Well-being and equity indicators, and (5) Complex indicators. Within each of the remaining 11 sector categories, the identified indicators were further grouped into four classes, as follows: (1) General sector indicators, (2) Environmental issues and related indicators, (3) Green economy policy interventions and related indicators, and (4) Well-being and equity indicators.

2.2 Research Methodology

2.3 Sampling Methods

The study relied on both primary and secondary data sources. The secondary data was obtained from both physical and electronic libraries or data archives and included policy, legislative, industry sector, and other information relevant to green economy at the national and identified county levels. Primary data was collected through questionnaire surveys, focus group discussions, key informant interviews, and observations during the field visits. The study commenced with review of relevant information, identification of key informants/stakeholders and consultations at national level, and with discussions with GIZ that enabled the study team and client to select Nakuru (representative of mostly rural) and Mombasa (representative of mostly urbanised) Counties as possible pilot sites for the Green Economy Baseline Study.

The identification of which of the 11 sectors were deemed to be significant sectors was done at the first stakeholder meeting for each County (see Annex 2 for Stakeholder Meeting Minutes). The Mombasa meeting was held at the Pride Inn on 14th October 2015 while the Nakuru meeting was held at the Merica Hotel on 28th October 2015. The stakeholders also prioritised the shortlist of significant sectors that would be studied further during the baseline study. Present at the meetings were the relevant government departments and organizations including but not limited to: County Government of Nakuru; County Government of Mombasa; Geothermal Development Company; Kenya Airports Authority, KAA; KenGen; Kenya Pipeline Company,

KPC; Ministry of Environment and Natural Resources; Ministry of Water and Irrigation; Ministry of Energy and Petroleum; Ministry of Devolution and Planning; Ministry of Tourism; Ministry of Agriculture, Fisheries and Livestock; and Private sector organizations representatives, e.g. KEPSA, KMA. Others organizations were consulted as suggested during the subsequent field visits.

2.4 Data Collection Tools

Secondary data and information relevant to the study sites and issues were collected from existing literature and reports, as well as from recognised internet sources. Questionnaires provided by GIZ, and targeting planners and private sector players, were reviewed and customised on the basis of the outcomes of the first stakeholder meetings and secondary information collected. These were then pre-tested in Nairobi and refined for use in the study areas (See Annex 3 for Questionnaires). Some further slight modifications were made during the first field day to address some heretofore unseen ambiguities.

Key informant interviews: These formed a very important component of the evaluation process. Key stakeholders were interviewed to provide crucial information and guidance to the evaluation. This tool was used at both national and county levels, and, to a lesser extent, at the local level.

Interview guidelines were used for each stakeholder category and for each sector. This ensured that pertinent issues were addressed in a systematic, exhaustive manner.

The interviews were guided by standard questionnaires in the initial phase, followed by face to face using check list to fill on missing gaps and follow ups for detailed policy status.

Questionnaire survey: The consultants developed specific questionnaires for each sub-sector/stakeholder under review. Questionnaires were used to capture data/information. The field study was conducted in two phases.

- Phase 1 – Mombasa team worked from 2nd to 6th November 2015, while the Nakuru team worked from 5th to 14th November 2015.
- Phase 2 – This was a gap filling exercise informed by sort fall in data, comments and reviews of the interim report. The Mombasa team went out from 12th to 15th January 2016 and Nakuru team conducted theirs from the 6th to 7th January 2016.

2.5 Data Analysis Methods

Much of the data/information from the evaluation was qualitative in nature. However, a certain amount of quantitative data was generated. Data analysis was done using MS-Excel and SPSS to generate necessary measures of progress/achievement such as totals, averages, ranges, frequencies, percentages. These were presented graphically in tables, graphs and pie-charts to enhance reader friendliness.

2.6 Limitations/Constraints

Some of the limitations of the study included;

1. Scarcity of data, inconsistency (conflicting) in similar indicator data, and rarity of longitudinal or spatially and consistently monitored data in all sectors in both counties.

2. Dearth of direct greenhouse gas emissions data or carbon dioxide equivalents.
3. Available data at country level not disaggregated to county level.
4. Access to the right staff member for detailed information on certain indicators.
5. The time taken to answer the questionnaire was on average about 2.5 hrs and this was reduced to about 1.5 with the modifications made to the questionnaire.
6. Moving around from one place to another plus waiting for the interviewee to host the team added on average some 30 to 60 minutes, hence administration of each questionnaire including travel and waiting time was taking about 2.5-3 hrs - on this basis, it was only possible for each team to handle 4 questionnaires per day.
7. Rescheduling of appointments was common as the respondents would find it difficult to dedicate the time agreed
8. Access to information – Some institutions/companies (private sector in particular) would request further letters from the County Department of Environment. Some would request clearance letters from their head offices in Nairobi, which we were unable to procure
9. The institutions have not accessed historical data for their respective units due to the ongoing governance transition.
10. Some interviewees requested that they be left with the questionnaires to fill in and the team to follow up with them - the return rate under this scenario was extremely low
11. Some interviewees declined to fill in the forms either because they felt they did not have enough knowledge to fill in the forms, or unless sanctioned by top management.

3. NAKURU COUNTY FINDINGS

3.1 Overview

Nakuru County has an area of 7,496.5 Km². The population census of 2009 puts Nakuru County at a population of 1,603, 325 (Male - 50.2 percent, Female - 49.8 percent) and a population density of 213.9 people per Km². Nakuru County has nine (9) and Fifty-Five (55) wards. It is the fourth largest County in Kenya after Nairobi, Kakamega and Kiambu in that order in terms of population. It borders Narok, Nyandarua, Kajiado, Kiambu, Laikipia, Baringo, Kericho and Bomet Counties. Its strategic location endears it as the gateway to the North Rift Valley of Kenya.

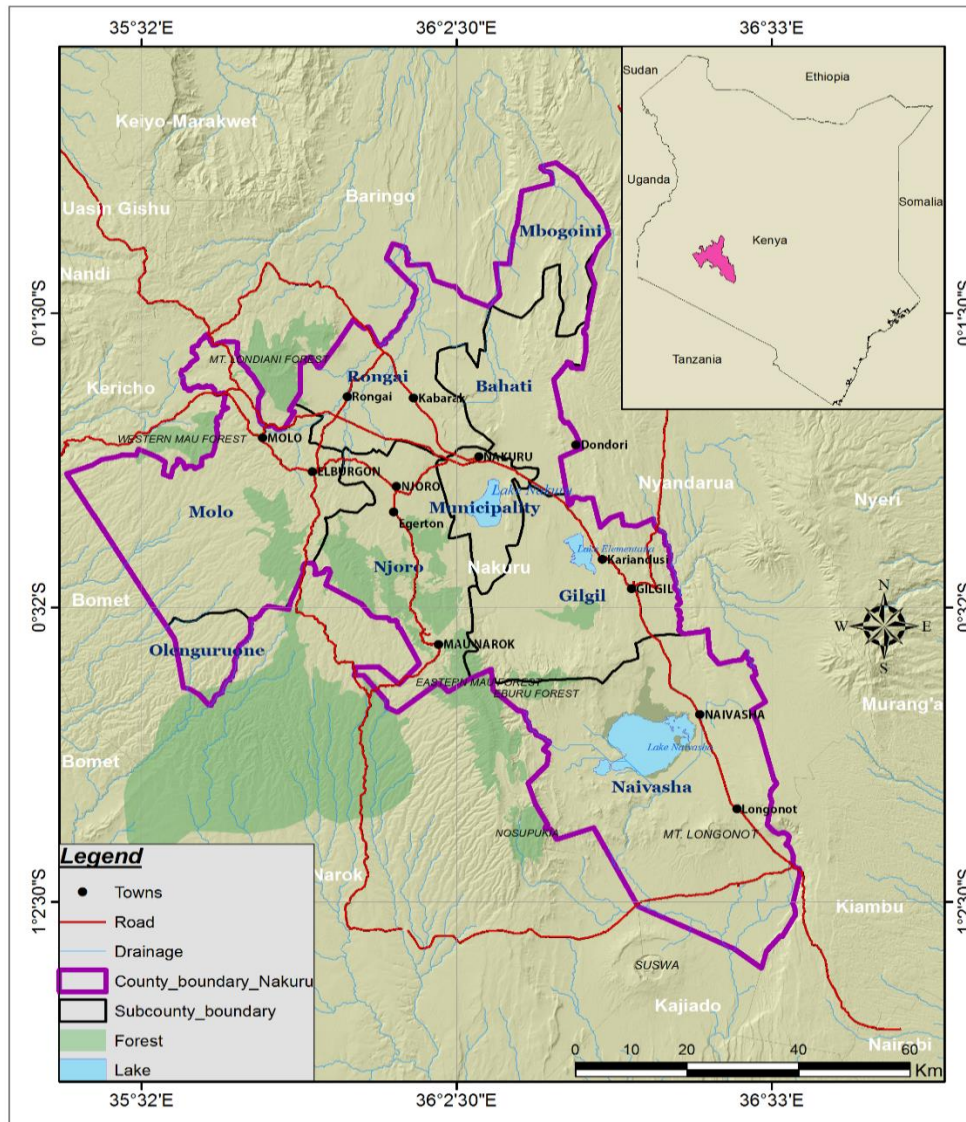


Figure 3-1: Location Map Showing Nakuru County and its Neighbours

Nakuru is strategically located along the A104 road that links Nakuru town to the International, National and Regional transportation systems. This Trans-African Highway (A104) links Mombasa to Lagos passes through Nakuru town.

3.2 Socioeconomic Setting

The County is divided into nine administrative Sub-Counties namely; Naivasha, Gilgil, Nakuru, Rongai, Nakuru North, Subukia, Njoro, Molo, and Kuresoi. Njoro and Kuresoi were hived off from Molo Sub-County, Gilgil from Naivasha, Rongai from Nakuru Town, and Subukia from Nakuru North. Table 3-1 shows the administrative units in the County with respect to Divisions, Locations, Sub Locations and number of households.

Table 3-1: Administrative Units and Area of Nakuru County

<i>Sub-County</i>	<i>Area in Km²</i>	<i>Divisions</i>	<i>Locations</i>	<i>Sub- Locations</i>	<i>Households</i>
<i>Nakuru Town</i>	297.2	3	7	21	91,110
<i>Naivasha</i>	1960.2	3	12	20	73,457
<i>Mollo</i>	478.7	4	14	28	30,783
<i>Njoro</i>	702.0	5	13	27	41,583
<i>Kuresoi</i>	1,191.0	4	22	51	51,085
<i>Rongai</i>	993.3	4	18	27	34,021
<i>Nakuru North</i>	374	2	6	17	37,525
<i>Subukia</i>	424.2	3	6	16	18,409
<i>Gilgil</i>	1074.4	3	8	12	31,861
<i>Total</i>	7495.1	31	106	219	409,836

The total population of Nakuru County stood at 1,867,461 in year 2014, comprising of 937,131 males and 930,330 females as per the projections of Kenya National Population and Housing Census of 2009. The population was projected to increase to 1,925,296, comprising of 966,154 males and 959,142 female in 2015 and to 2,046,39 comprising of 1,026,924 males and 1,019,471 females at the end of 2017 (Table 3-2). This remarkable growth in the population implies that the County will have to invest in more social and physical infrastructure to match the needs of the growing population.

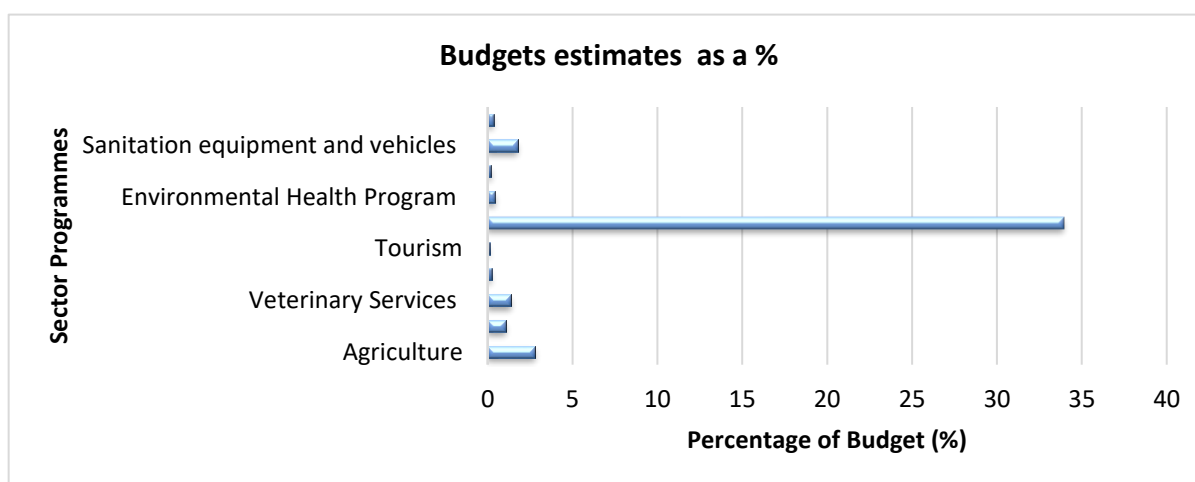
Table 3-2:County Population Projections

	2009		2014		2015 Projections		2017 Projections	
<i>Gender</i>	Male	Female	Male	Female	Male	Female	Male	Female
<i>Total</i>	804,582	798,743	937,313	930,330	966 154	959 122	1,026,924	1,019,471

Based on the County's development plan (2013-2017), the working - age population in 2012 (15-64 years) in the county was 968,745 accounting for 55.1 per cent of the total population of whom 484,378 are male while 484,366 are female. The primary working -age population comprises the employed and the unemployed.

It is expected to increase from 968,745 persons in 2012 to 1,128,338 persons in 2017. Given a labour force population which is more than half of the total population, measures will need to be put in place to provide adequate employment opportunities.

The 2015/2016 Nakuru County Annual Development Plan (ADP) forecasts for all the sectors represented at the County level has an approximate combined budget of KES 13,412,114,791.27 including entire development vote and part of operation and Maintenance services. The County treasury projects that a significant proportion of the County budget shall be financed through County internal resources while it is expected that the County Government's development partners shall bridge the gaps. The analysis of the estimates shows that more than 42 percent was intended to be spent in the priority green sectors as shown in figure 3-2.



(Source: Nakuru County Annual Development Plan, 2015-2016)

Figure 3-2: Nakuru County Budgets Estimates 2015/16

The study did not get significant response on the individual sectors on their economic significance as GDP value and economic significance as employment figures. This is a confirmation of challenge and incoherence in data acquisition and storage at the new county administration units.

The Nakuru County Fiscal Year (FY) 2014/2015 programme based budget shifted focus from input, as was the case with the 2013/2014 budget to outputs and outcomes. Programme based budgeting added performance metrics to budgets whereby programs and functions were to show output results. The County budget for the FY 2014/2015 was estimated at Ksh 9,872,659,542. The proposed budget balances with the estimated revenue for the financial year was Ksh 9.8 billion. The Government targeted raising Ksh 2 billion from local sources. Shillings 7.2 billion was to be obtained from national government transfer out of which Ksh 228 million was conditional for rural electrification (Nakuru County CIDP).

The County was to receive Ksh 138 million as loans and grants and projected Ksh 620 million conditional transfers to level 5 hospitals. The County also targeted to raise Ksh 540 million from FIF a form of Appropriation in Aid (AIA) from health facilities. The CIDP indicates that the recurrent expenditure for the

County in the FY 2014/2015 was estimated at Ksh 6,788,616,022. This includes Ksh 900 million allocated to the County Assembly. Ksh 4.6 billion (47%) of the allocation was towards recurrent expenditure to go towards employee compensation, whereas Ksh 2,149,036,783 consisted of operation and maintenance costs across the board. Allocation towards development expenditure in the FY included Ksh 1.375 billion towards Ward development projects. This was towards among other things water works, development of road infrastructure, construction and rehabilitation of health facilities.

An estimated amount of Ksh 1,029.1 billion (33%) of allocations towards development expenditure was proposed for infrastructure development including roads, water and Information Communication Technology (ICT). Shillings 38.692 million was proposed for education infrastructure for early childhood education while Ksh 26 million was proposed for youth polytechnic and resource centre.

The County health sector was estimated to receive the largest share of allocation of the budget at Ksh 3,475,968,141 accounting for 35.2 percent of the total expenditure of Ksh 9,872,659,542. This is followed by Roads, Transport and public works taking Ksh 1,115,827,945 i.e. 11.3 percent of the total expenditure. Agriculture, livestock and fisheries received 570,209,839 i.e. 5.8 percent while Environment, Water and Natural Resources was to be allocated Ksh 367,764,899 i.e. 3.7 percent.

Total allocation proposed towards development expenditure in the financial year was Ksh 3,087,043,520. The County health sector was second in terms of allocations towards development expenditure receiving Ksh 442,390,520 accounting for 14.3 percent of the allocations. The study established that Kenya shillings 784,550,000 accounting for 25.4 percent of budgetary allocation towards the development of roads in the County was largely to contribute to the opening of rural economies making it easy for farmers to trart agricultural produce as well as increase access to services such as health.

3.3 Sector Analysis

3.3.1 Agriculture

I. General economic sector indicators

Small scale and vast agricultural farms is characteristic of the county. Main crops grown and marketed include coffee, wheat, barley, maize and beans and massive silos have been established at the outskirts of the city by National Cereal Board and Lesiolo Grain Handlers Ltd. The types of fruits and vegetables grown are tomatoes, peas, carrots, onions, French beans, citrus, peaches, apples, cabbages, strawberries, asparagus and leeks. Most of these are grown in Bahati, Njoro, Molo, Rongai, Olenguruone, Nakuru Municipality, Gilgil and Mbogo-ini Divisions. There are three canning factories: Kokoto, Njoro Cannery and Kabazi Cannery. Other cash crops include tea, flowers, and pyrethrum. Most of all the barley needed for beer production is grown around Molo and Mau Narok. It is processed in Molo Town, where Kenya Malting Ltd has a factory and depots.

Table 3-3: Agriculture Sector Statistics

<i>Item</i>	<i>Number</i>
-------------	---------------

<i>Categories of Agricultural Land (in '000 Hactares)</i>	
<i>High Potential</i>	Medium Potential
291	39
	Low Potential
	231
<i>Crop Farming</i>	
<i>Average land size (Small scale in Hectares)</i>	0.77
<i>Average farm size (Large scale in Hectares)</i>	263
<i>Percentage of farmers with title deeds (%)</i>	72.5
<i>Total area under food crops (Ha)</i>	243,711.06
<i>Total area under cash crops (Ha)</i>	71,416.35
<i>Population working in agriculture</i>	942,168
<i>Livestock Farming</i>	
<i>Number of ranches</i>	17
<i>Average size of ranches (Hectares)</i>	1,011.7
<i>Main livestock bread: Cattle, Pigs, Goats, Sheep, Poultry, Rabbits and bee keeping</i>	
<i>Land carrying capacity</i>	16.5
<i>Milk production</i>	Quantity ((litres) (millions)
	182.66018
	Value (Ksh) (Billions)
	4.08975
	<i>Population working under livestock (%)</i>
	24
<i>Fisheries production</i>	Fishermen
	150
	Fish farm families
	3,500
	Population of fish farmers
	Approx 3600
	Number of fish ponds
	Approx 3500
	Landing beaches (Naivasha)
	1
	Fishing beaches (Naivasha)
	1

This company also runs an extension scheme for its farmers under contract. Pyrethrum is a natural insecticide and the County is one of the most important production areas in the world. These crops are raw materials for manufacturing industries found in Nakuru and Nairobi. Daily farming is Key economic activity and provides the inputs for the various milk processing plants.

The land area under food crops and cash crops in Nakuru County is 243,711.06 (Ha) and 71,416.35 (Ha) respectively covering approximately 3,151.240 Km² of the total area of the county. A number of factors have contributed to the increase in land under cultivation namely; the rich volcanic soils of Nakuru County that give great potential for crops, reliable rainfall in most parts of the county, readily available labour force and the availability of ready market for crop produce both in the urban centres and the proximity to other major urban centres such as Nairobi, Naivasha, Gilgil and Narok which offers incentives for the sector to flourish. There is however a substantial size of land in the county that is either arid or semi-arid but can be made productive through irrigation. The County share the national Vision of increasing land under cultivation through sustainable irrigation projects.

The most common livestock kept in the large-scale commercial farms are dairy and beef cattle, goat and sheep, with milk, meat, hide and skin, wool and mutton as their main products. The diversity and success of agriculture

in the County is underscored by the presence of even horse-breeding, at Suswa Farm in Njoro Division. The majority of the large commercial farms are found around Naivasha, Molo, Rongai, Bahati and Njoro Divisions. The dairy industry is also well developed. An increasing number of farmer households are entering the dairy business, knowing that this can bring financial stability.

Naivasha Sub-County is the great agricultural centre in Nakuru County. Horticulture is a major contributor to the County's economy, (adding to the 22% stake in the Gross Domestic Product (GDP)). Of this, the flower farming subsector contributes 1.6%. Table 4 shows the types of flowers grown in Nakuru County. The climate is very favourable for flower farming, contributing highly to employment and economic development of Nakuru County.

Table 3-4: Flower Production

<i>Flowers Grown</i>	<i>Percentage</i>
<i>Roses</i>	53.6
<i>Easter Lilies</i>	26.5
<i>Arabicum</i>	4.1
<i>Carnations</i>	3.1
<i>Hypericum</i>	1.98
<i>Others: Gypsophilla, Lilies Eryngiums, Arabicum, Hypericum, Statice</i>	

According to the Lake Naivasha Growers Group (LNGG), there are 73 flower farms in the county, with Naivasha town accounting for 48 of these. Invalidated statistics indicate that floriculture sub sector is contributing up to 1.6% to the country's GDP, has seen a great increase both in exports, local sales and profits. Unverified records from the Kenya Flower Council shows that the sub-sector exported 123,511 tons of flowers valued at KES 42.9 billion with over 500,000 people (including over 90,000 flower farm employees) directly or indirectly depending on the floriculture industry.

II. Environmental issues and related indicators

The ministry of agriculture, Nakuru County identified some of the environmental issues currently being addressed as;

- Excessive use of agrochemicals beyond the recommended rates
- Deliberate use of wrong chemicals on vegetables mainly to prolong shelf life-e.g. tomatoes
- Excessive soil erosion leading to siltation of water bodies
- Pollution of water bodies by agrochemicals
- Deforestation- clearing of forests for more cultivation
- Burning of crop residue during land preparation- common in Molo sub county

In Naivasha, floriculture has come under pressure from conservationists due to concerns surrounding overdrawing of water from Lake Naivasha for farm use, chemical pollution, poor employee working conditions and remuneration. The mounting problems has seen Karuturi, one of the largest cut rose producers in the county, at 6 billion stems of cut rose for export a day, come under receivership and face management hardships. The use of chemicals has been a contentious issue. However, due to the sensitivity of information around floriculture, the team was not able to get data on specific chemicals used in Nakuru County. According to the Pests Control Products Board the volume of fungicide, herbicides is as presented in Figure 3-3. The Table 3-5 shows volume of pesticides imported from financial year 2003/2004- 2009/2010

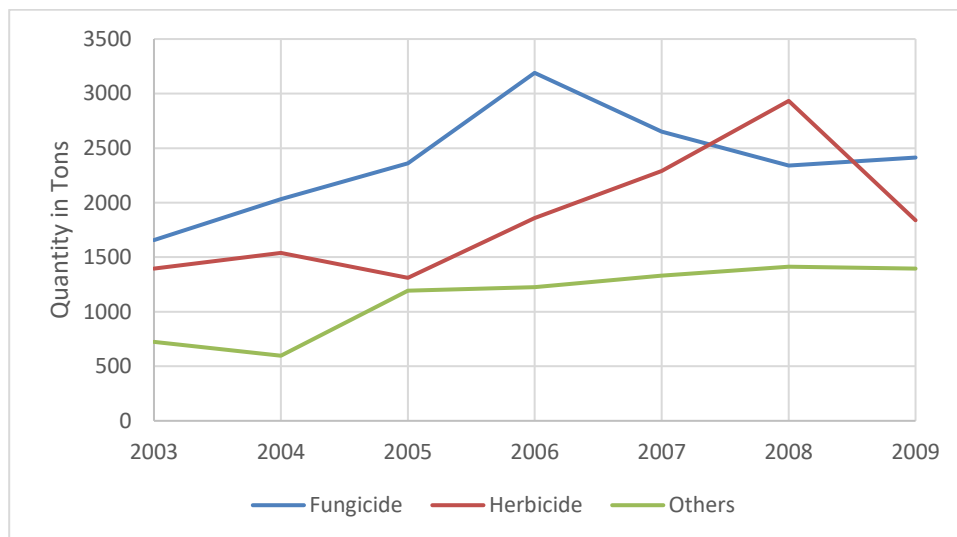


Figure 3-3: Volume of Pesticides Imported from Financial Year 2003– 2009 (PCPB, 2010).

Table 3-5: Value of Pesticides Imported from Financial Year 2003/04 – 2009/10

	2003-2004	2004-2005	2005-2006	2006/2007	2007/2008	2008/2009	2009/2010
	Value in '000 000'Kshs	Value in '000 000'Kshs	Value in '000 000'Kshs	Value in '000 000'Kshs	Value in '000 000'Kshs	Value in '000 000'Kshs	Value in '000 000'Kshs
<i>Insecticide</i>	2,411	2,077	2,031	1,181	3909	2,079	2,493
<i>Fungicide</i>	925	1,113	1,506	1,251	602	3,153	3,874
<i>Herbicide</i>	571	650	620	324	206	944	939
<i>Others</i>	142	133	337	362	191	1,167	918

(Source: PCPB, 2010)

III. Well-being and equity indicators

There has been a general increase in both crop yields and acreage under has presented in the figure 3-4 to figure 3-6

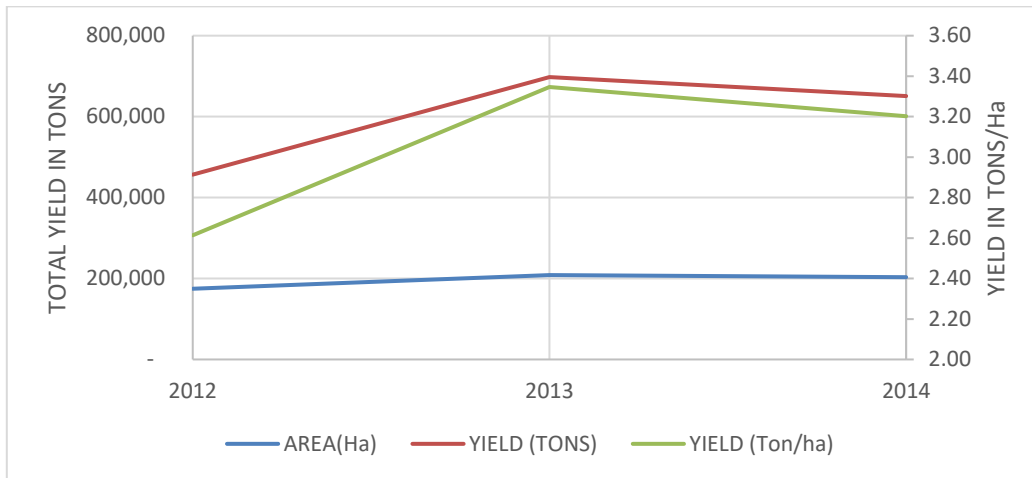


Figure 3-4: County Total food Crop Production 2012 - 2014

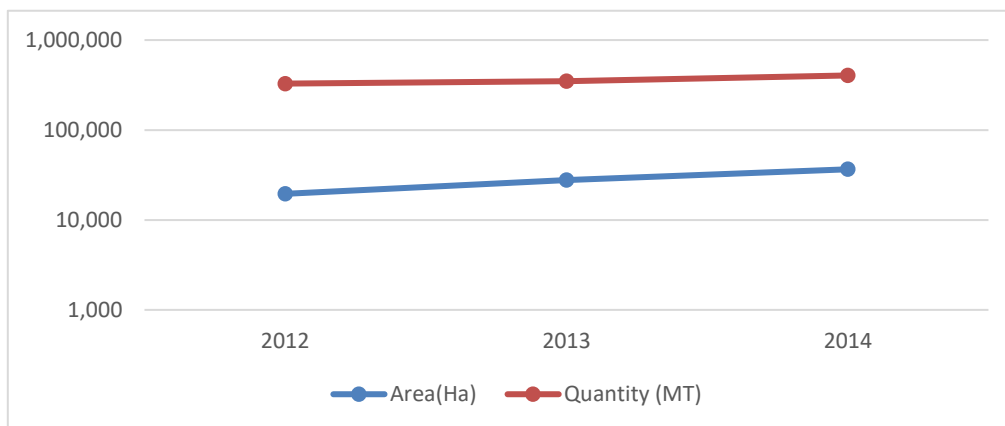


Figure 3-5: County Horticulture Production 2012 - 2014

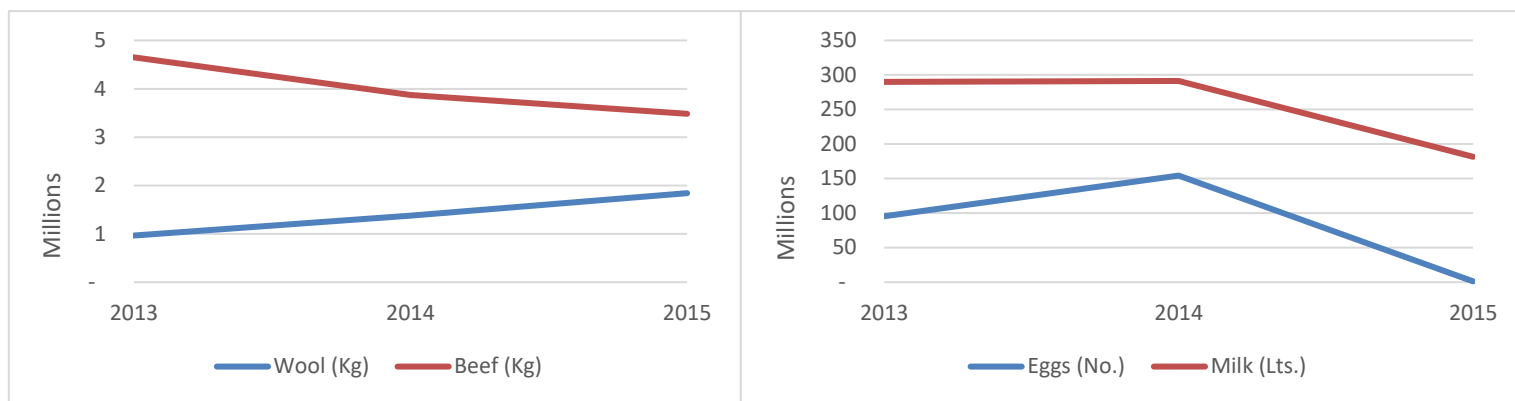


Figure 3-6: Nakuru County Beef and Wool Production; Milk and Eggs Production 2013 to 2015

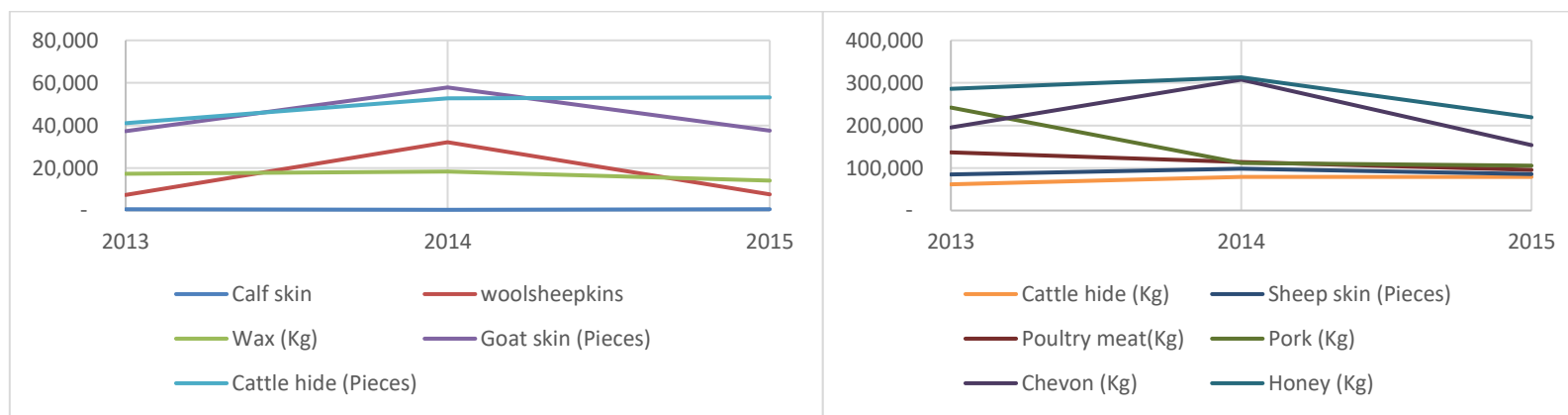


Figure 3-7: Nakuru County Livestock Production 2013 to 2015

IV. Green economy policy interventions

Nakuru County Integrated Development Plan (2013-2017) has identified projects and programmes in agriculture in accordance with Medium Term Expenditure Framework. The strategy for intervention in the agriculture sector include;

- the adoption of improved land and water management practices;
- adoption of renewable energy for agricultural activities, especially in effective utilisation value addition and preservation;
- the utilisation of farm biomass for conservation and energy production;
- adoption of high value crops both for subsistence and commercial farming is being encouraged;
- Crop development sub-sector to sensitize farmers to shift from subsistence farming to commercial farming under the slogan Kilimo Biashara.
- The livestock sub-sector may train farmers on proper animals husbandry practises, disease control for improve livestock productivity and further liaise with the community and the private sector on provision of veterinary clinical and extension services.
- Fisheries sub sector may use a complete fish value chain approach through training of farmers and fishermen to increase production and also markets infrastructures establishment to reduce postharvest losses.
- The irrigation sub sector can facilitate small scale farmers to put more land under irrigation and ensure mitigation on changing climate and overreliance on rain fed agriculture.
- The forestry Sub-Sector can focus on afforestation and reforestation programmes in order to increase the tree cover towards the 10% recommended national target.

3.3.2 Water Sector

I. General economic sector indicators

Nakuru County is endowed with natural water resources including four major lakes, Nakuru, Naivasha, Solai and Elementaita. In addition, there are rivers, shallow wells, springs, dams, pans and boreholes spread all over the county especially in drier parts of Naivasha, Gilgil, Molo, Njoro and Rongai. As shown in table 3-6, the boreholes have boosted water supply in the county. Some of the major rivers include, Malewa, Njoro, Molo and Igwamiti. The county is also endowed with springs found in Subukia, Nakuru North, Molo and Kuresoi areas. Rain water is another major source of water in the county with about 80 per cent of households harvesting rain-water.

Table 3-6: Water Sector Statistics

<i>Item</i>	<i>Numbers</i>
<i>Households with access to piped water</i>	150,608
<i>Households with access to potable water</i>	258,619
<i>Number of permanent rivers</i>	6
<i>No. of shallow wells</i>	7,800
<i>No. of protected springs</i>	8
<i>No. of un-protected springs</i>	16
<i>No. of water pans</i>	40
<i>No. of Dams</i>	10
<i>No. of Bore holes</i>	60
<i>Households with roof catchment systems</i>	80%
<i>Number of Water Resource User Associations (WRUA)</i>	3
<i>Distribution of Households by main source of water (%)</i>	
Piped into dwelling	8.0
Rain/harvested	28.8
Borehole	3.1
Protected well	14.2
Protected spring	5.8
Stream	2.7
Jabias	16.9
Water Vendor	0.2
Pond	14.2
Dam	0.4
Lake	1.6
Others	0.4
<i>Percentage distribution of Households by Time taken to fetch Drinking Water (in minutes for one way)</i>	
0	13.3
1-4	35.5
5-14	29
15-29	10.4
30-59	5.9
60+	6.1

II. Environmental issues and related indicators

Water pollution remains a major environmental issue especially in all water sources in Nakuru. There is high input of solid and chemical pollutants in the water systems. The increase in water levels of all lakes is a worrying trend which is associated with changed rain pattern, with heavy rains occurring during the short rains season. The lake level increase is threatening the ecosystem of Lake Nakuru Park. Lake Nakuru is a tourist attraction point due to flamingos feeding ground. The alteration of the water pH may alter the

amount and types of algae that grow in the lake and are food for the flamingos. With increase in population river flow and morphology is affected by the upstream land use change (Figure 8).



Figure 3-8: Land use change in upper Gilgil area

Consequences of this upstream changes are heavy sediment loads into the flow streams and high treatment costs for potable supplies to towns downstream.

III. Wellbeing and equity indicators

To counter the effects of drought and inadequate supply of clean drinking water, the community in different parts of the county have formed groups and established vibrant water schemes. The schemes are funded through members' contribution, government through Rift Valley Water Service Board (RVWSB), Constituency Development Fund (CDF) and donors' funds such as Community Development Trust Fund (CDTF) and World Vision Kenya. The schemes mainly provide water to the urban centres in the county. Malewa- Nakuru East Water Project is a main source of water supply to Nakuru and Gilgil towns. Nguzo Springs supplies water to Molo, Rongai and Njoro areas. Solai Wells near Bahati area supplies water to Nakuru town, and also the adjacent Bahati areas (Nakuru CIDP 2013).

The 2009 Population and Housing Census indicated that 85 per cent of the residents had access to private improved sanitation. In rural areas, open defecation was estimated to be still practiced by 0.03 per cent of the population. Lack of affordable housing in the major towns in the county has led to mushrooming of informal settlements (slums) in these urban areas resulting in poor sanitation and poor management of both solid and liquid waste. There will be need therefore, for enhanced measures to ensure proper physical planning and management of waste disposal in the county.

Almost all the population in Nakuru County have access to sanitation facilities including pit latrines and flush toilets. Sewerage Treatment Works located in Nakuru and Naivasha are manned by the respective Water Service Providers.

<i>WSP</i>	NAKURU TOWN	NAKURU RURAL	NAIVASHA
<i>Total population in service area</i>	383,169	511,586	275,000

<i>Total population in service area served</i>	357,696	123,978	178,402
<i>Total no. of connections</i>	48,157	19,028	2,164
<i>Total no. of active connections</i>	43,918	7,682	2,042
<i>NRW</i>	46	63	36
<i>Water coverage</i>	93	24	65
<i>Sanitation coverage</i>	77	71	75
<i>Hours of Supply</i>	18	9	14

Source WASREB, IMPACT Report 2014

Some of the activities undertaken by Department towards improvement of Sanitation include: Fencing of Treatment Works, Rehabilitation of existing sewerage works in Naivasha, Nakuru, Molo, extension of the existing sewerage network within Nakuru Town that can be served by the Sewerage Network. There will also be a major sanitation improvement in Gilani, Kaptembwo, Kwa Rhonda in Nakuru and Kihoto, Karagita, Kamere Tarabete/Kasarani in Naivasha under KISIP (Kenya Informal Settlements Improvement Project) being undertaken by the Ministry of Housing.

The Nakuru County Strategic plan spells the changes they have put in place to change water sector: Prepare a master plan on water development; incorporate all the water undertakers in the planning process; rehabilitate, drill and equip boreholes; construct dams along rivers- Kuresoi dam; construct pans; promote community management of water projects in order to:

- Ensure increase in water sources.
- Increase access to piped water to 60 per cent by 2017

The water Sub sector will strive to increase coverage of water supply, sanitation and drainage services in Nakuru County by 20 percent across all the service levels in the next five years ending in 2017.

Main challenges include: Inadequate water per capita; inadequate piped / clean drinking water; low investment in water development; and low and unreliable rainfall patterns.

IV. Green economy policy interventions

Several initiatives that can endear the green economy agenda, have been documented in the Nakuru CIDP. The water Sub sector will strive to increase coverage of water supply, sanitation and drainage services in Nakuru County by 20% across all the service levels in the next five year.

Table3-7 highlights some of the water sector priorities, constraints and proposed development strategies that Green Economy programme can be anchored upon.

Table 3-7: Green Economy Initiatives in the Water Sector

Priority	Constraints	GE Initiatives
<i>Increase coverage of water supply, sanitation and drainage services by 20% across all the service levels</i>	Water Scarcity; very low coverage of sewerage services; high production costs; poor water quality; inadequate funds and high cost of capital investment; infancy of the County water department; conflicting water uses; stringent legal and regulatory framework	Deepen technical, human and financial capacity Minimize conflicts and disputes among water service users; Mainstream cross -cutting issues on gender, legal, environmental and poverty reduction strategies De-silting of dams; rehabilitation of treatments works; pipeline extensions and rehabilitation.

3.3.3 Forestry

I. General economic sector indicators

The natural environment is such a system, which is viewed as a source of human well-being (Dasgupta 2001). Ideally, an index may be constructed to serve as a measure that informs about the state and trends of the environment. However, over the past decade or so, efforts to design credible indicators of environmental quality have been wrought with difficulty for the simple fact that a widely accepted set of indicators capable of capturing the state of the environment is hard to come by. At the level of a jurisdiction, forest cover is often used to indicate the soundness of the natural environment.

Nakuru County has a number of special but dispersed forest areas. These include Menengai forest and Mau complex.

- **Menengai Forest:** This is an urban forest situated within Nakuru town. The epicenter of the forest is the Menengai Crater. It was gazetted as a forest in the 1930s. It is surrounded by residential areas of Milimani Estate in the South, Ngachura and Bahati in the East, Solai in the North and Olo-Rongai in the West. Various Government of Kenya facilities have been hived off from the forest; these include the Kenya Broadcasting Corporation and the Nakuru G.K Prison.
- **Mau Complex:** This forest is of particular importance to the tea and tourism sectors of the county. It is also the source of water supply to many urban centres in the county. It was one of the two largest forest blocks in the Mau Complex, covering 65,921 hectares, of which 35,301 were excised in 2001. It is the main upper catchment of the four rivers (Makalia, Naishi, Nderit, Njoro) flowing into Lake Nakuru.

The UNEP report cited 35,301 hectares representing 54 percent of forest area were excised in 2001 but contested before the High Court. The excised area is the main catchments for Lake Nakuru. It also covers main ridges and peaks along the top of the Mau Escarpment, including areas between 2,800 and 3,000 above sea level that were covered with bamboo forests, a vegetation cover with high catchment values.

In the excised area, there were still some indigenous forest areas with very low settlement density near and over the summit of the Mau Escarpment that should be recovered.

Individual farmers are encouraged to adopt agroforestry right from the small scale of 2 Acres in Olenguruoni (Figure 9). NGO's such as Golden Scape Tree Africa provide Eucalyptus seedlings to farmers and assist in digging holes and planting.



Figure 3-9: Smallholder Agroforestry Plot

A joint research by University of Wyoming and Egerton University on land use cover for the period 1989 to 2003 showed that from 1986 to 1989 the landscape was relatively static with small scale changes in all informational classes. The greatest land cover changes occurred after 1989, with rapid loss in plantation forest and the conversion of forested areas into small-scale (mixed) agriculture. Additionally, several larger agricultural areas were transformed into smallholdings. By 2003 the majority of plantation forest are converted to small-scale agriculture and there is large net loss in large agricultural systems. Overall the amount of grasslands increased through time, primarily in the uppermost reaches of the watershed. The trend in forest loss was unsurprising given the historical knowledge of the area. The Mau forest has been undergoing significant losses throughout its extent (Kenya Forest Working Group 2001). It was somewhat surprising to see rapid plantation forest losses between 2000 and 2003 in the uppermost regions of the watershed, given that the area was not converted to small-scale agriculture. We hypothesize that the plantations are not being maintained under sound agroforestry practices and are instead being transformed into more natural forests, rendering them less spectrally distinct from the indigenous forest.

II. Environmental issues and related indicators

The stability of the forest sector and secured provision of ecological goods and services are essential to the long term prosperity of the county's economy. The Mau Complex has been impacted by extensive illegal, irregular and ill-planned settlements, as well as illegal forest resources extraction. The widespread encroachment and excursions of the Mau Complex has led to the destruction of some 104,000 hectares representing over 24 percent of the Mau Complex area over the last 10 years. In addition, some 43,700 hectares have been encroached in the remaining protected forests of the Mau Complex (UNEP 2008). In

the forests of Marmanet, some 11,000 to 12,000 hectares have been lost to illegal and irregular settlements. Such an extensive and on-going destruction of key natural assets for the country is a matter of national concern. It presents significant environmental and economic threats and underlines a breakdown of law and order, with ramification to internal security and conflicts.

Forest sector is faced with such environmental challenges such as: high demand for timber; high demand for charcoal and fuel wood; forest encroachment for farming and plantations; non adherence to existing policies; population pressure and settlements; forest and bush fires; poor planning for settlement and lack of prioritization.

III. Wellbeing and equity indicators

An estimate of the value of goods and services provided by the Mau Complex is well over Ksh 20 billion (UNEP 2008). Eastern Mau Forest Reserve is located in Nakuru District and managed by the Kenya Forest Service. The table 3-8 offers an overview of the forestry sector.

Table 3-8: Forestry Sector Statistics

ITEM	NUMBER
<i>Number of gazetted forests</i>	2
<i>No. of Non-gazetted forests</i>	6
<i>Size of gazetted forests (Km²)</i>	679.643
<i>Main forest products</i>	Timber, Poles(transmission), Charcoal, Firewood, Bamboos, Withies
<i>No. of people engaged in forestry</i>	5,000
<i>Seedlings production</i>	350,000
<i>No. of farms engaged in farm forestry (farm families)</i>	45,000
<i>Average no. of trees per farm</i>	500
<i>Non-timber forest products harvested</i>	Honey, Resin, Herbs, Minerals
<i>No. of Community Forest Associations (CFA)</i>	11
<i>Number and names of endangered plant species found within forests</i>	Cedar, Podo, Prunes, Sandal wood
<i>Quantity of timber produced (tonnes)</i>	5,000

IV. Green economy policy interventions

Kenya aims to be a nation that has a clean, secure and sustainable environment by 2030. This will be realized through increased forest cover and reduction in all environment-related diseases. Through promotion of environmental conservation Kenya can achieve the Millennium Development Goals (MDGs) by addressing environmental degradation and pollution. The Njoro River Rehabilitation Project by the Egerton University is one such initiatives in line with the main objective of the Kenyan Government

of achieving a clean and secure environment by year 2030. Medium term goals identified by the government are: (i) to increase forest cover from less than 3% at present to 4%; and (ii) to lessen by half all environment-related diseases. Table 3-9 presents the identified priority projects, constraints and Green Economy opportunities within Nakuru County.

Table 3-9: GreenEconomy Initiatives in the Forestry Sector.

<i>Priority</i>	<i>Constraints</i>	<i>GE Initiatives</i>
<i>Restoration of 10% forest cove</i>	Uncontrolled tree cutting, bush clearing, logging and deforestation	Promotion of agro-forestry and afforestation; forest demarcation; restoration of riverbanks and sustainable forest resources management; adherence to land policies; and local stakeholder's involvement in policy formulation.
<i>Recovery of the grabbed forest land</i>	Increased settlement in forested lands.	Full enforcement of the Forest Act and EMCA

3.3.4 Energy Sector

I. General economic sector indicators

Access to reliable energy is critical for growth of all other sectors. According to the Population and Housing Census of 2009, 34 percent of the County households which translates to 139,430 households, use electricity for lighting. However, 85.7 percent of these connections are in urban areas. Within the rural areas, 36.1 percent and 25.3 percent of the rural households utilize lantern and tin lamp respectively as common sources of lighting. The Tables 10 and 11 show sources of lighting and cooking fuel in Nakuru County.

Table 3-10: Sources of Lighting Fuel in Percentage

<i>Main Source of Lighting Fuel</i>	<i>County</i>	<i>National</i>
<i>Electricity</i>	34.0	22.7
<i>Solar</i>	2.3	1.6
<i>Gas Lamp</i>	0.5	1.0
<i>Pressure Lamp</i>	0.6	0.6
<i>Lantern</i>	36.1	30.5
<i>Tin lamp</i>	25.3	38.5
<i>Fuel wood</i>	0.8	4.5
<i>Others</i>	0.5	0.7

Source: Nakuru CIDP 2013

Table 3-11: Sources of Cooking Fuel in Percentage

Households by main type of Cooking fuel	Nakuru County	National
Gas (LPG)	1.1	0.8
Biogas	4.6	5.0
Solar	0.7	0.7
Paraffin	0.1	0.1
Firewood	7.8	11.6
Charcoal	85.6	64.6

Source: Nakuru CIDP 2013

Charcoal remains the main source of energy majorly relied on for cooking at 85.6 percent against the national rate of 64.6 percent. This is because charcoal is easier to access in terms of cost. Further, due to over-reliance on hydro-electricity, the frequency of power outages in the County is high and frequently leads to production losses. The high production costs associated with unreliable and expensive energy makes the County's agricultural exports mainly pyrethrum and horticulture less competitive in global markets. It also weakens the potentiality of the urban centres in the County to become industrialized.

Nakuru County boasts of a high geothermal potential due to its strategic geographic location at the floor of the Rift Valley that has seismic activities. Nonetheless, the economic potential of this geothermal energy as an alternative source of clean and green energy in addition to the potential revenue resources are yet to be fully exploited. There are more than 6 high temperature geothermal potential sites within Nakuru County with an estimated potential of more than 5,000 MW. They include: Menengai; Eburru-Badlands; Arus-Bogoria; Olkaria; Longonot; and Suswa. The Menengai Geothermal project is expected to provide great impetus to the development of Nakuru town and has the potential to transform Nakuru into a mega city.

II. Environmental issues and related indicators

The sector faces diverse environmental challenges that include: Lack of attention to local governance in most approaches; lack of access to clean sources of energy; lack of access to sufficient and sustainable supplies of energy; widespread inefficient production and use of traditional energy sources; and highly uneven distribution and use of modern energy sources, such as electricity, petroleum products, and liquefied or compressed natural gas, which pose important issues of economics, equity, and quality of life. The Mau Complex catchments is estimated at 189.4 megawatts with an average annual energy production of 960 gigawatts-hour. The sale value of the average energy production on these sites will be in the range of Kshs 10 billion per year. In order to secure the installed hydropower capacity and the yet to be developed potential of the Mau Complex catchments, it is imperative to secure and conserve their forests. However, the county is taking a leading role in renewable energy installations, particularly geothermal energy (Table 12).

Table 3-12: Renewable Energy Installed Capacity 2004 – 2012 (MW)

TYPE	2004	2005	2006	2007	2008	2009	2010	2011	2012
Geothermal	128	128	128	128	128	158	189	190.6	199.6

Source: KNBS Statistical Abstract 2013

III. Wellbeing and equity indicators

A total of 34 percent of residents in Nakuru County use electricity as their main source of lighting. A further 36 percent use lanterns, and 25 percent use tin lamps. One percent use fuel wood. Electricity use is mostly common/dominantly in male headed households at 35 percent as compared with female headed households at 31%.

The study looked into stakeholder use of energy and perception on different factors that determine the access. These findings are presented in figures 3-10, 3-11, 3-12, and 3-13

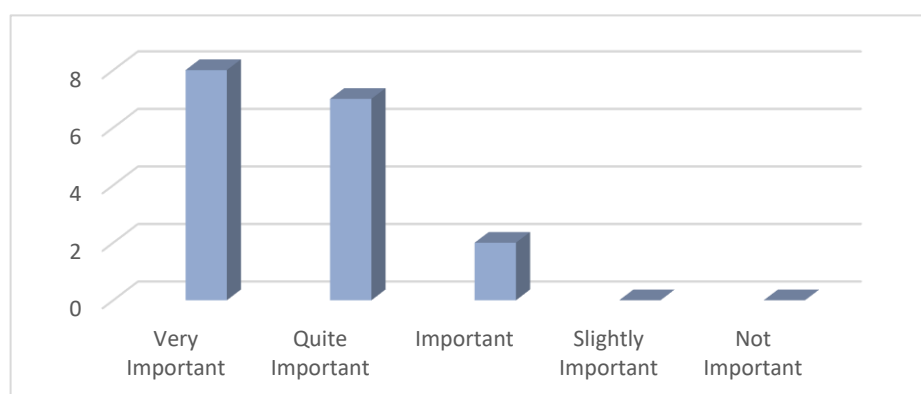


Figure 3-10: Reliability as an important factor in energy source

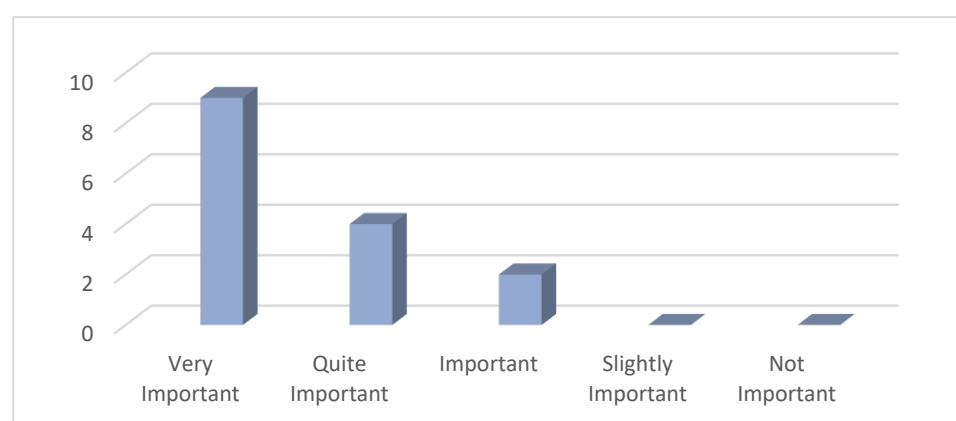


Figure 3-11: Cost as an important factor in energy source

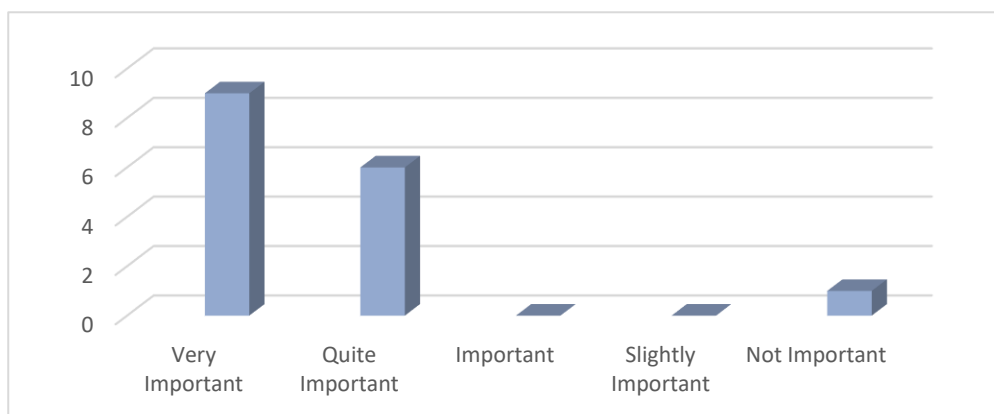


Figure 3-12: Eco-friendly (low carbon footprint)

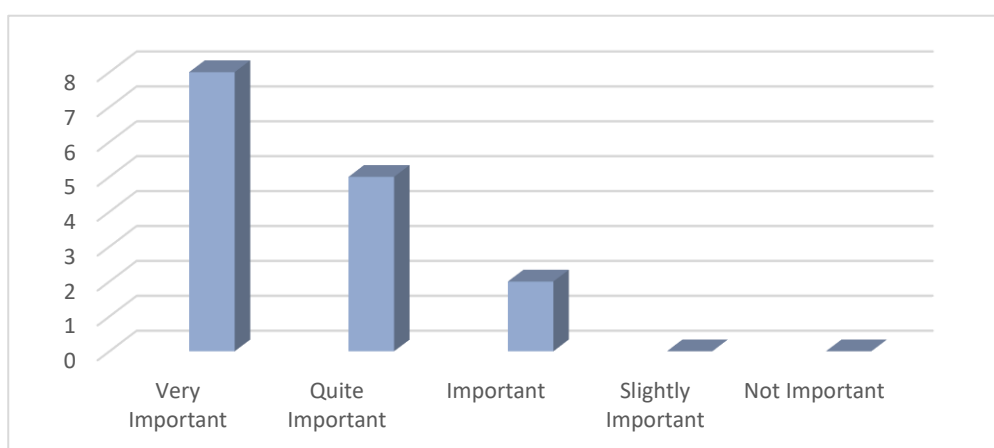


Figure 3-13: Availability as an important factor in energy source

Only 5 percent of residents in Nakuru County use Liquefied Petroleum Gas (LPG), and 8 percent use paraffin. 46 percent use fire wood and 40 percent use charcoal. Firewood is the most common cooking fuel by gender at 44 percent in male headed households and 50 percent in female headed households.

IV. Green economy policy interventions

Major issues of concern on energy access in the County in the context of what needs to be done to enhance development of energy resources and technologies:

- **Geothermal:** Resource management is at the national level; the high cost of putting up geothermal power stations is prohibitive; and resource is often found on private land, which means people are displaced for its development.
- **Solid waste:** Although solid waste offers a good opportunity for energy production, its use is inhibited due to socio-cultural issues—negative attitudes associated with using human waste
- **Biogas:** Installation is costly; there is limited capacity to construct and control quality of installation; private ownership of land could be a hindrance particularly for the

Development of centralized systems. On a positive note, however, the Nakuru County government is already embracing the technology

- **Solar Home Systems:** High initial cost of installation hinders uptake; installation standards not maintained; there is lack of awareness on potential for solar energy and lack of quality control on equipment.
- **Improved Cook Stoves:** Low levels of public awareness amongst households and institutions, there is limited distribution of ICS across the county, and quality control of locally manufactured ICS remains a key concern.

There are a number of projects committed to improve generation in the immediate to mid-term. Estimated committed generation between 2012 and 2018 is 2,639MW. An extract for the Nakuru county projects are presented in Table 13.

Table 3-13: Energy sector commitment for Green Economy in Nakuru County

KenGen	Type	Output (MW)	Est Commissioning
Olkaria I unit 6	Geothermal	70	Jun-2015
Olkaria V	Geothermal	140	Jun-2017
Eburru	Geothermal	25	Jun-2017
Well head Generation	Geothermal	10	2014
Well head Generation Units	Geothermal	30	2014
Bogoria - Silali Well Head Generation Units	Geothermal	75	2017
Bogoria - Silali Well Head Generation Units	Geothermal	75	2018
Menengai Phase I Power Project	Geothermal	400	300MW May-2017 100MW Sep-2017

Table 3-14: Green Economy interventions in the energy sector

GE Initiatives	Constraints	Activities
Connecting of all county public primary schools.	Way leave acquisition; vandalism; inadequate funding; high connection charges.	Public awareness; source for more funding, seek for government subsidies
Harness alternative energy sources	High acquisition cost; limited skills and manpower	Promotion and establishment of solar, wind mills and biogas.

The following were identified as emerging issues in the energy sector within Nakuru County:

- Energy as a key driver of economic growth-Access to reliable energy is critical for growth of all other sectors of the economy
- Energy for domestic use characterised by inefficient consumption patterns- still heavily dependent on wood fuel and charcoal, leading to illegal logging, charcoal burning and harvesting of indigenous forests.
- Presence of high potential green energy sources-apart from the Menengai Geothermal Project which is underway, the county is yet to fully exploit other high potential geothermal sites.

3.3.5 Transport and Infrastructure

I. General economic sector indicators

Nakuru County seeks to be a world class provider of cost-effective physical and ICT infrastructure facilities and services. It endeavors to provide efficient, affordable and reliable infrastructure for sustainable economic growth and development through construction, modernization, rehabilitation and effective management of all infrastructure facilities (Nakuru County CIDP 2013-2017).

Table 3-15: Transport Sector Statistics

Items		Numbers
Road length in KM	Bitumen	911.9
	Gravel Surface	1,110.8
	Earth Surface	2,326.6
	Total	4,249.3
Condition of roads	Good	20%
	Fair	35%
	Poor	45%
Railway line length (km)		192
Railway stations		10
Airstrips		5
Mobile Network coverage (%)		91

The County government's efforts come against a backdrop of a road network of only 20% rated good with old and dilapidated bridges. The state is due to lack of regular rehabilitation and maintenance. Fortunately, the creation of Kenya Urban Roads Authority, Kenya Rural Roads Authority and Kenya Highways Authority is expected to improve efficiency and reduce procurement bottlenecks that have slowed the subsector in the past. The main types of transport include: Land transport: animal and human portage, road, pipeline and railway.

The County has a railway line length of 192 Km connecting major urban areas of the county namely; Naivasha, Gilgil, Nakuru, Njoro, Molo and Rongai. It has ten railway stations serving as drop/ collecting points for agricultural and industrial good as well as providing public transport from Nakuru to the Mombasa via Nairobi and Nakuru to Uganda via Eldoret, Kisumu, Busia and Malaba.

II. Environmental issues and related indicators

Despite the optimism and the growing demand for transport services, Nakuru County road sector is currently facing environmental challenges which include:

- **Loss of public space:** The increasing traffic within Nakuru Town has adverse impacts on public access to markets, churches, farms, community interactions and public amenities.
- **Pollution:** Energy consumption by vehicles within and transiting through major urban areas of the County is a major contributor to noise, air, water and soil pollution. The problem is associated with health concerns in the urban centres.
- **Accidents and safety:** Growing traffic in Nakuru County is linked with a growing number of accidents and fatalities especially along the Naivasha –Nakuru highway. Accidents account for a significant share of recurring delays.
- **Land consumption:** The increasing traffic within and across the County has increased demand for parking and holding spaces. This limits land available for other developments.
- **Funding:** The inadequate funding to the sector is hampering effective enforcement of environmental regulations and standards in the road sector. This has resulted into questionable road and transport related projects.

III. Wellbeing and equity indicators

Statistics from KNBS show that approximately 911.9 Km of roads in the County have bitumen surface, 1,110.8Km are of gravel surface and 2,326.6Km of earth surface. The Nairobi – Uganda highway runs across the County thus promoting cross-border interconnections within the three East African Countries. The County has a large network of unclassified roads that are maintained through funds disbursed from various agencies including County Government, Kenya Urban Roads Authority, Kenya National Highway Authority, Kenya Rural Roads Authority and Constituency Development Fund.

IV. Green economy policy interventions

The matrix highlights of priority areas guiding the counties transition to a Green Economy and selected development initiatives to anchor the Green Economy programme is presented in Table 3-16.

Table 3-16: Green Economy Interventions

Priority	Development Initiatives
----------	-------------------------

Street lights and high mast floodlights	Construct lights that are environmentally friendly e.g. solar powered
Storm water drainage networks	Preparation of maintenance master plan and policy
Design and construction of parks	Source for PPP and donors in funding the parks
Elaborate road furniture project	Construct road furniture, traffic lights and signals
SMART urban management	Develop traffic management plan and policy
Road safety and business opportunity	Source for PPP and donors for funding
Vehicle fleet mobility and service delivery	Development of county fleet management plan and policy
The physically challenged	Disability mainstreaming in public places
Communication and economic activities	Airport construction and airstrip rehabilitation
Transport of farm products	Open new roads, construct bridges and maintain roadside drainages
Provision of shelter	Boda boda sheds for easy revenue generation
Information access and communication	Install internet infrastructure
Improved service delivery	Automation of county Operations, revenue collection and initiate digital villages and ICT centres

3.3.6 Tourism

I. General economic sector indicators

Nakuru County boasts of some of the world's only attractions. These spread across the vast lands of the county from Naivasha to Mau Summit attractions such as Lake Naivasha, Mount Longonot, Hell's Gate National Park, Crater Lake, and Longonot National Park. Other sites of interest around Nakuru include Menengai Crater, an extinct volcano 2,490m (8,167 ft.) high. The views of the crater itself, as well as the surrounding countryside, are spectacular.

Table 3-17: Tourism Sector Statistics

Items	Numbers
Staff of KWS	154
Camps	14
Number and names of endangered animal species	White Rhino (68), Black Rhino (42)
Unclassified hotels	171
Number of tourism hotels	220
Hotel bed capacity	12,000 beds

The Kenya National Bureau of Statistics categorises Nakuru County in the Central Kenya tourist zone, whose statistics is presented in Figure 3-14. The zone enjoys a huge opportunity for growth from its boasts of East African guests. The visitors' trends to the Lake Nakuru National Park is also presented in Figure 3-15.

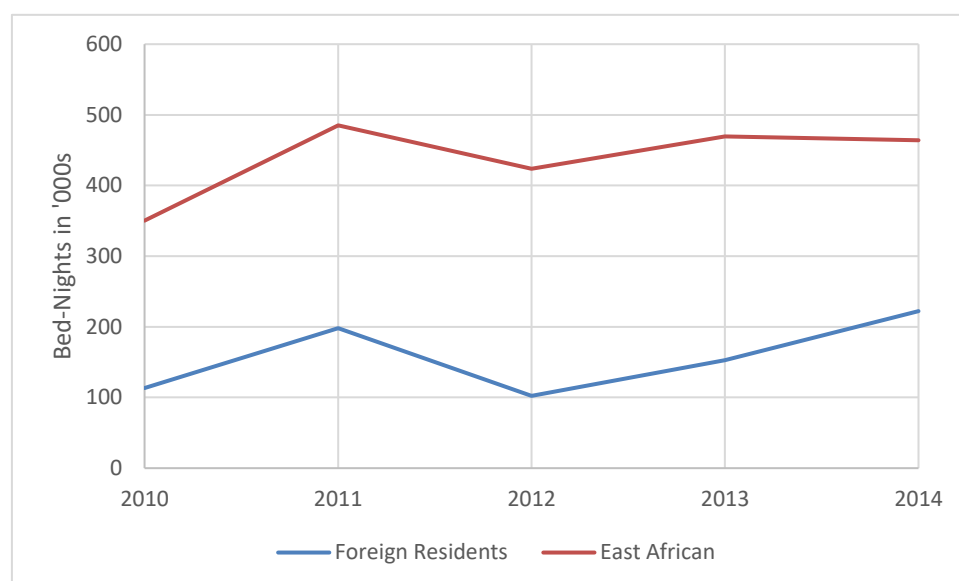


Figure 3-14: Beds Occupied, 2010 - 2014

Some of the key parks and tourist attractions in Nakuru are herein described:

- **Hyrax Hill:** Prehistoric Site, discovered by the Leakey's in 1926, is considered a major Neolithic and Iron Age site. The adjoining museum features finds from various nearby excavations. The second largest surviving volcanic crater in the world, the Menengai Crater is 2,242 meters above sea level at its highest point. The crater plunges 483 m down from the rim and the summit is accessible by foot or vehicle 8 km from the main road. The mountain is also surrounded by a nature reserve. Wildlife enthusiasts can enjoy the value of conservation and rich natural capital the ecosystem provides.
- **Lake Nakuru Park:** Gives shelter to thousands of flamingos and many other species of birds. The Park has as many as 56 varied species of mammals, including rhinos, buffalos and giraffes roaming freely in this vibrant ecosystem. Its potential in promoting the Green Economy agenda is enormous. Tourists from all over the world come not only to sight wild animals, but also to explore rough terrains of the region. Some of the famous view points in the park are Baboon Cliff, Lion Hill and Out of Africa. This national park also houses Makalia, which is a far-famed waterfall. Since there is so much to explore in the park, tourists relish each and every moment of their stay here. Nakuru has many tourist places to visit. There are very nice sightseeing, attractions in which are must see.

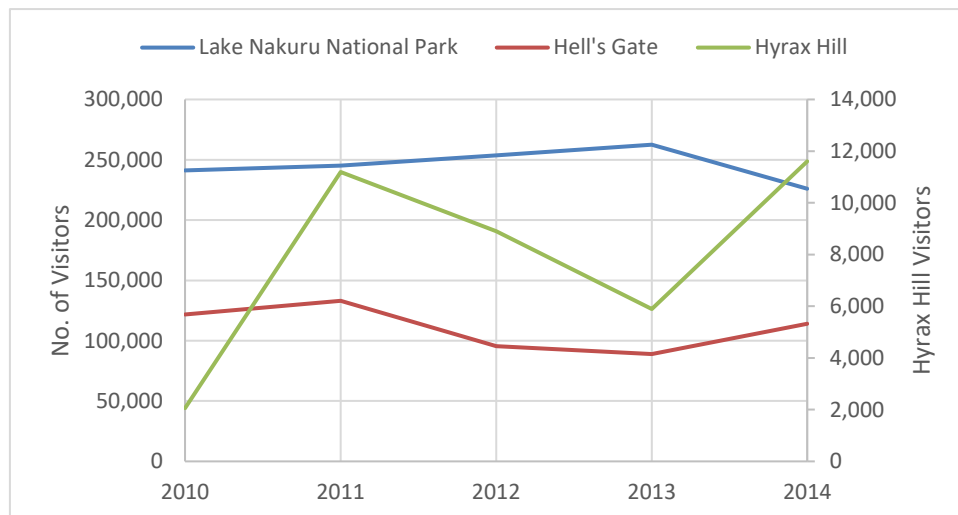


Figure 3-15: Number of Visitors to Nakuru National Park, 2010 - 2014

- **Hell's Gate National Park:** The Park lies south of Lake Naivasha, Northwest of Nairobi. The stunning park is famous for its natural rock towers scaling high, cliffs, flat plains and eroded gulleys. Olkaria and Hobley's are two extinct volcanoes located in the park. In the Hell's Gate Gorge, lined with red cliffs are two volcanic plugs: Fischer's Tower and Central Tower. The Central Tower is the smaller gorge, which extends to the south where a path descends into the hot springs. It has abundant plains game such as buffalo, zebra, eland, hartebeest, giraffe, baboons, Thomson's gazelles, the rare Chandler's mountain reedbuck, lion, leopard and cheetah. Over 100 species of birds have been recorded including vultures, Verreaux's eagles, augur buzzard and swifts. The cliffs once hosted the now almost extinct population of the Lammergeyer or the bearded vulture that used the cliffs to drop animal bones from a height to break them open and scoop out the marrow. The cliffs are home to a shy antelope, the Klipspringer that is adapted to living on the rocks.

Nakuru County has over 70 world class hotels and lodges, which are famous for their unique sites and comfort, including:

- **The Sleeping Warrior Lodge:** The Sleeping Warrior Lodge offers an intimately relaxed tented camp and cottages popular with local families and overseas visitors alike.
- **ARR Kongoni Lodge:** This Lodge is located on the Kongoni Ranch providing a scenic view of Lake Naivasha.
- **Malewa Wildlife Lodge:** The Lodge is on the shores of the Malewa River in the Kigio Conservancy, Naivasha. Accommodation is in traditionally styled buildings.
- **Naivasha Kongoni Lodge:** The Lodge offers the ultimate personal comfort, luxury and innovation – for business and pleasure.
- **Lake Naivasha Country Club:** Perfect for active and leisure travelers. Lake Naivasha Country Club offers an abundance of activities to please all visitors.

- *Lake Naivasha Sawela Lodge*: The Lodge is set on the shores of Lake Naivasha in Kenya. The lodge offers accommodations and conference facilities in Naivasha.
- *Malu Lodge*: The Lodge is located in a private forest reserve in Naivasha Kenya. The lodge is ideal for weekend getaways.
- *Chui Lodge*: The Lodge is set inside Oserian Wildlife Conservancy in Naivasha Kenya. The lodge has 8 cottages, dining room, lounge and bar area.

II. Environmental issues and related indicators

The combined impacts of global and local environmental forces are apparent as you enter Lake Nakuru Park. It is rapidly becoming submerged as the lake swells past its normal size. Over the last two years, at the end of the dry season, just at start of the short rains, half of the buildings at the park entrance are normally underwater which confirms genuine concern when the height of the long rainy season hits. The expansion of the lake is also a fairly new phenomenon which makes everyone a bit concerned. While the lake is expanding ever outward, the land is being submerged under a wave of different sorts.

Over the last ten or so years the park has seen an expansion of some local plants to the point where they are considered invasive and filling up the different grass, shrub, and woodlands which can have far reaching ecological consequences as other seedlings are shaded out. Add that to the fact that Nakuru Town is constantly expanding outward, often to the park fence, and that the park's buffalo population is constantly growing and degrading the habitat area of other more habitat-specific species, and you have a real problem. Already species like the Bohor reedbuck have gone locally extinct and other species like waterbuck might be soon to follow if the local buffalo population isn't reduced.

III. Wellbeing and equity indicators

The hotels engage guests through various initiatives; these include philanthropy and guests donations. Through philanthropy, the facilities make donations of stationery, clothing etc. to orphanages and local primary schools. Lake Naivasha Country Club for example supports Karagita primary school which also benefits from firewood obtained from fallen trees within the hotel. The most recent donation was playground items for kids such as slides, swings etc. to one of the orphanages. The facilities also purchase products including fruits and vegetables, staff meat from local suppliers in the county. Specifically, vegetables are obtained from local individual farmers near Longonot, Nakuru town.

Hotels in Naivasha are part of Lake Naivasha Water Association. They are involved in conservation initiatives that help create awareness to farmers on usage of agricultural chemicals which affects the lake during rain seasons, and training them on best agricultural practices to reduce such effects such as strip/terrace/contour ploughing.

At least 50% of the employees are from the locality and they are provided with medical cover, uniform, food and transport (to and from the hotels).

IV. Green economy and policy interventions

A summary of interventions by stakeholders in the sector is presented in Table 3-18.

Table 3-18: Green Economy interventions in the Tourism Sector

Priority	Constraints	GE Strategies
Enhance tourism activities	- Poor planning - Poor community involvement in sports; - Poor road infrastructure and urban zoning - Underutilized tourist sites - Underdeveloped sport tourism - Low numbers of local tourists and - slow exploitation of alternative tourism potential - Inadequate accommodation facilities	- Give incentives to local tourists - Carryout promotional campaigns - Development of high altitude sports such as golf - Expansion of sports infrastructure and rest parks - Sports training; - Promote annual days such as highland games

3.3.7 Waste Management

I. General economic sector indicators

Kenya urbanization rate is fast, and as it does so, the problems of solid waste management are escalating. A recent report (World Bank 2012) showed that the amount of solid waste is growing at an even faster rate than urbanisation globally, as urban residents on average begin to consume more and therefore generate more waste. In medium-sized towns such as Nakuru, the problems can be even more acute. The population continues to rise rapidly (at even higher rates than national averages), and the capacities and resources to manage and the associated problems of solid waste management (Mwanzia and Kimani) escalate.

Approximately 250 tonnes of solid wastes are generated every day within Nakuru Municipality. When the Municipal Department of Environment was established in 1997, only about 20 percent of this was being collected. The system was entirely run by the Municipality, although a small number of private companies also collected waste from higher income settlements in the town. These were not regulated, and operated outside of Municipal controls. In common with many cities in the developing world, a high proportion of the waste generated is organic material (46 percent) (MCN 2010).

II. Environmental issues and related indicators

In areas not covered by sewer network effluent management remains the biggest environmental challenge. This is extremely serious in relatively flat terrains, where pools are witnessed in rainy seasons. Hotels in Naivasha have effluents from the cottage kitchen flows through grease traps before draining in a soak pit.

Solid waste at the urban facilities are collected in bins then later stored at a waste collection centre for disposal by the County while organic waste is at times utilised internally. Table 3-19 shows a summary of the waste management in the county.

Table 3-19: Sector Statistics for Waste Management

Item		Numbers
Households by main mode of human waste disposal-	Total number of household	409, 836
	Main sewer	32, 864
	Septic tank	14,507
	Cess pool	3,148
	VIP pit latrine	26,746
	Pit latrine(covered/uncovered)	320,416
	Bucket	554
	Bush	11,262
	Others	339
Community distribution by type of waste/garbage disposal (%)	Collected by local authority	9.4
	Collected by private firms	2.5
	Garbage pit	29.0
	Burning	12.1
	Public garbage heap	14.7
	Farm Garden	31.2
	Neighbourhood community groups	0.6
	Others	0.4

III. Wellbeing and equity indicators

Practical Action is working closely with a group of solid waste collectors known Nakuru Waste Collectors and Recyclers Management (NAWACOM), which has a membership of 146. Such community-based organisations play an important role in contributing to the management of solid waste generated in the municipality. They complement the municipal council's efforts by reducing the amount of waste to be collected as they retrieve the reusable materials. Their activities are also a source of income, as they sell the recyclable materials to manufacturers. Practical Action sponsors a waste management project among people who live on a rubbish dump a few kilometres outside Nakuru. This project uses vermiculture to compost organic waste, which gives some of poorest people the means to earn a living.

IV. Green economy and policy interventions

The study shows that Nakuru County has chosen a growth path that is determined to drive its green growth agenda. Tables 3-20 highlights some of the sector priorities, constraints and opportunities.

Table 3-20: Green Economy intervention in waste sector

Priorities	Constraints	Opportunities
- Increase provision of solid waste management services - Maintenance of hygiene and aesthetic qualities - Improvement of grey water management. - Improve construction and maintenance of sanitation facilities - Control air, land, and water pollution. - Improve storm water management	- Inefficiencies in solid waste management; - inadequate capacity; - low infrastructural development; - flooding; - minimal awareness.	- Modernize waste management and avail equipment. - Encourage Waste recycling and use of alternative systems - Partial privatization of SWM and zoning of refuse collection areas. - Enforcement of available regulations. - Expansion of the sewer line by NAWASSCO - Adopt Resources Oriented Sanitation Concepts for re-use of human wastes. - Structural storm water management strategies. - Streamline institutional arrangements enhance public participation - Allocate adequate funding

3.3.8 Prioritisation of Analysed Sectors

The analysis made on the response forms were scored for each sector against the 10 point criteria for sector prioritisation (1- low; 2- medium; 3 - High). The results are presented in Tables 3-21 to 3-26. The scores were then used to prioritize the sector as summarized as follows:

- 1) Water (Score 2.7)
- 2) Agriculture (Score 2.5)
- 3) Energy (Score 2.4)
- 4) Waste (Score 2.2)
- 5) Tourism (Score 1.8)
- 6) Forestry (Score 1.8)

Table 3-21: Assessment of the agriculture sector

(1) AGRICULTURE		
CRITERIA	REMARKS	MARKS
Importance of the sector according to the respondents	- Considered by respondents to be <i>important</i> for GDP and employment	3
Adequacy of GDP related data	- Good sector and sub-sector data available at county, and at national level	3
Adequacy of other General Sector data	- Very good	3
Adequacy of data on GHG emissions	- Not available	1
Adequacy of other Environmental & related data	- Some snapshot data exist	3
Adequacy of Green Economy policy interventions and other related data	- There are some structured GE-related initiatives within the sector	3
Adequacy of well-being and equity data	- Data available but some indicators can be estimated from national data	3
Level of renewable energy use excluding on-grid geothermal and hydro power, based on (1) existing documents and (2) respondent analysis	- Moderate	2
Level of waste treatment and recycling	- The sector is generally a major land and water polluter (particularly from horticulture activities)	1
Level of funding available to the sector <i>vis à vis</i> (1) sector funding needs and (2) proportion of budgetary allocations to the sector compared to all sectors combined	- Funding available for mega projects nested within Vision 2030 Flagship projects. - Sector ranks well in budgetary allocation (2015/2016 fiscal year)	3
TOTAL SCORE		25
AVERAGE SCORE		2.5

Table 3-22: Assessment of the energy sector

(2) ENERGY		
CRITERIA	REMARKS	MARKS
Importance of the sector according to the respondents	- Considered by respondents to be <i>veryimportant</i> for GDP and employment	3
Adequacy of GDP related data	- No data available but maybe can be estimated from national data	1
Adequacy of other General Sector data	- Some data available but mostly at national level	3
Adequacy of data on GHG emissions	- No data available but maybe can be estimated from national data	1
Adequacy of other Environmental & related data	- Some data available, and data for all indicators available at national level	3
Adequacy of Green Economy policy interventions and other related data	- Sector is engaged in GE initiatives which are not well structured	3
Adequacy of well-being and equity data	- No data available but indicators can be estimated from available data at national level	3
Level of renewable energy use excluding on-grid geothermal and hydro power, based on (1) existing documents and (2) respondent analysis	- Energy sector is generally moving towards a higher mix of renewable vs non-renewable energy sources	3
Level of waste treatment and recycling	- There is little information on waste treatment and recycling from energy generation	1
Level of funding available to the sector <i>vis à vis</i> (1) sector funding needs and (2) proportion of budgetary allocations to the sector compared to all sectors combined	- Energy sector is generally a national service sector so not many projects under County government. - Sector ranks medium in budgetary allocation, under Trade, Energy and Industry (2015/2016 fiscal year)	3
TOTAL SCORE		24
AVERAGE SCORE		2.4

Table 3-23: Assessment of the water sector

(3) WATER		
CRITERIA	REMARKS	MARKS
Importance of the sector according to the respondents	- Considered by respondents to be <i>important</i> for GDP and employment	3
Adequacy of GDP related data	- Good data available at county	3

Adequacy of other General Sector data	- Good data available but mostly at national level	3
Adequacy of data on GHG emissions	- No data available but some indicators can be estimated from national data	2
Adequacy of other Environmental & related data	- Some snapshot data, some can be derived from national datasets	3
Adequacy of Green Economy policy interventions and other related data	- Sector is low on GE initiatives which are not well structured and lack diversity	1
Adequacy of well-being and equity data	- Service provision data available but some indicators can be estimated from national data	3
Level of renewable energy use excluding on-grid geothermal and hydro power, based on (1) existing documents and (2) respondent analysis	- Used for pumping	3
Level of waste treatment and recycling	- The sector generally has high treatment	3
Level of funding available to the sector <i>vis à vis</i> (1) sector funding needs and (2) proportion of budgetary allocations to the sector compared to all sectors combined	- Funding available for mega projects (roads/railway) nested within Vision 2030 Flagship projects, County undertaking major infrastructure works. - Sector ranks high in budgetary allocation (2015/2016 fiscal year)	3
TOTAL SCORE		27
AVERAGE SCORE		2.7

Table 3-24: Assessment of the tourism sector

(4) TOURISM		
CRITERIA	REMARKS	MARKS
Importance of the sector according to the respondents	- Considered by respondents to be <i>important</i> for GDP and employment	3
Adequacy of GDP related data	- Not available	1
Adequacy of other General Sector data	- Snapshot data available but mostly at coast region level, no longitudinal data	3
Adequacy of data on GHG emissions	- No data available	1
Adequacy of other Environmental & related data	- No data available	1
Adequacy of Green Economy policy interventions and other related data	- There are some structured GE-related initiatives within hotels in Naivasha and Nakuru	3
Adequacy of well-being and equity data	- No data available but some indicators can be estimated from national data	1

Level of renewable energy use excluding on-grid geothermal and hydro power, based on (1) existing documents and (2) respondent analysis	- Moderate	2
Level of waste treatment and recycling	- The sector generally does not treat waste products in the desired fashion - Few establishments have taken up recycling and reuse	2
Level of funding available to the sector <i>vis à vis</i> (1) sector funding needs and (2) proportion of budgetary allocations to the sector compared to all sectors combined	- Tourism sector is generally facilitated by public private partnerships rather than directly by County. - Sector has no specific budgetary allocation is under a conglomerate Ministry of Trade, Industrialization, Tourism and Wildlife Management (2015/2016 fiscal year)	1
TOTAL SCORE		18
AVERAGE SCORE		1.8

Table 3-25: Assessment of the waste sector

(5) WASTE SECTOR		
CRITERIA	REMARKS	MARKS
Importance of the sector according to the respondents	- Considered by respondents to be moderate to important for GDP and employment	2
Adequacy of GDP related data	- No sector data both at county and national seegregation	1
Adequacy of other General Sector data	- Snapshot data available, no longitudinal data	3
Adequacy of data on GHG emissions	- No data available	1
Adequacy of other Environmental & related data	- Snapshot data available, no longitudinal data	3
Adequacy of Green Economy policy interventions and other related data	- GE initiatives are few and uncoordinated, but there are policy reviews ongoing provide opportunity for embedding GE principles	3
Adequacy of well-being and equity data (Annex 4)	- Local data moderate but Not centralised	2
Level of renewable energy use excluding on-grid geothermal and hydro power, based on (1) existing documents and (2) respondent analysis	- Initiative in biomass utilisation, recycling and biogas	3
Level of waste treatment and recycling	- Recycling is undertaken on individual an institutional in-house processes in varying levels depending on waste product type	2

Level of funding available to the sector <i>vis à vis</i> (1) sector funding needs and (2) proportion of budgetary allocations to the sector compared to all sectors combined	- Funding available to rehabilitate existing facilities but not for expansion to meet demand. - Sector ranks moderate in budgetary allocation under Water, Environment and Natural Resources (2015/2016 fiscal year)	2
TOTAL SCORE		22
AVERAGE SCORE		2.2

Table 3-26: Assessment of the forestry

(6) FORESTRY		
CRITERIA	REMARKS	MARKS
Importance of the sector according to the respondents	- Considered by respondents to be <i>important</i> for GDP and employment	3
Adequacy of GDP related data	- No data available but some indicators can be estimated from national data	1
Adequacy of other General Sector data	- Some data available but mostly at national level	3
Adequacy of data on GHG emissions	- No data available	1
Adequacy of other Environmental & related data	- Some available at county	2
Adequacy of Green Economy policy interventions and other related data	- Sector is active on GE initiatives which are run both by local, national and global initiatives	3
Adequacy of well-being and equity data	- Some data available on results of concluded programs, but some indicators can be estimated from national data	2
Level of renewable energy use excluding on-grid geothermal and hydro power, based on (1) existing documents and (2) respondent analysis	- Low level of renewable energy use	1
Level of waste treatment and recycling	- Low level of mostly <i>ad hoc</i> cleaning of stream pollution	1
Level of funding available to the sector <i>vis à vis</i> (1) sector funding needs and (2) proportion of budgetary allocations to the sector compared to all sectors combined	- Low funding - Sector ranks average in budgetary allocation, under Agriculture, Livestock, Veterinary and Fisheries (2015/2016 fiscal year)	1
TOTAL SCORE		18
AVERAGE SCORE		1.8

3.4 County Planning Cycle

Nakuru County planning process is sector based. There are 8 Sector Working Groups (SWG) in Nakuru County. The SWGs prepare sector plan reports which inform the County Fiscal Strategy Paper (CFSP). The CFSP sets the budgetary ceilings.

The County planning process; guided by the Public Finance Management (PFM) Act Sectoral planning through SWGs, prepares sector plans and reports which inform the CFSP, county Medium Term Expenditure Framework (MTEF) budget. The CFSP informs the sector allocations, while budgeting follows a Performance Based Budget (PBB) approach. The process also adopts an incremental budgets principle (at 10%) without clear interrogation of sector requirements.

Challenges with the current status

- CPU not fully constituted
- Due to sector ceilings allocated through the CFSP, other sectors don't incorporate environmental issues
- The MTEF budget is prepared by the executive who prepares the draft estimates (through public participation), BUT
- The county assembly undertakes its own public participation process on the budget, and can sometimes do a budget overhaul
- Non adherence to the CIDP
- Budget deficiency
- Meeting constitutional requirements
- The planning process is guided by the PFM Act. Policy documents such as MTEF budgets, ADP, CBROP and CFSP are adhered to.
- Adherence to the 30% development expenditure has however been a challenge.
- CPU not fully constituted as required by the CGA (2012). There is a county macro working group from the county treasury instead.

Opportunity for Mainstreaming of GE

- Through the ADPS, sector level planning
- Incorporating environmentally friendly technologies e.g. on building codes, provision of green spaces

- Policy on GE and enforcement of the same. E.g. production and consumption tendencies
- Undertaking Strategic environment Assessment (SEA) on CIDP, Urban and spatial plans

Areas where Support is required

- Capacity building through training and awareness creation on green economy
- Target policy makers, development partners and other stakeholders(private)
- Review of development plans to green them
- Consideration for SEA on CIDP (2018-2023) technical and financial support.
- Linkage to environmental support funds

3.5 Institutional arrangements for annual development planning

County governments are structures with almost similar institutional frameworks with autonomous and differentiated roles and responsibilities. The various institutions and there mandates are discussed here below.

I. Governor

Some of the relevant functions of a Governor include: Representing the county in national and international fora and events; appointing, with the approval of the county assembly, the county executive committee; submitting the county plans and policies to the county assembly for approval; considering, approve and assent to bills passed by the county assembly; chairing meetings of the county executive committee; submitting to the county assembly an annual report on the implementation status of the county policies and plans; providing leadership, promoting citizen participation, democracy, good governance, unity and cohesion within the county; and promoting the competitiveness of the county and be accountable for the management and use of the county resources.

II. Council of County Governors

The Council provides a forum for: Consultation; sharing of information; considering matters of common interest; dispute resolution; capacity building for governors; receiving reports and monitoring the implementation of projects.

III. Intergovernmental Relations Technical Committee

Its functions include: taking responsibility for the day to day administration of the Summit and of the Council and in particular facilitate the activities of the Summit and of the Council; implementing the decisions of the Summit and of the Council of governors

IV. Senate

The main role of Senators is to: represent the counties, while also serving to protect the interests of the counties and their governments; participating in the law-making function of Parliament by considering,

debating and approving Bills concerning counties; determining the allocation of national revenue among counties, as provided in Art. 217, and exercises oversight over national revenue allocated to the county governments.

V. County Assembly

The County assembly is mandated to make any laws that are necessary for, or incidental to, the effective performance of the functions and exercise of the powers of the county government; exercise oversight over the county executive committee and any other county executive organs; and receive and approve plans and policies for the management and exploitation of the county's resources; and the development and management of its infrastructure and institutions.

VI. County Executives

There functions include: Implementing county and national legislations; managing and coordinating the functions of the county administration and its departments; preparing proposed legislation for consideration by the county assembly; providing the county assembly with full and regular reports on matters relating to the county.

VII. County Public Service Board

Their functions are: Establishing and abolish offices in the county public service; appointing persons to hold or act in offices of the county public service; exercising disciplinary controls; preparing regular reports for submission to the county assembly; facilitate the development of human resource planning and budgeting for personnel emoluments; and advising the County Government on human resource management and development.

VIII. City and Municipality Boards

The City and Municipality boards are mandated to: oversee the affairs of the city or municipality; develop and adopt policies, plans, strategies and programmes, and may set targets for delivery of services; formulate and implement an integrated development plan; control land use, land sub-division, land development and zoning by public and private sectors for any purpose; develop and manage schemes; maintain a comprehensive database and information system; implement applicable national and county legislation; monitor and, regulate city and municipal; settle and implement tariff, rates and tax and debt collection policies as delegated by the county government.

3.6 Strategic Documents guiding development and decision making

There are a number of key documents, both at National and County levels that influence development planning at the County level. At the national level, some of these documents include:

- The Public Finance Management Act 2012 which provides for the establishment of a County Budget and Economic Forum (CBEF). The CBEFs provide a platform for the County and public to consult on preparation of a County Fiscal Strategy Paper and County Budget Review and Outlook Paper (BROP).

- The Urban Areas and Cities Act of 2011 - makes provisions for public participation.
- The National Climate Change Response Strategy (NCCRS) 2010.
- The National Integrated Monitoring and Evaluation system (NIMES).
- EMCA 1999 that exercise general supervision and coordination of lead agencies that execute environmental functions.

There are a number of key documents, both at National and County levels that help to guide development planning at the County level. Some of these documents include:

- The Nakuru County Integrated Development Plan (CIDP) 2013-2017 where stakeholders gave their views in six stakeholder forums
- Nakuru County Annual Development Plan 2015-2016 Fiscal Year. Budget Office of the County Government of Mombasa, Finance and Economic Department.
- The Nakuru County Monitoring and Evaluation system, aligned to the National Integrated Monitoring and Evaluation system (NIMES), to serve the needs of the County. The system takes cognizance of the projects and programmes included in the County Integrated Development Plan as well as indicators facilitating the MTEF process and development of the country.

The following tools/methods/platforms are used by Planners:

- Tools
- Public Finance Management Act, IFMIS, Monitoring Committees, Master Plans, Strategic Plans, Zoning Plans, Land Use Plans.
- Methods: these include Information Communications Technology (ICT) forum with a toll free number, Policy Briefs, SWOT, Cost Benefit Analysis.
- Engagement platforms include: Community meetings, workshops, conferences, public fora, mass media, intergovernmental committees, notice boards, e-citizens, websites.

3.7 Relevant actors

Key players in greening Nakuru county economy

Stakeholder	Role
State departments	Provide policy guideline and finance the implementation
County Government	Regulation, coordination, advice and finances
Private Sector	Provide markets for goods and services; bridge skills and knowledge gaps and facilitate technological transfer.
CBOs	Provide labour and local materials; determine ownership and sustainability; and safeguard public interest.
NGOs	Mobilize resources and lead advocacy for rights.
Research institutions and universities	Research, surveillance, training and extension and transfer of research finding.
Regulatory bodies	Setting standards, regulations, quality control and certification

4. MOMBASA COUNTY FINDINGS

4.1 Overview

Mombasa County lies between latitudes 3°56' and 4°10' south and between longitudes 39°34' and 39°46' east and is located in the central-south portion of Kenya's Indian Ocean coastline. It covers an area of 229.9 km² excluding 65 km² of water mass which is 200 nautical miles into the Indian Ocean, and borders Kilifi County to the North, Kwale County to the South West and the Indian Ocean to the East (Mombasa County Government, 2013).

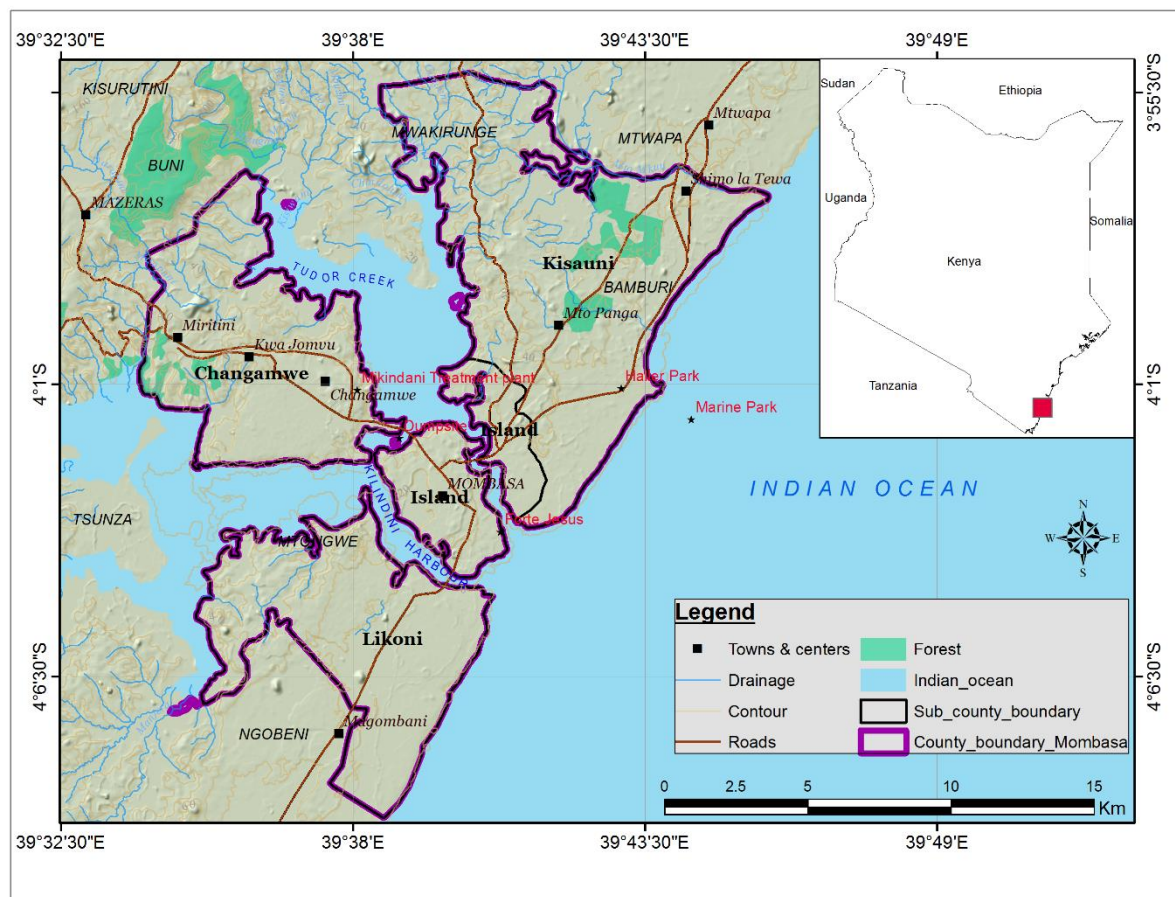


Figure 4-1: Mombasa County (Source: NK/PADECO/EJEC, 2016).

The County has a population of about 1 million, an annual population growth rate of 3.45% (1999-2009) and a population density of 6,131 people per km² based on the 2009 national population census (Commission on Revenue Allocation, 2013; Mombasa County Government, 2013). The County has six Sub-Counties that also serve as parliamentary constituencies: Changamwe, Jomvu, Kisauni, Nyali, Likoni, and Mvita. The Sub-Counties and their Wards are presented in Table 4-1.

Table 4-1: Administrative/political jurisdictions in Mombasa County.

Sub-County/ Constituency	# of Wards	Area (Km ²)	Locations	Sub- Locations	Names of Wards
Changamwe	5	16.00	3	5	Port Reitz, Kipevu, Airport, Miritini, Chaani
Jomvu	3	29.00	2	4	Jomvu Kuu, Magongo, Mikindani
Kisauni	7	106.12	1	6	Mjambere, Junda, Bamburi, Mwakirunge, Mtopanga, Magogoni, Shanzu
Nyali	5	22.88	1	3	Frere Town, Ziwa La Ng'ombe, Mkomani, Kongowea, Kadzandani
Likoni	5	41.10	4	6	Mtongwe, Shika Adabu, Bofu, Likoni, Timbwani
Mvita	5	14.80	7	11	Mji Wa Kale/Makadara, Tudor, Tononoka, Shimanzi/Ganjoni Majengo
TOTALS	30	229.90	18	35	

(Source: Mombasa County Government, 2013, referenced therein as having been sourced from Regional Commissioner's Office Mombasa in 2013).

4.2 Biophysical Setting

The County lies within the Coastal lowland which rises gradually from the sea level in the East to about 132m above sea level in the mainland, and is characterised by three distinct physiographic features, which include the coastal plain, the hilly areas mainly found within the Western part of the County, and the shoreline-ocean zone with fringing coral reefs and tidal flats (Mombasa County Annual Development Plan, 2014).

A synopsis of the climate is given in the Mombasa County Annual Development Plan, 2014. The report notes that the rainfall pattern is characterized by two distinct long (April - June) and short (October to December) seasons, with an annual average of 640mm. The annual mean temperature in the County is 27.9°C with a minimum of 22.7°C in July and a maximum of 33.1°C in February. It has an average humidity at noon of about 65 per cent.

The County's ecosystem has both marine and terrestrial components: both ecosystems are characterized by diverse species of flora and fauna, the most common being coconut trees and different species of fish, which have different cultural, social and financial values (Mombasa County Annual Development Plan, 2014). The fringing coral reefs, creeks and tidal flats with extensive mangrove forests are breeding grounds for fish, while the fringing coral reefs in North Coast are an important marine conservation area

hosting the Mombasa Marine National Park and Reserve (Mombasa County Annual Development Plan, 2014).

The percentage of agricultural land is negligible compared to other counties in Kenya, at just 21,000 hectares or 0.04% of the country total of 56,914,000 hectares (KNBS, 2015). The report, however, rates the quality of the coastal agricultural land areas as ‘high potential’ i.e. with annual rainfall at over 980mm per annum.

4.3 Socioeconomic Setting

The coastal region is a key contributor to the National Gross Domestic Product with 45% from tourism and 15% from the shipping activities which are important to the coastal economy (WRI, 2007; Government of Kenya, 2009). The main economic activities include tourism, shipping, commerce and subsistence farming (Mombasa County Annual Development Plan, 2014). The community’s sources of livelihood include: small scale fisheries, marine fisheries, mariculture in community based farms, tourism, agriculture and forestry, mineral processing such as oil and gas, cement manufacturing at Bamburi Cement, coral limestone, shale and sand mining (Mombasa County Annual Development Plan, 2014). At 60 per cent, the formal sector provides majority of employment, and the major employers are the hotel industry, shipping industry, Government of Kenya and various private institutions (Mombasa County Annual Development Plan, 2014). It was noted some years ago that Mombasa should be a prosperous city, offering plentiful economic opportunities to its population, but that it had not yet realised its potential since services and infrastructure were becoming increasingly inadequate and about a quarter of its households fell below the absolute poverty line (Rakodi et al., 2000).

The population distribution and settlement patterns in the County (Tables 4-2, 4-3) are influenced by proximity to vital social and physical infrastructure networks such as roads, housing, water and electricity (Mombasa County Annual Development Plan, 2014). Other factors that influence settlement patterns include accessibility to employment opportunities, and security. There were 268,700 households estimated in 2009 (Table 4-3). Highly populated areas are in Majengo, Bamburi, Bangladesh, Mikindani, Jomvu, Miritini, Migadini, Port Reitz, Mishomoroni and Bombolulu among others.

Table 4-2: Population distribution and density by Sub-County, Mombasa County.

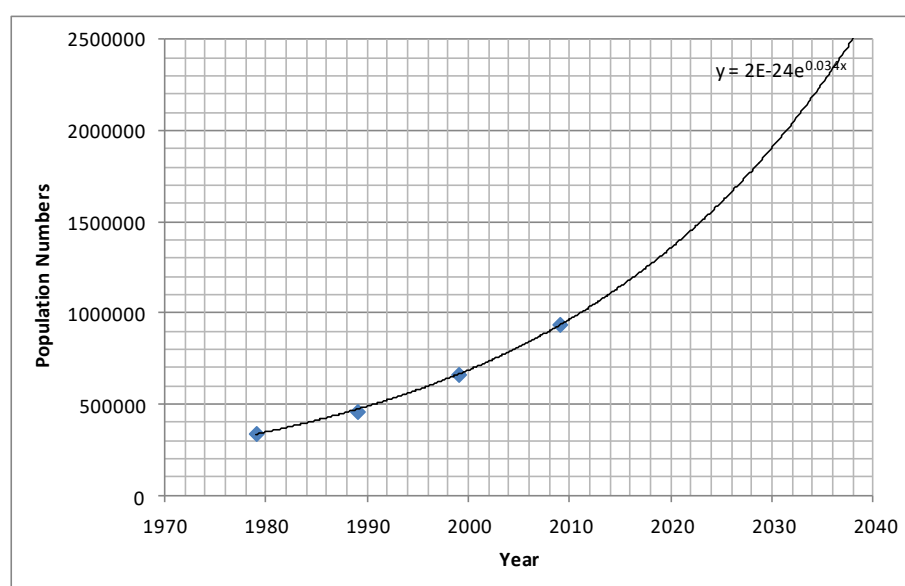
	2009 (Census)		2012 (Projections)		2015 (Projections)		2017 (Projections)	
Constituency	Population	Density	Population	Density	Population	Density	Population	Density
Changamwe*	147,613	9,226	165,438	5,705	185,415	6,394	200,005	6,898
Jomvu	102,566	3,537	114,951	7,184	128,832	8,052	139,005	8,688
Kisauni	194,065	2,188	217,499	2,246	243,762	2,293	263,010	2,474
Nyali	185,990	8,129	208,449	9,103	233,619	10,202	252,066	11,007
Likoni	166,008	4,040	186,054	4,527	208,520	5,073	224,986	5,474
Mvita	143,128	9,671	160,411	11,078	179,781	12,416	193,977	13,396
County Totals	939,370	6,131	1,052,802	6,640.5	1,179,929	7,405	1,273,049	7,989.5

(Source: From Mombasa County Annual Development Plan, 2014, referenced therein as Kenya National Bureau of Statistics, Mombasa, 2013).

Table 4-3: Mombasa County household statistics based on KNBS

No. of Households	Land Area (Km ²)	Density (persons per Km ²)	Male Population	Female Population	Total Population
268,700	218.9*	4,292	486,924	452,446	939,370

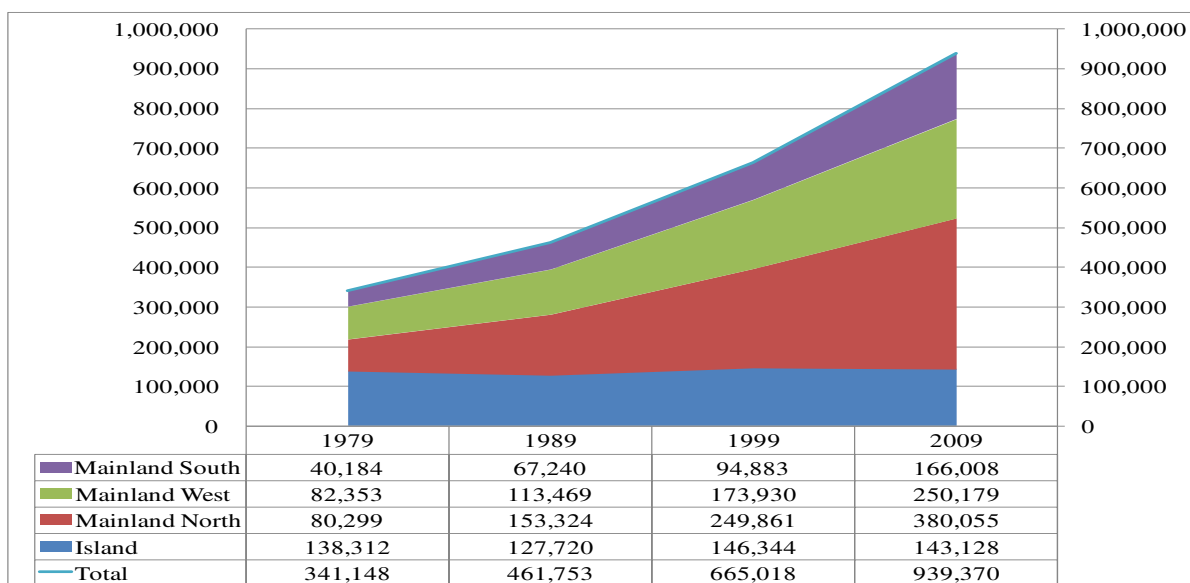
NB: *Land area as provided in the Mombasa County Government (2013) first CIDP is 229.9 km² in contrast to 218.9 km² in the KNBS (2015) report.



(KNBS census data constrained by the 1979-2009 data.

Figure 4-2: Population growth of Mombasa County

Assuming that the population of the County continues to grow at the rate of 3.45% annually (for the period 1999-2009), then the population will be 1,318,687 in 2019, 1,851,171 in 2029 and 2,598,672 by 2039 (Fig. 4-2). However, the population growth rate by zone over the period 1979 to 2009 shows a differentiated pattern (Fig. 4-3): the population in the island has remained more or less static, while that in the mainland has been growing. It is envisaged that these differentiated trends will continue into the future, and it is likely that the population resident on the island may decrease as it becomes more and more of a business district.



(Source: NK/PADECO/EJEC, 2016)

Figure 4-3: Population growth in each zone, 1979-2009

Despite efforts being made to settle people, the County still has a very large number of landless people most of whom live in the city's slums of Mishomoroni, Junda and Kisumu ndogo in Kisauni Sub-County, Shika-Adabu and Ngomeni in Likoni Sub-County, and Bangladesh in Changamwe Sub-County (Mombasa County Annual Development Plan, 2014). The land adjudication process is ongoing for Shika-Adabu and Vyemani settlement schemes, but there are also other proposed schemes in the County namely; Maweche, Kibundani, Ujamaa-Shonda and Kidungunyi (Mombasa County Annual Development Plan, 2014). The sparsely populated areas in the outskirts of the County, which include Mwakirunge-Maunguja, Mwangala, Mreroni and the Mkupe Jetty area, are least developed in terms of infrastructure such as road network, electricity and water supply (Mombasa County Annual Development Plan, 2014). Education and health facilities are also scantily available in these areas making the inhabitants highly prone to poverty and disease incidences (Mombasa County Annual Development Plan, 2014).

The total recorded employment at national level is presented in Table 4-4. This data indicates that roughly about 20% are waged employees, and about 80% are in the informal sector. Given the Mombasa County total labour force (Table 4-5), estimates of the employees in Mombasa County have been made as shown in Table 4-5. These data tally well with the author derived estimates for waged employees in Mombasa using the KNBS (2016) data and with the corresponding assumptions used to derive the data presented.

Table 4-4: Total recorded employment¹ at national level in thousands, 2011 – 2015

	2011	2012	2013	2014	2015*
Modern Establishments - Urban and - Rural Areas:					
- Wage Employees					
- Self-employed and unpaid family workers	2,084.1 73.8	2,155.8 76.9	2,283.1 83.8	2,370.2 103.0	2,478.0 123.2
Sub -Total	2,157.9	2,232.7	2,366.9	2,473.2	2,601.2
Informal Sector ²	9,958.3	10,548.4	11,150.1	11,846.0	12,559.6
TOTAL	12,116.2	12,781.1	13,517.0	14,319.2	15,160.8

(Source: KNBS, 2016). * Provisional; 1 - Figures refer to employment stock as at 30th June and excludes small scale farming and pastoralist activities; 2 - Estimated.

Table 4-5: County labour force, required total employment (in numbers) by year.

Year	2013 (Estimate)	2015 (Projections)	2017 (Projections)
Total Labour Force (15-64 age cohorts)	545,303	611,150	830,525
15% Unemployment	463,508	519,478	705,946
Number of jobs to be generated for NUR of 4%	59,983	586,704	797,304
Number of jobs to be generated for NUR of 6%	49,077	574,481	780,694
<i>No. of Wage Employees, Self Employed and Unpaid#</i>	<i>92,702</i>	<i>103,896</i>	<i>141,189</i>
<i>No. of persons working in Informal Sector#</i>	<i>370,806</i>	<i>415,582</i>	<i>564,757</i>

(Source: Partially modified (data in italics) from Mombasa County Government, 2013). NB: NUR = Natural Unemployment Rate. # - Based on 15% unemployment rate, and assuming 20:80 split between "Wage Employees, Self Employed and Unpaid" versus "Informal Sector".

In Mombasa County, 28% of the residents have no formal education, 37% of those with primary level of education and 46% of those with secondary level of education or above are working for pay (Ngugi, 2013) (Table 4-6). Mombasa County's Gini index is 0.365 compared with Turkana County, which has the least inequality nationally (0.283) (Ngugi, 2013). The total labour force in 2015 is estimated at 611,150 people, and a significant number of jobs needs to be generated per year to keep the inflation rate down (Table 4.5). The adoption and implementation of green economy initiatives can contribute significantly to the generation of new jobs and hence to reduction of the unemployment rate.

Table 4-6: Overall employment by education levels in Mombasa County

	LoE	Work for pay	Family Busine ss	FAH	I/V	R/ H	FS	Incap .	No work	No. of Individu als
Kenya Total	-	23.7	13.1	32.0	1.1	9.2	12.8	0.5	7.7	20,249,800
Kenya Rural	-	15.6	11.2	43.5	1.0	8.8	13.0	0.5	6.3	12,984,800

	LoE	Work for pay	Family Busine ss	FAH	I/V	R/ H	FS	Incap .	No work	No. of Individu als
Kenya Urban	-	38.1	16.4	11.4	1.3	9.9	12.2	0.3	10.2	7,265,012
Mombasa County	None	28.4	15.5	6.8	2.6	28.2	1.1	1.2	16.3	56,308
	Primary	37.3	16.3	3.4	1.1	19.1	8.0	0.3	14.6	239,944
	Secondar y	46.2	15.2	2.2	1.3	10.3	13.3	0.1	11.4	301,769
	Total	41.0	15.7	3.1	1.3	15.5	10.0	0.3	13.1	598,021

(Source: Ngugi, 2013)NB - There is a rounding off error of 0.1% in the "none" and "primary" entries so row percentages are 100.1%; however, for lack of access to the primary dataset this error is only noted, not corrected). (LoE - Level of Education; FAH - Family Agricultural Holdings; I/V - Intern/Volunteer; R/H - Retired/Homemaker; FS - Fulltime student; Incap. - Incapacitated).

Based on the sectors outlined for this report (General, Agriculture, Buildings, Cities, Energy, Fishery, Forestry, Manufacturing, Tourism, Transport, Waste, Water), data were extracted from the KNBS (2015) Kenya Facts and Figures Report and were combined and re-tabulated, and estimates of number of waged employees in Mombasa County were derived based on some assumptions as explained in the table captions (Tables 4-7 and 4-8). A large part of the information derived from KNBS (2016) did not fall into any specific sector and were therefore listed under the "General - cross sectorial and complex" category, and thus the estimates provided should be considered as underestimating the real figures. The General sector in 2014 comprised 59.53% of the waged employees (Table 4-8). The other sectors had the following percentages of waged employees in 2014 (Table 4-8: Buildings and Cities: 7.54%; Agriculture, Forestry, Fishery: 17.03%; Manufacturing: 15.33%; Tourism: 3.08%; Transport: 3.36%, and Waste, Water: 0.53%.

More generally, total employment (national) outside small-scale agriculture and pastoralist activities increased by 5.9 per cent to 15,160.8 thousand persons; wage employment within the public sector increased from 700.8 thousand persons in 2014 to 718.4 thousand persons in 2015; and employment in the informal sector grew by 6.0 per cent from an estimated 11,846.0 thousand persons in 2014 to 12,559.6 thousand persons in 2015 (KNBS, 2016). The informal sector, therefore, still comprises by far the largest source of income for Kenyans, and should, therefore, be taken very seriously, along with public and private sector groups, in planning for green economy initiatives.

Table 4-7: Estimated Mombasa wage employment (numbers in thousands) by industry: 2011 – 2014.

Estimated Mombasa Wage Employment (in 000's) By Industry: 2011 - 2014 (Mombasa employment numbers are based on a multiplier of 0.0511 which represents 5.11% of the National work-for-pay population)					
	Year	2011	2012	2013	2014

Employment By Industry (population count in '000s)	Main Sector Category	Employee Count	Employee Count	Employee Count	Employee Count
Construction	Cities / Buildings	5.42	5.93	6.63	7.34
Real Estate Activities	Cities / Buildings	0.18	0.19	0.19	0.20
Electricity, Gas, Steam and Air Conditioning Supply	Energy	0.58	0.73	0.75	0.78
Agriculture, Forestry and Fishing	Agriculture, Forestry, Fishery	16.88	17.25	17.50	17.03
Information and Communication	General	4.11	4.38	4.72	5.06
Financial and Insurance activities	General	2.97	3.15	3.34	3.45
Administrative and support service activities	General	0.21	0.24	0.25	0.25
Education	General	18.90	19.66	21.63	22.99
Wholesale and Retail Trade; Repair of motor vehicles and motorcycles	General	9.73	10.12	10.84	11.25
Professional, scientific and technical activities	General	3.13	3.20	3.33	3.40
Public administrative and defence; compulsory social security	General	10.52	10.60	11.36	11.93
Human Health and social work activities	General	5.00	5.35	5.74	6.05
Arts, Entertainment and recreation	General	0.31	0.33	0.34	0.34
Other Service Activities	General	1.39	1.44	1.49	1.56
Activities of households as employers; undifferentiated goods and services-	General	5.35	5.43	5.61	5.73

producing activities of households for own use					
Activities of extraterritorial organisations and bodies	General	0.05	0.05	0.06	0.06
Mining and Quarrying	Manufacturing	0.44	0.46	0.48	0.65
Manufacturing	Manufacturing	13.80	13.85	14.27	14.68
Accommodation and food service activities	Tourism	3.35	3.52	3.76	3.73
Transportation and storage	Transport	3.72	3.84	3.88	4.07
Water Supply; sewerage, waste management and remediation activities	Waste / Water	0.39	0.43	0.49	0.53
Totals in '000s		106.476	110.139	116.642	121.092

(Source of primary data - KNBS, 2015 - Kenya Facts and Figures; Secondary data calculated by authors).NB: Calculation process and assumptions (1) 41% of Mombasa County labour force (598,021 - 15-64 yr age group) worked for pay (2009 census), or a total of 245,189 persons (Ngugi, 2013). (2) The percentage of the Mombasa work for pay group in relation to total national figure (4,799,203 - i.e. 23.7% of 20,249,800 [Ngugi, 2013]) of work for pay group is 5.11%. (3) This percentage is constant from 2011-2014. (4) The data for national employee count per sector, taken from the KNBS (2015) Facts and Figures Report, is multiplied by 5.11% to derive the "Employee Count" in the table. Generally slight underestimation.

Table 4-8: Estimated Mombasa wage employment (%) by industry: 2011 – 2014.

	Year	2011	2012	2013	2014
Employment By Industry (Percentages)	Main Sector Category	% Employees	% Employees	% Employees	% Employees
Construction	Cities / Buildings	5.09%	5.39%	5.68%	6.06%
Real Estate Activities	Cities / Buildings	0.17%	0.17%	0.17%	0.16%
Electricity, Gas, Steam and Air Conditioning Supply	Energy	0.55%	0.66%	0.64%	0.65%

	Year	2011	2012	2013	2014
Agriculture, Forestry and Fishing	Agriculture, Forestry, Fishery	15.85%	15.66%	15.00%	14.06%
Information and Communication	General	3.86%	3.98%	4.05%	4.18%
Financial and Insurance activities	General	2.79%	2.86%	2.86%	2.85%
Administrative and support service activities	General	0.20%	0.21%	0.21%	0.21%
Education	General	17.75%	17.85%	18.54%	18.99%
Wholesale and Retail Trade; Repair of motor vehicles and motorcycles	General	9.14%	9.18%	9.29%	9.29%
Professional, scientific and technical activities	General	2.94%	2.91%	2.86%	2.81%
Public administrative and defence; compulsory social security	General	9.88%	9.62%	9.74%	9.85%
Human Health and social work activities	General	4.70%	4.86%	4.92%	5.00%
Arts, Entertainment and recreation	General	0.29%	0.30%	0.29%	0.28%
Other Service Activities	General	1.31%	1.31%	1.28%	1.29%
Activities of households as employers; undifferentiated goods and services-producing activities of households for own use	General	5.03%	4.93%	4.81%	4.73%
Activities of extraterritorial organisations and bodies	General	0.05%	0.05%	0.05%	0.05%

	Year	2011	2012	2013	2014
Mining and Quarrying	Manufacturing	0.42%	0.42%	0.41%	0.54%
Manufacturing	Manufacturing	12.96%	12.57%	12.24%	12.13%
Accommodation and food service activities	Tourism	3.15%	3.20%	3.22%	3.08%
Transportation and storage	Transport	3.50%	3.49%	3.33%	3.36%
Water Supply; sewerage, waste management and remediation activities	Waste / Water	0.36%	0.39%	0.42%	0.44%
Totals in '000s		100.00%	100.00%	100.00%	100.00%

(Source of primary data - KNBS, 2015 - Kenya Facts and Figures; Secondary data calculated by authors).

NB: Calculation process and assumptions (1) 41% of Mombasa County labour force (598,021 - 15-64 yr age group) worked for pay (2009 census), or a total of 245,189 persons (Ngugi, 2013). (2) The percentage of the Mombasa work for pay group in relation to total national figure (4,799,203 - i.e. 23.7% of 20,249,800 [Ngugi, 2013]) of work for pay group is 5.11%. (3) This percentage is constant from 2011-2014. (4) The data for national employee count per sector, taken from the KNBS (2015) Facts and Figures Report, is multiplied by 5.11% to derive the "Employee Count" in the table. Generally slight underestimation.

4.4 Sectoral Analysis

Mombasa County lies within the larger coast region. The management of the Kenyan coast is sectoral with the formulation of legislation to regulate fisheries, water, tourism, forests and biodiversity conservation through various Acts of Parliament (land planning, tourism industry, tourism industry licensing, tourism development and cooperation, wildlife conservation and water) (Omuombo et al., 2013b). These acts focus on the management and conservation of the various resources and play a role in the management of the coastal resources. The Environmental Management and Co-ordination Act (EMCA) of 1999 provides a legal and institutional framework for integrated coastal zone management (ICZM) that can bring together all the sectoral management frameworks (Omuombo et al., 2013b).

During the inception stakeholder workshop held in Mombasa on the 14th October 2015, the stakeholders prioritized the following sectors for Green Economy initiatives: Waste management, Tourism, Cities and Built environment, Transport and infrastructure, and Fisheries. Following further consultations with County officials, as well as the views of respondents to the questionnaire survey, it was agreed that Energy sector should also be included in the analysis.

4.4.1 Waste Sector

Waste management can generate valuable resources and be a source of energy and jobs for more sustainable cities (AEO, 2016). In the long run, African sustainable cities can develop a zero waste approach (whereby a zero waste city combines social, political, economic and technical interventions in targeted urban areas that complement one another) adapted to the local context (AEO, 2016). Such an approach could eventually achieve 100% recycling through better waste management (Zero Waste Europe, 2014).

I. General economic sector indicators

There was no direct estimate in the literature of the contribution of the waste sector to the national GDP, instead it is grouped together with water supply and the sector is described as follows "water supply; sewerage, waste management" (KNBS, 2016). Nationally, this sector's contribution to national GDP is low, ranging between 0.8 and 0.9 % over the past five years or so (Table 4-9). The respondents to the baseline survey, however, felt that the sector contribution to the County GDP and employment was, at the very least, important (Table 4-10).

Table 4-9: Contribution of "water supply; sewerage, waste management" sector to GDP at national level

ACTIVITY	YEAR				
	2011	2012	2013	2014	2015
Contribution to GDP (%)	0.9	0.9	0.9	0.8	0.8
GDP Constant 2009 Prices (Ksh Million)	28,489	29,358	29,658	30,873	31,458
GDP % Changes (Growth)	3.6	3.1	1.0	4.1	1.9
Contribution to overall growth - Percentage Changes (Growth)	0.5	0.6	0.2	0.6	0.3

(Source: KNBS, 2016)

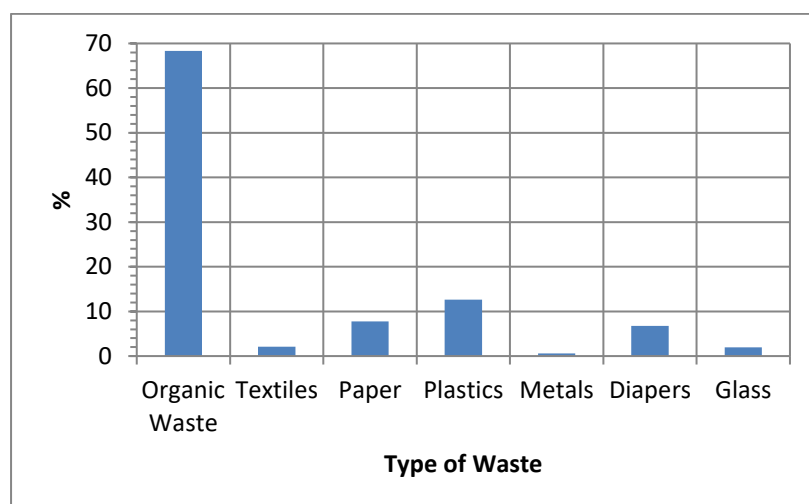
Table 4-10: City planners' views on the waste sector contribution to the County GDP and employment.

Indicators	Remarks
Economic significance - GDP	Important to very important
Economic significance - Employment	Slightly important to quite important
Key sources of energy for the sector	Electricity, fossil fuels, solar power

(a) Waste generation

Mombasa County's solid waste generation rates and types are influenced by economic development, the degree of industrialization, public habits, local climate and the presence of industries like tourism (NEMA, 2016). Organic wastes form the largest component of the total amount of solid waste generated, contributing 68.35 percent; others are - textiles (2.06 %), paper (7.73 %), plastics (hard, soft and fibre) (12.6 %), metals (0.59 %), diapers (6.72 %) and glass (1.95 %) (Fig. 4.4) (NEMA, 2016). The island, being the main commercial area of the county, generates 55% of the total solid waste which is formally

managed (Draft ISUDP, 2015). Mombasa County has been struggling to establish an efficient waste management system, partly due to difficulties in finding land for waste disposal and stakeholder resistance to establishment of a new dumpsite or landfill.



Source: NEMA, 2016

Figure 4-4: Percentage of solid waste generated by type in Mombasa County

The amount of solid waste generated per day is about 875 tonnes (Palfreman and Clark, 2015; Mombasa County Government, unpublished data), and the average waste per capita is estimated at 0.79 kg (Palfreman and Clark, 2015). In the hospitality sector, organic waste contributes 76.67% and commercial 68.29% and hotels 76.67% of the total waste generation per sector (Draft ISUDP, 2015). It is projected that the amount of waste produced will rise to 1,735.87 tons per day by the year 2035, based on a projected population of about 2.3 million and using a conservative per capita generation rate of 0.76 kg per person per day (Figure 4-5) (World Bank, 2012). NEMAs State of Environment Report for Mombasa County notes that presently, the ratio of organic to non-organic waste in low-income residential areas is 3:1, while in high income areas it is 2:1. Organic waste production in low income areas is about 72.29% of total waste generated within the income group while in the middle income areas is 72.30% and high income areas is 48.41% (Draft ISUDP, 2015). Textile waste per income level is about 2.60%, 1.75% and 1.84% of the total waste produced in low, middle and high-income area respectively. The highest waste generated in the high income group is paper waste while the least waste produced is metal waste (Draft ISUDP, 2015).

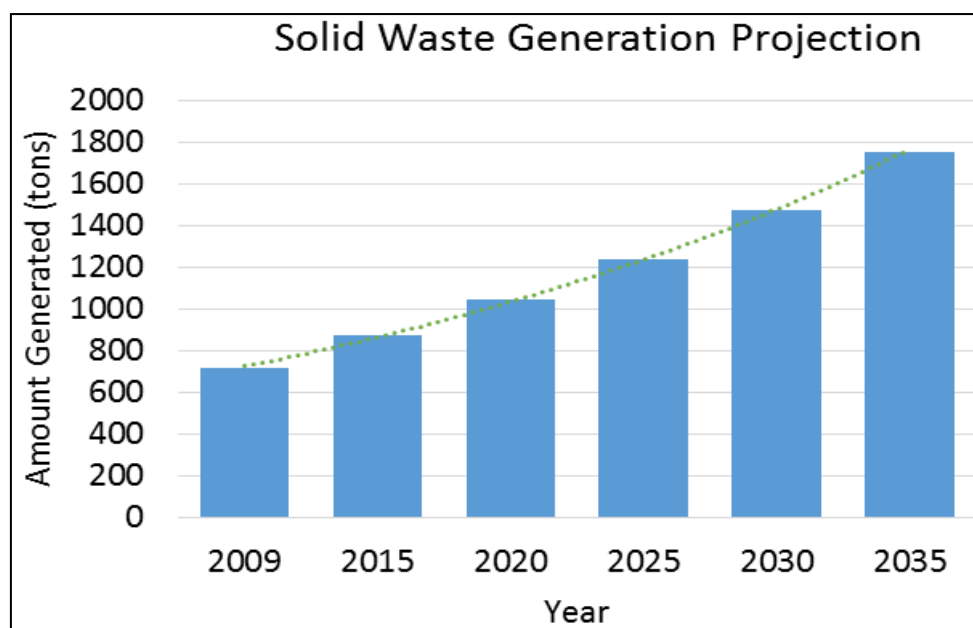


Figure 4-5: Solid Waste Generation Projection Mombasa County Government
Source: draft ISUDP, 2015 report

The households in Mombasa generate about 337 million litres of waste a day but the town faces a major challenge in disposal as only residents of Changamwe mainland have access to the service (CWSB, 2012). Less than 300,000 of Mombasa's one million residents have access to sewerage services (CWSB, 2012). Mombasa has two sewerage treatment plants of Kizingo and Kipevu, but it's the latter that is fully functional after undergoing rehabilitation at Ksh 297 million in 2014. Due to lack of a functional sewerage system on the island, some residents dispose effluent into the Indian Ocean. The only functional sewage treatment plant at Kipevu serves residents of Changamwe, Port Reitz, Magongo and Jomvu. The estimated sewage flows in Mombasa in future years is summarized in Table 4-10, with the assumption that the sewage flow/wastewater return is 80% of metered water entering the households (Draft ISUDP, 2015). The total generation of wastewater / sewage will be about 219,865 m³/d in horizon year 2035 including flows due to floating population (Draft ISUDP, 2015). There is an earlier existing plan (National Water Master Plan 2030) to enhance the Mombasa sewerage infrastructure (Table 4-11), and, therefore, there is opportunity to develop links and synergies with national government to implement these plans. There is, however, need to verify the estimated parameters to resolve differences between the numbers provided by the NWMP and the ISUDP as these will have implications on implementation costs.

Table 4-11: Future Sewage Generation in Mombasa, computed based on population, water supply and sewage generation rates (2015)

S. No.	Description	2015	2025	2035
1	Population	1,155,891	1,633,244	2,307,729
2	Estimated Water Supply (excluding 20% UFW) (M ³ /d)	92,355	191,099	249,847
3	Sewage Flow (80%) (M ³ /d)	73,884	152,879	199,878
4	Sewage Flow including infiltration (M ³ /d) (10%)	7,388	15,288	19,988
5	Total Sewage Flow (MLD)	81,272	168,167	219,865

6	Current Sewage Collection & Treatment related Infrastructure (M ³ /d)	27,100	27,100	27,100
7	Gap in infrastructure for collection/ treatment (M ³ /d)	54,172	141,067	192,765
8	Infrastructure Required	Sewage Collection, Treatment and Disposal system		

(Source: from ISUDP Report)

Table 4-12: Proposed sewerage development plan for Mombasa - National Water Master Plan, 2030

Major Urban Area	Service Population in 2030	Required Capacity in 2030 (m ³ /day)	Capacity in 2010 (m ³ /day)	Under Construction (m ³ /day)	Proposed Projects		
					Rehabilitation Works (m ³ /day)	Expansion Works (m ³ /day)	New Construction (m ³ /day)
Mombasa	2,644,591	247,005	17,100	0	17,100	229,905	0

(b) Waste collection and transport

Waste collection is mainly done by the county government in conjunction with individual waste pickers, CBOs and other private sector companies (Draft ISUDP, 2015). Mombasa County has 72 transfer stations; 32 stations within the Island, 21 in Mainland North, 9 in Mainland South and 10 in Mainland West. Of these 72 stations, 65 are trucks while 7 are formally designated waste transfer spots (Table 4-13). The County Government and private enterprises normally handle bulk/primary waste from the transfer stations or large properties (e.g. shopping centres, hotels) while CBOs and smaller individual waste pickers do door-to-door collection of waste (Fig. 4-6) (Draft ISUDP, 2015). Waste collection services are paid for almost entirely by the residents, with most households or businesses paying between Kshs. 200 and Kshs. 2000 per month, per property. Waste collected is then dumped at these official waste transfer stations and is transported to the official dump site in trucks (Draft ISUDP, 2015).

Table 4-13 Distribution of Transfer Stations within the County

Area	Population (2009 Census)	No. of Collection Points	Collection Points/Population Ratio
Mainland North	380,055	21	1:18,098
Mainland South	166,008	9	1:18,446
Mainland West	250,179	10	1:25,018
Island	143,128	32	1:14,792
Total	939,370	72	1:13047

Source: Field Survey, 2014 (NEMA, 2014)

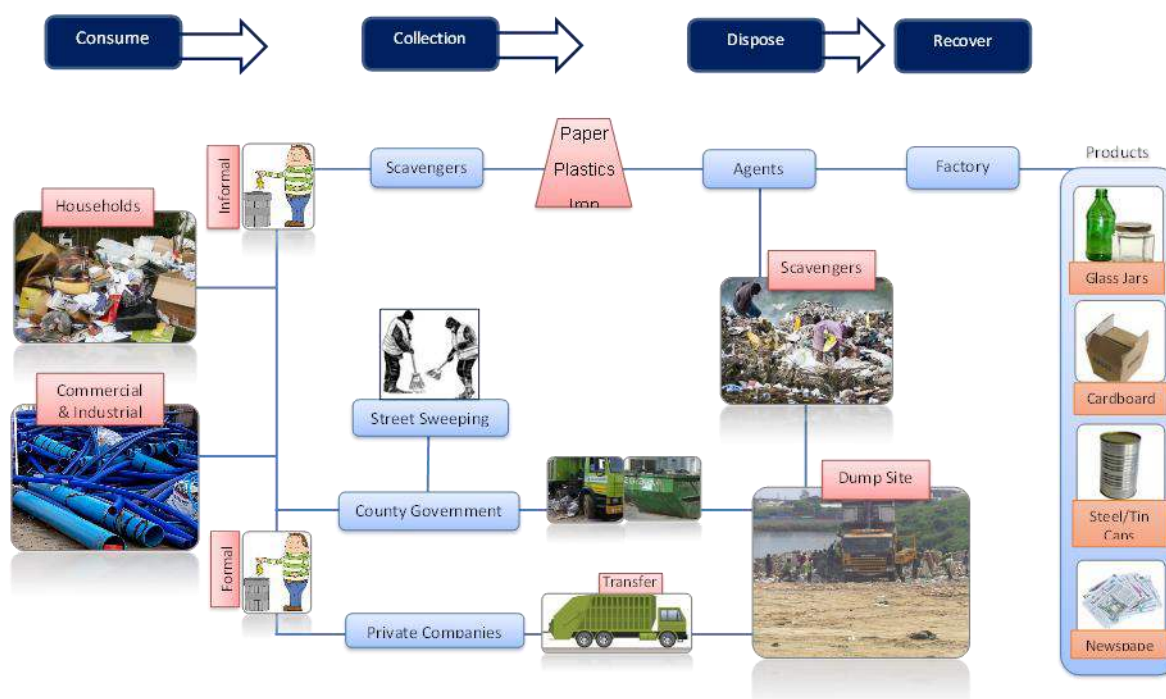


Figure 4-6: Solid waste streams and scavengers role in Mombasa (Draft ISUDP, 2015)

(c) Waste Disposal

Methods of waste disposal in use include the following; septic tank and soak pits, illegal and official dumping sites, incineration, composting, and discharge into the ocean without treatment (NEMA, 2012). Mombasa County has three dumping sites, namely, Kibarani, Mwakirunge and Likoni dump sites. About 460 tons (53 percent) of the solid waste is handled by MOWASCO while the rest (47 percent) is informally disposed of through methods such as burning, burying, incineration, composting, septic tanks and soak pits, and illegal dumping, thus contributing to environmental pollution (NEMA, 2016). Kibarani, the largest dumping site in the County, is located 7 km from the CBD and occupies approximately 8 acres of land in Mainland West. It was receiving about 90 percent of all formally collected waste which averaged about 386.5 tonnes, but has recently been closed and waste is now being taken to Mwakirunge dump site. Mwakirunge and Likoni (Ujamaa Shonda) dumping sites were estimated to receive about 15.65 and 18.22 tonnes, respectively (NEMA, 2016), prior to the recent closure of Kibarani dump site. About 55 percent of the solid waste that is formally managed comes from the island (draft ISUDP).

(d) Waste Recycling

A limited proportion of waste collected is recycled. It is estimated that 23 per cent of the waste is recycled or incinerated by commercial industries for export, while 16 per cent and 2 per cent are used for energy and heat recovery, respectively (Palfreman and Clark, 2015).

II. Environmental issues and related indicators

There are 13,286 of the 183,540 households who rely almost wholly on groundwater for their potable water needs, distributed as follows: wells - 6,245 households, boreholes - 6,941 households (GOK, 2011). Groundwater supplies in Mombasa are vulnerable to salinity intrusion and pollution from pit latrines and septic tanks as the region lacks sufficient sewage treatment to manage the human waste generated in the region (WRI, 2007). There is thus need for better management of solid and liquid wastes, including provision of adequate sanitation facilities and waste treatment plants, to ensure that such water is not contaminated, as well as treatment of water in already contaminated areas to a level where it is safe for human consumption.

Renewable energy is generally not used in the waste sector, but there was mention of a self-help group that generates biogas from waste at Kongowea, but these tend to be informal, small-scale and unstructured (Table 4-14). Energy use is considered to be only slightly efficient, but waste products such as plastics are recycled at a small scale and scavenging at dump sites is common. There are many methods that are used for wastewater treatment and management at the county facilities and also in business premises and private residences depending on whether or not they are connected to the main sewer system (Table 4-14). The respondents to the baseline survey were not aware of any waste sector products, such as recycled ones, being eco-labelled.

Table 4-14: County planners' views on issues relating to the energy, wastewater treatment, and ecolabelling indicators in the waste sector.

Indicators	Remarks
Renewable energy used	- None, but there was one mention for biogas production at Kongowea by a self-help group with the possibility of rolling this out to other area.
Resource efficiency - energy use	- Not efficient to slightly efficient
Resource efficiency - products/materials recycled	- Plastics, metals, bottles. - Uncontrolled recycling of metals and plastics at the dumpsite (scavenging). - Water (end product released to the ocean) - Sludge from sewage used for manure
Waste water treatment and management methods	- Conventional sludge activated system in Kipevu (Mainland) - Rehabilitation of the Kizingo underground waste management plant on the Island - Likoni (south), Nyali and Kisauni (north) use septic tanks and pit latrines - Collected and transmitted to the treatment facilities - Separation of materials such as: soil and silts (preliminary), aeration process (bacterial breakdown), sedimentation (sludge) and water.
Ecolabelling of products from the sector	- None/Not aware of any being eco-labelled.

The main environmental issues are indiscriminate dumping, ocean pollution from effluent/waste emanating from coast-based enterprises and users, lack of a sanitary landfill for Mombasa City, spillages of raw sewage, and air pollution (Table 4-15). The respondents to the baseline survey had good knowledge of the ramifications of these issues, but faced two main challenges, that of enforcement, and lack of a designated sanitary landfill (Table 4-15).

Table 4-15: County planners' views on the environmental issues and impacts, mitigation measures in place and challenges faced in the County - waste sector

Environmental Issues	Related Societal Problems	Major Impacts on Sector	Mitigation Measures in Place	Challenges faced
Indiscriminate dumping	- Blocked drainages leading to flooding, diseases, displacement	- Negative impacts on the budget - Non-compliance to service delivery	- Capacity building for environmental committees, MCAs, Senior Staff - Awareness creation on proper disposal methods	- No laws for enforcement - Enforcement - No designated landfill
Ocean pollution	- Decrease in fish stocks due to damage of breeding grounds - Loss of work and income from fishing has led to insecurity - Pollution overflows (due to design flaws) leads to pressure on the workforce.	- Negative impacts on the budget - Routine works (unblocking) -	- Civic education	- Enforcement
Lack of a landfill	- Street scavengers -	- Pollution, littering - Pollution and compliance	- Plan in place to construct a landfill (CIP) - By laws exist - Inspectorate unit exists - Improving collection of wastes	- No designated landfill
Raw sewage	- Diseases -	- More work (unblocking) for the workforce - Routine works	- Vaccinations - Soap and milk provided	- Harmful/toxic gases & microorganisms - Health issues
Air pollution	- Air pollution	- Air pollution	- Improving collection	- None stated

There are as yet no studies that have quantified carbon dioxide and methane gas emissions from both solid and liquid waste streams in the County.

III. Wellbeing and equity indicators

The County Government is in charge of waste handling in conjunction with individual waste pickers, Community Based Organisations (CBOs), and other private sector companies. While the County Government and private enterprises normally handle bulk/primary waste from the transfer stations or large properties (e.g. shopping centres, hotels), CBOs and individual waste pickers collect wastes from door to door. The cost of waste collection is borne by the residents, who spend between Ksh. 200 and Ksh. 2,000 per month per property. The waste collected is then dumped at official waste transfer stations with only limited quantities being recycled (NEMA, 2016).

The transfer stations do not adequately serve settled areas on the periphery especially in Mainland North and Mainland West. Unfortunately, considerable distances between formal waste collection points and residential areas lead to dumping or indiscriminate burning of waste. High income areas located at a distance of 1km from formal collection points such as Mainland North are well served by door-to-door waste collection services which takes waste directly to the Mwakirunge dumpsite (NEMA, 2016). Different strategies may be required to address waste production among different socio-economic groups as the per capita amounts generated are distinctly different (Table 4-16), and types of waste (by amount) are also different, e.g. the highest waste generated in the high income group is paper waste, while that in the low income group is organic.

Table 4-16: Estimated waste generation rate by socio-economic group for Mombasa County

Socio-economic type	%	Population (2009)	Population (2015)	Per Capita (kg/day)	Waste Production Per Day in Tonnes (2015)
Low	53.6	503,446	590,343	0.55	324.7 (37%)
Medium	23.9	224,613	302,458	0.89	269.2 (31%)
High	22.5	211,311	211,199	1.33	280.9 (32%)
Total	100	939,370	1,104,000	0.80*	874.8 (100%)

(Source: Palfreman and Clark, 2015)

Table 4-17: Contribution of "water supply; sewerage, waste management" sector to wage employment at national level

WATER SUPPLY, SEWERAGE, WASTE MANAGEMENT SECTOR	YEAR				
	2011	2012	2013	2014	2015
Wage employment in Private Sector* (in 000s)	1.3	1.3	1.4	1.4	1.4
Wage employment in Private Sector* (in 000s) - ALL SECTORS	1,440.8	1,493.6	1,599.8	1,669.4	1,759.6
Wage employment in Public Sector* (in 000s)	6.3	7.2	8.1	9.0	10.1
Wage employment in Public Sector* (in 000s) - ALL SECTORS	643.3	662.1	683.3	700.8	718.4
<i>Estimated number in wage employment in Private Sector (Waste/Water)* for Mombasa County (persons) - Using 5.11%</i>	66	78	79	82	83
<i>Estimated number in wage employment in Public Sector (Waste/Water)* for Mombasa County - Using 5.11%</i>	322	979	1099	1259	1461
<i>Total number</i>	388	434	485	531	
Table 4-7 Estimated Number	390	430	490	530	

(Source: Partially modified from KNBS, 2016). Data for 2011-2015, in relation to a total of 21 industry sector categories; 2015 data are provisional (KNBS, 2016). Assumption for the county estimate is that the same

proportions as for national level apply, and based on total labour force for Mombasa County assuming a 15% unemployment rate (refer to Table 4.5 above). * - includes remediation activities. NB: Official estimates (July 2016, D. Onyango, pers comm.) are: (a) County Govt (Public Sector) = 331; MOWASCO (Private Sector) = 75; Total = 406; County Govt (water supply) = 4; Private Sector = ?? (was not available).

Mombasa County loses 548 million KES each year due to poor sanitation, and this includes losses due to access time, premature death, health care costs and productivity (Ministry of Health, 2014). This estimate does not include some costs that could be significant (such as water pollution and tourism) and is therefore likely to under-estimate the true cost of poor sanitation (Ministry of Health, 2014). The estimated number of people in formal wage employment¹ in the Water / Waste sector in Mombasa County is quite low, with about 1550 people employed and with slightly more than 90% of them working in the public sector (Table 4-17).

The national ratio of male to female employees in the sector is 2:1 (Kenya National Bureau of Standards: Kenya Facts and Figures, 2015). The Mombasa County baseline survey respondents were uncertain about the ratio of male to female employees in the sector as there was no available data from official sources, but it is not unreasonable to assume that the ratio at national level is similar to that at the county level. Similarly, since there was no reliable figure on the percentage of people employed in the sector at County level relative to the total workforce within the County, an estimate was derived as indicated in Table 4-17. The respondents generally agreed that amongst the low income earners, work income was not the sole livelihood for the employee household, and that the contribution of the sector to the household budget ranges from very low to medium.

IV. Green economy policy interventions

(a) National Level

At the national level, the Kenya GESIP Report (2015), Objective 1.7, proposes to "Promote integrated waste management in all the counties by 2020", and has the following two strategic actions to achieve this with a timeframe of 2020 and through strong public private partnerships:

- Roll out pollution prevention programs across manufacturing and service industries
- Roll out recycling and industrial symbiosis projects through private public partnerships

The National Solid Waste Management Strategy purposes to guide sustainable solid waste management in Kenya and to ensure a healthy, safe and secure environment for all. The Strategy is a deliberate and visionary commitment for the country in the management of solid waste. Its guiding principles are geared towards addressing the following:

- The Current situation (Where are we now?)
- The Preferred state (Where do we want to go?) and
- Implementation of the Strategy (How do we get there?)

¹Wage employment figures include casual employees, part-time workers, directors and partners serving on a regular basic salary contract. Self-employed persons and family workers who do not receive regular wages or salaries are excluded. Employment in the informal sector of the economy is not included (KNBS, 2015).

In the Constitution of Kenya, Article 42 on the Environment provides that every person has the right to a clean and healthy environment, which includes the right to have the environment protected for the benefit of present and future generations through legislative and other measures, particularly those contemplated in Article 69; and (b) to have obligations relating to the environment fulfilled under Article 70.

(b) County Level

The County GE initiatives in Table 4-18 below are derived from three County documents: the Annual Development Plan (ADP) 2015, the Integrated Strategic Urban Development Programme (ISUDP) Report, and the County Integrated Development Plan (CIDP) 2013 to 2017. The challenges and opportunities indicated are also derived from these reports. They do signify that the County Government is serious about improving the current wanting situation in the waste sector.

Table 4-18: Existing Green Economy initiatives, challenges, and opportunities in the waste sector.

GE Initiatives	Challenges	Opportunities
- Augment infrastructure for sewerage, drainage, solid waste management - Sewerage Coverage: from 17% (2015) to 35% (2017), 70% (2025) & 100% (2035), - Separate wastewater sewerage and storm water drainage: 70% (2025) & 100% (2035). - Develop integrated waste management facility - Waste segregation at source, recycling rate: 15% (2025) & 50% (2035) - Restrict illegal dumping and open burning of waste - Garbage recycling plant in Mwakirunge - Wastewater recycling (Kipevu)	- Inadequate solid waste segregation & management infrastructure - Inadequate incinerators for the health facilities - Dumpsite locations on flight path (Mwakirunge), near marine environment. (Kibarani) - Utilization of recycled water only for non-potable commercial use - wastewater recycling versus community traditional perceptions - wastewater recycling requires dual infrastructure and piping systems, with the resultant capital investment	- Landfill development - Install gravity wastewater system, a treatment plant and a recycling plant – operational 2025 – meet 10-15% non-potable demand by 2035 - Develop Sewage Master Plan for Mombasa – IWMS - Wastewater Master Plan ongoing (IDA supported), with STP (2020), MTP (2025) & LTP (2040) - Rehabilitation of existing sewerage system - On-site sanitation & bio-degradation technologies at scale - PPPs, Policy/Regulations, Capacity enhancement

(Source: CIDP, ADP, ISUDP)

(c) County Baseline Survey, November 2015

The respondents to the county baseline survey provided information on a number of GE initiatives that relate to the waste sector, and outlined the challenges and opportunities as described below and summarised in Table 4-19. Many of the initiatives cited relate to better waste handling, but clearly there are opportunities for recycling of waste materials for reuse or to generate energy from them and this can have significant positive impacts for environment and employment (Table 4-18). The County is in the

process of enacting waste management and environmental policies and regulations and this creates opportunities for green economy activities.

Table 4-19: Existing Green Economy initiatives, challenges, and opportunities in the waste sector according to the County planners - baseline survey.

GE Initiatives	Challenges	Opportunities
- Civic education on good disposal methods - Use recycled materials in offices - Waste segregation promotion by developers - Waste management and environmental policies/regulation for the County by-laws	- Vandalism of infrastructure - Low resource allocation - Vandalizing of litter bins - Compliance failure - Limited technical skill - Inadequate capacity - Sludge produced is not fit for use (heavy metals) - Perception of sludge is negative	- Sludge recycling - Revenue from recycling material - Revenue from energy production - Conversion of sludge to briquettes - Fertiliser/manure from organic waste - Fuel production from waste

(Source: Baseline survey, November 2015)

The County takes the waste sector serious and have a number of activities with stakeholders to help manage the waste and intend to improve on access to WASH for its inhabitants. To do this they have a suite of development objectives and tools to guide the implementation, as well as consultations and partnerships with stakeholders.

(d) Proposed Options, Approaches and Considerations

The low contribution of the waste sector to the national (and therefore presumably also the county) GDP reflects both low investment in sector and huge growth potential. The latter is also underpinned by the Kenya Green Economy Strategy and Implementation Plan (GESIP) which identified waste management as one of the key sectors for green economy. This study has identified a number of opportunities and benefits which can accrue to the sector if green economy principles are factored into the county development plans, and these include: solid & liquid waste recycling including tiered waste segregation at source and development of a circular economy; improved health and environment; increased productivity; enhanced tourism incomes; framework for structured involvement of formal and informal sector players; targeted funding revenues for operations, maintenance, and infrastructure development; synergies with national government, public private partnerships, and international donors for large infrastructure solutions, etc.

In addition to the key options proposed in sections (a), (b), and (c) above, other options may include:

- Development of waste management and environmental policies, regulations, and standards for waste treatment and disposal - some of these are already underway but the scope, depth and range of issues can be expanded;
- Strengthen relevant institutions to improve on regulatory enforcement and compliance;

- Create a system of targeted subsidies and taxes to support adoption of IWMS's and overall development;
- Promote the trade and purchase of green products and services;
- Encourage transparency, accountability, and solid fiscal management as this will ensure buy-in and attract funding for projects from local, national and international donors;
- Enhance opportunities for recycling of waste materials for reuse or to generate energy from them, adopting the principles of industrial symbiosis & circular economy; and
- Develop integrated waste management systems and managed landfills that can be used to also generate energy.

Factors that should be considered when incorporating the green economy options into the county development plans include:

- **Hazardous wastes:** These should be included in the integrated waste management system – protocols for storage, handling, transportation, and disposal (includes electronic and hospital waste);
- **Environmental protection and monitoring:** Monitoring of air, groundwater and coastal waters, sensitive marine and land ecosystems, and heritage sites;
- **Participatory approaches:** Community involvement in waste management;
- **Job creation:** Interlinked options for improvement of livelihoods and (youth) employment, gender equity;
- **Climate proofing:** Ensuring climate change resilient waste-related infrastructure based on projected changes in e.g. sea level rise (SLR), storm surges, rainfall/runoff amounts and intensity;
- **Public health:** Risk assessments/ evaluations and mitigation measures;
- **Non-reticulated system solutions** for integrated waste management – decentralised low cost and effective systems;
- **Technologies:** State-of-the-art, eco-friendly and resource efficient; and
- **Adequate financing and capacity:** Achieving the planned targets through a combination of county budget resources, synergies with national government, private-public partnerships, and human and institutional capacity building to develop implementation, maintenance and monitoring capacities.

4.4.2 Tourism

Mombasa County hosts several tourist attractions and world heritage sites including, as outlined in the Mombasa County Annual Development Plan (2014): (i) the historic Fort Jesus Museum which is also a UNESCO World Heritage site; (ii) the gigantic “Elephant Tusks” along Moi Avenue; (iii) several buildings in the old town including the Old Port; (iv) The white sandy beaches (v) Mombasa Marine Park, which is home to a variety of fishes and other sea creatures, and (vi) two private nature trails, Haller Park and Butterfly Pavilion, operated by Bamburi Cement factory.

The industry has faced strong competition from other destinations, and is hampered by poor tourism infrastructure, lack of capital to improve infrastructure, population pressure on land, adverse security advisory notices by countries of tourist’s origin, and security issues in general (Kenya Water Report, 2005) and the situation is today still pertinent to Mombasa County. The recent security challenges have greatly affected the tourism sector leading to low tourist arrivals, general loss of jobs and negative multiplier effect on other sectors of the County economy like trade and investments (County Government of Mombasa, 2015b). In general, while there exist a lot of data relevant to the sector at the national level, there is very little data at the County level (see Annex 4 - Mombasa County Data Sheets).

I. General economic sector indicators

The direct contribution of Travel & Tourism to GDP was KES220.6bn (4.1% of total GDP) in 2014, and was forecast to rise by 3.8% in 2015, and to rise by 5.1% pa, from 2015-2025, to KES376.5bn (4.0% of total GDP) in 2025 (Fig. 4-7; World Travel and Tourism Council, 2015). Tourism growth has been on an upward trajectory for the past 20 years or so. A drop in tourist arrivals in 2007/2008 can be attributed to a period of political instability, followed by a recovery in 2009/2010. Tourist numbers have steadily dropped since then and this appears to be related primarily to travel advisories issued by Western nations in view of the terrorism threats facing the country. This has affected earnings from tourism (Fig. 4-8) and Mombasa County has been particularly hard hit. The County respondents to the baseline survey ranked the sector contribution to the County GDP and employment as important (Table 4-20).

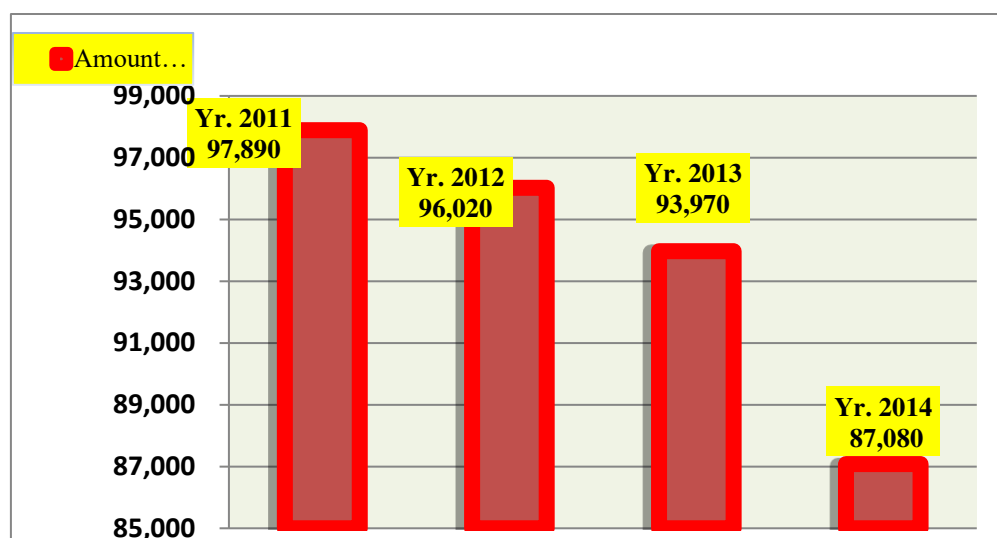


(Source: from Kenya International Tourism data – Indexmundi.com, 2014)

Figure 4-7: National tourism arrivals, 1995-2014.

Table 4-20: City planners' views on the tourism sector contribution to the County GDP and employment.

Indicators	Remarks
Economic significance - GDP	Important to very important
Economic significance - Employment	Quite important to very important
Key sources of energy for the sector	National electric power grid; gas for cooking



Derived from KNBS Kenya Facts and Figures Report, 2015

Figure 4-8: National Tourism Earnings

Table 4-21 Hotel rooms and beds available and occupied by Coastal Zone, 2014

Geographical Area of Mombasa	# of Hotels	ROOMS		BEDS		
		Available	Occupied	Available	Occupied	
Coast Beach	152	4320.6	1670.6	8248.2	2527.7	
Mombasa Island	29	186.6	88.7	304.5	95.9	
Coast Hinterland - East	44	171.7	47.2	287.5	65.1	
Coast Hinterland -West	46	179.5	49.3	300.6	68.1	
Totals	271	4858.4	1855.8	9140.8	2756.8	
MOMBASA BEDS OCCUPIED BY					BED	ROOM
	Foreign Residents	Uganda/ Tanzania Residents	Kenya Residents	Permanent Occupants	Occupation Rate % - Bed	Occupation Rate % - Room
Coast Beach	1425.5	21.4	1076.3	4.5	30.6	38.7
Mombasa Island	14.6	2.5	73.4	5.3	31.5	47.5
Coast Hinterland - East	32.4	0.1	28.5	4.1	22.6	27.5
Coast Hinterland -West	33.9	0.1	29.8	4.3	22.7	27.5
Totals	1506.4	24.1	1208	18.2	30.2	38.2

(Derived from KNBS Statistical Abstract, 2015).

In 2013, the number of tourism enterprises in Mombasa County was 201 (Mombasa County Annual Development Plan, 2014). As indicated in Table 4-22 above, most of the hotels are located along the beach, and KNBS (2015) data indicate that the highest occupancy rates (Table 4-22) are by foreigners,

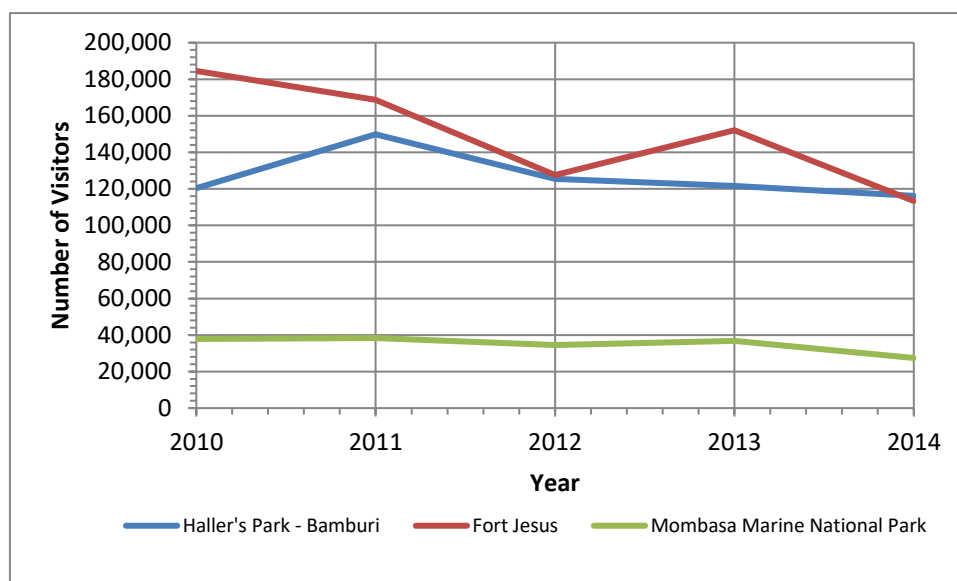
closely followed by Kenyans. The occupancy rate has fluctuated between 30 and 50% from 2010 to 2014, suggesting that much remains to be done to ensure higher occupancy rates of the hotels (Table 4-22).

Table 4-22: Mombasa/Coastal Hotel Rooms – Count of Beds Available and Occupied

2010	Beach	Other	Coast Hinterland
Room Available ('000 nights)	3,737	273	204
Occupancy Rate (per cent)	46	42	42
Bed Available ('000 nights)	7,294	442	339
Occupancy Rate (per cent)	44	34	35
2011	Beach	Other	Coast Hinterland
Room Available ('000 nights)	3,361	351	132
Occupancy Rate (per cent)	55	51	33
Bed Available ('000 nights)	6,594	590	281
Occupancy Rate (per cent)	48	48	29
2012	Beach	Other	Coast Hinterland
Room Available ('000 nights)	4,144	519	216
Occupancy Rate (per cent)	46	43	24
Bed Available ('000 nights)	7,880	867	411
Occupancy Rate (per cent)	40	30	22
2013	Beach	Other	Coast Hinterland
Room Available ('000 nights)	3,275	259	304
Occupancy Rate (per cent)	51	41	27
Bed Available ('000 nights)	6,446	389	530
Occupancy Rate (per cent)	43	32	24
2014	Beach	Other	Coast Hinterland
Room Available ('000 nights)	4,321	187	351
Occupancy Rate (per cent)	39	48	27
Bed Available ('000 nights)	8,248	304	588
Occupancy Rate (per cent)	31	31	23

(Source: KNBS Statistical Abstract 2015)

The historically rich and significant Fort Jesus in Mombasa, Haller's park, and Mombasa Marine National Park are three major and popular tourist attractions in Mombasa, drawing the attention of local and foreign tourists alike, but the numbers visiting have generally been declining from 2010 to 2014 (Fig. 4-9). A much higher number of visits can be realised for these sites, since, for example, in 2006, Haller Park received 4.7% of all visitors to Kenya's parks and reserves (WRI, 2007).



Source: Derived from KNBS Statistical Abstract 2015

Figure 4-9: Number of visitors to Fort Jesus, Haller's Park, and Mombasa Marine National Park

Gross and per capita emissions of CO₂ from the sector were generally unavailable. Dilapidated road and water transport (e.g. ferry) infrastructure has made Mombasa County unattractive to tourists. However, to help spur the growth of the sector conference tourism, marine-based tourism, hotel investment, and eco-tourism are proposed (Owinga and Datche, 2015).

II. Environmental issues and related indicators

Tourism is climate sensitive because climate dictates the type of natural attractions and resources like the sun and ecosystems, but also determines the nature of tourist activities, their location and the seasons to be spent by the tourists (Njoroge, 2015). Coastal towns such as Mombasa are particularly under threat of sea level rise (cf. Awuor et al., 2008; Ongoma and Onyango, 2014). In Mombasa County, it was observed that the tourism managers, even though having experienced some impacts of climate change such as water scarcity, coral reef bleaching, changes in rainfall, flooding, changes in beach structure, etc., seem not to have been able to link these local impacts with the global picture that they have of climate change (Njoroge, 2015). On the other hand, government officials, conservation and research institutes were more aware of climate change potential impacts at local scale: thus, while these stakeholder groups would plan for adaptation, the tourism managers would not have tangible adaptation plans (Njoroge, 2015). However, environmental related indicators for the tourism sector as per the data sheets (Annex 4) and which would help guide decisions on green economy interventions were not available, even though there have been many environmental studies carried out, where specific indicators have been evaluated either for specific firms, areas, or sub-sectors at scales lower than county level.

Predictably, the largest numbers of poor people live in the vicinity of parks near urban and other densely populated areas. For instance, more than 305,000 live near Mombasa Marine Park (WRI, 2007). Solid

and liquid effluent discharge from hotels, along with industry, residences among others, has contributed to negative impacts on the terrestrial and marine ecosystems of Mombasa. Addressing this and land degradation issues will help to greatly reduce the factors that negatively impact on the growth of the tourism sector. Youth activities are concentrated in tourism and recycling (scavenging for food and items that can be resold) (Hildebrand and Holst, 2015) and hence can play a great role in green economy issues within the County if better organised and formally incorporated in planning.

Within the tourism sector, there is scope for much increased use of renewable energy like solar and biogas, and there is a wide variation between establishments as to how efficiently they use existing energy (Table 4-23). The sector is quite aware of product ecolabelling and in many cases this is driven by the fairly recent rise in the uptake of eco-tourism products. During the field survey for this study, it was observed that the Severin Hotel has incorporated energy resource use efficiency and recycling, and it has established an elaborate waste water treatment and recycling system (Table 4-23). The hotel industry players are generally willing to get involved, in partnership with the County authorities and sector stakeholders, and play a significant role in advocating for and implementing green economy practices in their establishments.

Table 4-23: County planners' views on issues relating to the energy, wastewater treatment, and ecolabelling indicators in the tourism sector.

Indicators	Remarks
Renewable energy used	- National grid (HEP, Geothermal) high; Solar, Biogas (very low)
Resource efficiency - energy use	- Not efficient to highly efficient
Resource efficiency - products/materials recycled	- Waste water (e.g. Severin Hotel), biomass, waste water/sewage, building waste, plastic bottles, linen, building poles.
Waste water treatment and management methods	- Water recycling (e.g. Severin Hotel), septic tanks, cess pits, treatment plant channelling to gardens/irrigation, biodigester, environmentally friendly chemicals.
Ecolabelling of products from the sector	- Lighting system bulbs and KAM on energy, eco-jiko, coco-briquettes, environmentally friendly detergents, ecotourism, and waste separation.

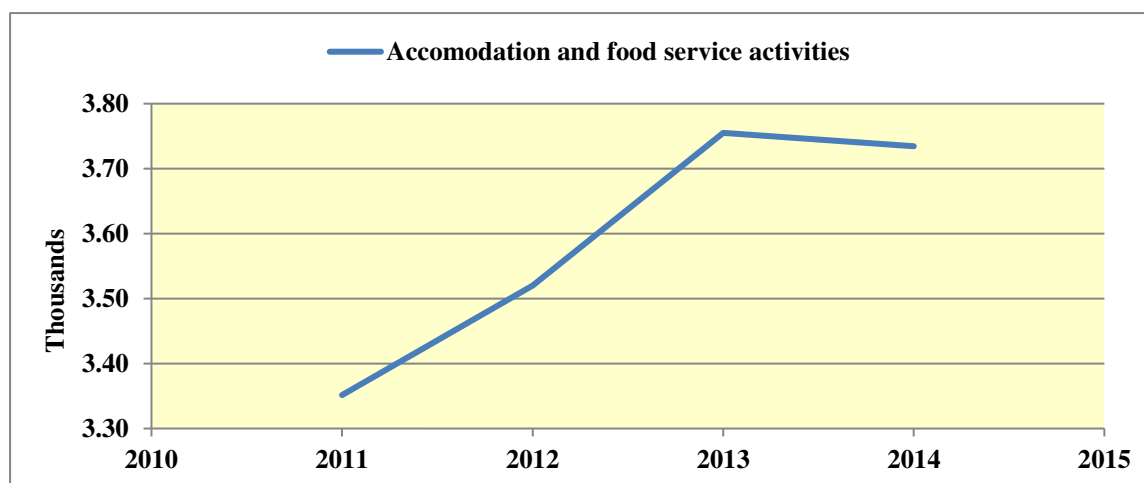
The County respondents to the baseline survey indicated that the environmental issues affecting the tourist sector included: poor sewerage and waste management with much of the effluent are being discharged into the adjacent ocean, environmental degradation in the coastal zone driven by anthropogenic activities, and global warming which raises the costs of running hotels as more energy is needed to run air conditioning systems (Table 4-24). Some of the key challenges hampering the efforts to ameliorate the environmental impacts include inadequate sewerage treatment and disposal infrastructure, lack of coordination to address the issues, inadequate technical capacity to manage waste, as well as financial and land constraints to set up the necessary infrastructure (Table 4-24).

Table 4-24:County planners' views on the environmental issues and impacts, mitigation measures in place and challenges faced in the County tourism sector.

Environmental Issues	Related Societal Problems	Major Impacts on Sector	Mitigation Measures in Place	Challenges faced
<i>Poor sewerage/waste management</i>	- Increase in effluent discharge to the ocean - Lack of knowledge on proper waste disposal - Attitudes towards proper disposal - Treatment of diseases related to waste control e.g. typhoid. - Buildings collapsing	- Polluted beach and ocean - Eyesore to tourists and local people - Possibility of disease outbreak - Extra cost to the operation	- Waste management and recycling - Bins - Waste disposal licenses - Health benefits issued to staff	- Lack of coordinated efforts - Lack of infrastructure - Lack of awareness of environmental stewardship - Lack of technical capacity - Financial constraints for conservation education and training programmes. - No land for disposal and recycling plant. - Follow up of environmental audits. - Cost of putting up plants and machinery very high. - Staff turnover
<i>Global warming</i>	- Cost of running hotels is higher because of the power bills	- Expense on power bills - Intense warm weather making it unbearable to work in good conditions	- Tree planting - Proper planning and building of houses to enable air circulation in town. - Solar and wind energy	- Financing on training and greening initiatives at school level
<i>Environmental degradation</i>	- Lack of knowledge and mitigation to environmental issues	- Compromise on standards needed in Environmental management	- SOP refreshers	- Maintaining and monitoring the standards is difficult. - High cost

III. Wellbeing and equity indicators

It is noted that tourism planning within a region provides for socio-economic development, environmental protection and conservation (Mombasa County Government, 2015). The number of people employed within the accommodation and food service activities between 2011-2014 has been on the rise between 2011 and 2014 (Fig. 4-10), and given the decline noted in national tourism figures over the same period, it is likely that this is driven more by demand for accommodation from domestic business and also domestic tourism.



Source: Kenya National Bureau of Standards: Kenya Facts and Figures, 2015

Figure 4-10: Mombasa wage employment (in thousands) in accommodation and food service activities

The baseline survey respondents estimated the ratio of male to female employees in the sector to be between 40 and 60 per cent (see Annex 3). However, 2014 statistics indicate that the ratio of male to female employees in the tourism industry at national level is 2:1, or 67% males to 33% females (Table 4-25). Women in Kenya constitute about a third of the estimated workforce in tourism.

Table 4-25: National wage employment in the tourism sector (number in thousands)

Year		2012		2013		2014	
Employment By Industry (population count in '000s)	Main Sector Category	Male	Female	Male	Female	Male	Female
Accommodation and food service activities	Tourism	44.5	24.4	49.9	23.6	49.0	24.1

(Source: Kenya National Bureau of Standards: Kenya Facts and Figures, 2015)

The baseline survey respondents indicated that, while for some workers in tourism sector work income is the sole livelihood, others have small scale income generating activities to supplement their earnings, and that the sector contribution to household budget ranges from low to very high. Msengeti and Obwogi (2015) carried out a study on employees in the hotel industry in Mombasa County, and concluded that pay as a financial component of reward has a weak influence on employee retention in hotels in Mombasa County tourism area as compared to the non-financial element of work environment.

It was further noted that there were also significant pay discrepancies between the players in the industry leading to employees constantly seeking better opportunities, further exacerbating the low employee retention issue (Msengeti and Obwogi, 2015). High staff turnover was noted to be a problem by the baseline survey respondents and that this was a contributing factor to unsatisfactory management of sewage and waste in general.

IV. Green economy policy interventions

(a) National Level

In the Kenya GESIP Report 2015, tourism focused interventions are highlighted in Thematic Area 2: Building Resilience. Resilience building efforts will focus on ensuring that the tourism dependent economies and livelihoods are less vulnerable to risks and challenges of climate change and changing growth dynamics. The aim of this building block is to embed resilience in the tourism growth strategy.

In addition, GESIPs thematic Area 3: Sustainable Natural Resource Management focuses on the economy-environment nexus to optimize the contribution of Kenya's natural resources to the economy, industrialization and livelihoods. It encompasses agriculture, forestry, water, fisheries, wildlife, land use, and extractive industries. Through this thematic area, GESIP seeks to address the drivers of change in both the biological and physical aspects of natural resources emphasizing the need for decoupling development from sustainable natural resources management and conservation of Kenya's natural capital.

The National Tourism Strategy, 2013-2018 is a fulfilment of the Tourism Act 2011, Section 3. The strategy addresses national issues confronting Kenya's tourism sector and focus the players in the sector on sustainable tourism. For long, the sector has been characterized by low and high seasons resulting in some instances in hotels and businesses closing down or cutting heavily on staff during the low season. This results in underutilized facilities, infrastructure, capacities, and human capital.

(b) County Level

The county GE initiatives shown in table 4.24 below are derived from County development plans i.e. CIDP (2013-2017), Annual Development Plans (ADPs) and Integrated Sustainable Urban Development Programme (ISUDP) project. The listed challenges and opportunities are a manifestation of the county governments' resolve to tackle tourism related challenges and revive the sector which is one of the key revenue contributors to the county.

Table 4-26: Existing Green Economy initiatives, challenges, and opportunities

GE Initiatives	Challenges	Opportunities
- Transfer of public land to county land specific for conference facilities - Allocation of adequate funds for outdoor marketing; e-marketing and media - An all inclusive county tourism strategic plan is under development. - Development of Community Cultural Centres - Establishment of Tourists Information Centres - Establishment of Beach Watch Inspectorate	- Taxation: The value added tax (VAT) of 16% is said by some to be harming the recovery of the sector - The monthly \$5 bed tax for all hotels and resorts, plus \$40 per month levy on tourist vehicles are contested - Inadequate Finances - only about 1 to 2% of budget goes to the Tourism and Culture Development department - Insecurity related to terrorist activities	- Community-private sector joint ventures to operate tourism-related enterprises e.g. development of premium parks, high value niche products, etc. - Increasing tourism demand from within and outside the country as well as improving the range of tourism products/services on offer - MICE (Meetings, Incentives, Conferences, Exhibits) markets: developing a specialized niche of group tourism - Enhancing tourism quality, sustainability, accessibility, skills, ICT use, and socio-economic base of the sector - Promoting Mombasa as a unique destination

Sources: CIDPs, ADPs, ISUDP and Mombasa Gate Master Plan

(c) County Baseline Survey, November 2015

The development and success of the hotel industry, for example, depends on the availability of natural resources since the latter are one of the main assets of the local hotel industry, but on the other hand, hotels are among some of the largest consumers of energy, water resources and consequently generate a large amount of waste (Gitobu and Njoroge, 2015). It has been shown, from a survey of 24 hotels in Mombasa County, that environmental conservation, profit increment and market advantage has to a very large extent driven hotels in Mombasa County, Kenya to adopt green marketing practices such as; product modifications, changes in the production process, elaborate advertising, change in packaging, and selling of products based on environmental benefits (cf. eco-labelling) (Gitobu and Njoroge, 2015). Mombasa County should strive, as a matter of priority, to ensure that all tourism market players in the County have access to the requisite finances needed so as to effectively compete with tourism players from other Counties (Owinga and Datche, 2015).

The Green Economy initiatives that were mentioned by the respondents to the baseline survey (Table 4-27) are for the most part not anchored within the sector, but would generate benefits for the sector if implemented, and offer opportunities for mutually beneficial partnerships with local communities, schools, community based organisations, governmental agencies, and private companies.

Table 4-27: Existing Green Economy initiatives, challenges, and opportunities in the tourism sector according to the County planners.

GE Initiatives	Challenges	Opportunities
- Coast Region Green Growth Initiative	- No corporate social responsibility to support GE growth as it is a non-profit making body - No formal monitoring and evaluation system - More funding is required	- Advocacy and collaboration projects
- Tree planting - Sensitization of community green economy - Clean up events - Green school project	- Funding: Most of the schools/ institutional based GE need funding - so as to undertake trainings to teachers and students - Space: in most town schools there is little space to undertake GE initiatives.	- The community welcomes the initiatives - Most schools and teachers and management of schools have keen interest and are willing - Proposals for support e.g. for tree nursery, planting, beautification, botanical and institutional gardens.
- Kenya Coastal Climate Change Adaptation Project	- None stated	- None stated
- Coastal Water Security and Climate Resilience Project	- None stated	- None stated
- Solar lamps for street lighting. - Construction of green building with a mega solar P.V. system to provide services instead of electricity.	- Solar system - initial cost too high	- Payback period is short - can be extended even to power lighting
- Water waste treatment plant	- Water purification very expensive	- Wildlife Clubs Kenya as an institution has staff and volunteers willing to undertake these initiatives. - Bottling of fresh water can be done in-house and same sold to clients of other institutions.

The information provided by respondents to the baseline survey on the development planning cycle for the tourism sector does not seem to capture the salient issues necessary for its proper management and monitoring with respect to potential related green economy ventures, and this could be partly due to the sector respondents not being fully conversant with the workings and range of activities being undertaken with the sector.

(d) Proposed Options, Approaches and Considerations

The County government's constitutional mandate on tourism is limited to local tourism services and policy development. This can be the factor behind the sector being given less prominence as most of the economically vibrant tasks are under the mandate of the national government. The County government then limits financial allocation to the sector with the understanding that national government will allocate funds and support the sector. In addition to the key options, strategies and opportunities heightened in the previous sections, other options may include:

- Domestic Tourism: Exploit the demand created by domestic tourism. Over the 2014 Christmas period, for example, occupancy rates in Mombasa's hotels were between 80% and 90%, with some achieving nearly 100% occupancy (Kenya Tourism Development Authority, 2016);
- Local Trade: capitalise on local trade by bolstering ancillary revenue streams, such as food, marine products;
- Marketing: Utilize platforms such as the Tourism Recovery Taskforce launched in July 2014. The taskforce was charged with developing a comprehensive strategy for reviving tourism, including initiatives to improve security, develop infrastructure and shift the perception of the country in foreign markets;
- Ecolabel products;
- Incentives for adoption of green economy principles in the sector e.g. green technology in buildings, eco-friendly waste treatment and management practices, and activities undertaken with consideration to environmental protection;
- Classification: Collaborate with national government in the implementation of a classification programme geared towards improving quality assurance and standardisation across all of the country's tourism facilities; and
- Business Travel or Business Events: utilise the "MICE" markets – meetings, incentives, conferences and exhibitions.
- Roll out "trees for pay" program for youth;
- Support and enhance existing Green Economy initiatives (see section 4.2.2d).

Factors that should be considered when incorporating the green economy options into the county tourism development plans include:

- **Alternative tourism:** In essence, tourism activities or development that are viewed as non-traditional. It is often defined in opposition to large-scale mass tourism to represent small-scale sustainable tourism developments. Alternative tourism is also presented as an 'ideal type', that is, an improved model of tourism development that redresses the ills of traditional, mass tourism;

- **Code of Ethics / Conduct / Practice:** Recommended practices based on a system of self regulation intended to promote environmentally and/or socio-culturally sustainable behaviour;
- **Cultural tourism:** Travel for the purpose of learning about cultures or aspects of cultures;
- **eCommerce:** Internet facilitated commerce, using electronic means for promoting, selling, distributing, and servicing products. This will reduce corruption and inefficiency;
- **Joint Venture partnerships-** using community-private sector joint ventures as a contractual partnership between a community or local institution and a private investor, to work together in establishing and operating a single tourism or hunting enterprise
- **Promotion and Investment:** Investors are urged to take advantage of several strategies and programmes that have been earmarked by the county government to attract investments into the industry such as establishment of resort cities, branding of premium parks, development of high value niche products, development of a replica of Kenya Utalii college at the Coast, and building of new high- end international hotels chains among others.
- **Market linkages-** increasing integration between tourism and other economic sectors including agriculture, fisheries, manufacturing, construction and crafts production.
- **Positioning:** process of ensuring potential customers have a desired perception of a product or service, relative to the competition from other counties.
- **Environmental management systems:** establish systems with the aim of mitigating negative environmental impacts; and
- **Green Globe/Green Globe 21:** GREEN GLOBE 21 is the worldwide benchmarking and certification programme which facilitates sustainable travel and tourism for consumers, companies and communities. It is based on Agenda 21 and principles for Sustainable Development endorsed by 182 governments at the United Nations Rio de Janeiro Earth Summit in 1992.

4.4.3 Cities and Built Environment

I. General economic sector indicators

The built environment includes all of a city's buildings, parks, and public spaces and its associated support infrastructure such as water and energy networks. From the proposed land use in the county, it can be seen that the current about 100km² of land will be fully developed by the year 2035 (Table 4-28), with significant developments expected with respect to residential, industrial, public utility, transportation, recreation & beach, and future development land uses. This huge increase within a time span of only 20 years, beyond which further growth opportunities will be limited, is indeed a critical growth period so the opportunity to carry out the proposed developments within a Green Economy context should not be missed.

Presently, land ownership in most areas is not guaranteed as most of the residents do not legally own land and the land they live on is owned by absentee landlords (Mombasa County Annual Development Plan, 2014). A number of informal settlements exist in the County, and the growing population creates a huge demand for quality and affordable housing (Mombasa County Annual Development Plan, 2014). There

has been an increase in real estate investment with the construction of beachfront apartments, hotels and resorts concentrated along the low-lying areas of the shoreline (Government of Kenya, 2010). These infrastructural developments increase shoreline instability and imperil the stability of seawalls or buildings due to high-energy waves from intensive erosion (Odada, 1995; Mwanje, 1997; Kairu and Nyandwi, 2000). The development of the sector will, therefore, not be without its challenges which should be anticipated, carefully considered, and mitigated so as not to derail the effort.

Table 4-28: Proposed Land use for Mombasa County

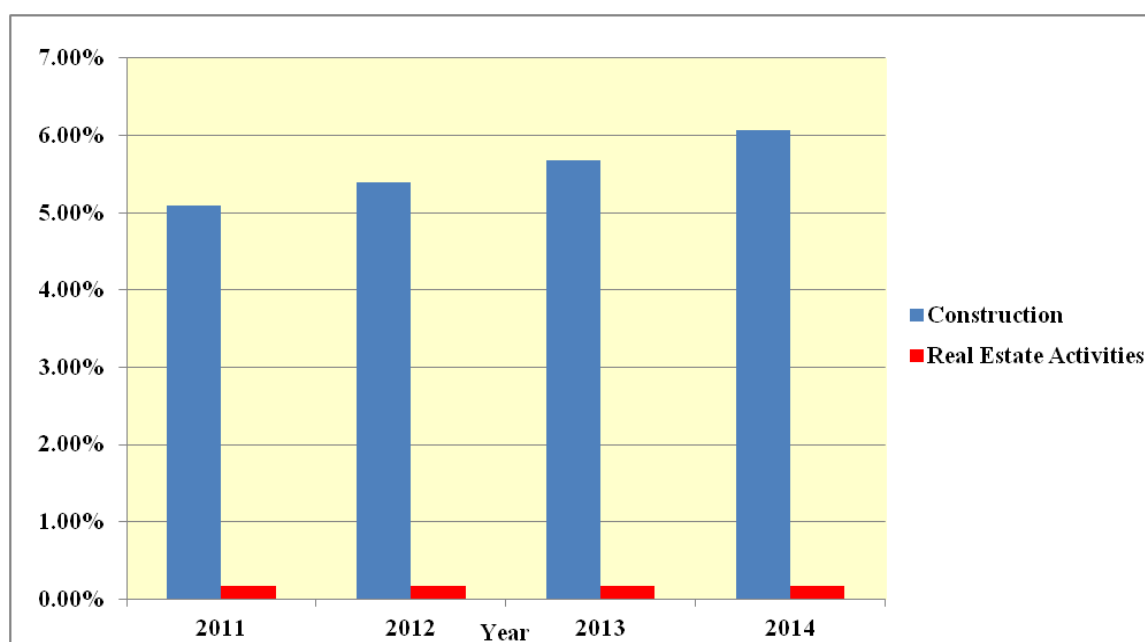
S/No.	Land Use	Existing Land Use 2015		Proposed Land Use 2035	
		Area (Km ²)	% Area	Area (Km ²)	% Area
1.	Residential	62.52	28.06	78.03	35.5
2.	Industry	12.17	5.46	27.82	12.7
3.	Educational	4.07	1.83	11.15	5.1
4.	Public Purpose	9.61	4.31	17.84	8.1
5.	Commercial	4.50	2.02	12.26	5.6
6.	Public Utility	0.67	0.30	8.91	4.1
7.	Transportation	13.84	6.21	27.87	12.7
8.	Undeveloped	98.99	44.43	0	0.0
9.	Agricultural	6.75	3.03	6.75	3.1
10.	Recreational & Beach	4.36	1.96	18.00	8.2
11.	Forest	2.82	1.27	2.82	1.3
12.	Water Pond	0.08	0.04	0.08	0.0
13.	Quarry	2.04	0.92	1.00	0.5
14.	Future Development	0.00	0.00	7.23	3.3
	Total Developable area		100	219.76	100.0
15.	Riparian	0.39		0.39	
16.	Other Non-Developable area	0		2.82	
	Total Non-Developable area			3.21	
Total	Land Mass	222.81	77	222.97	77
	Water Mass	65.12	23	64.97	23
	Area	287.94	100	287.94	100

Source: Draft ISUDP, 2015

The County planners view the cities and built environment sector as being at the very least important for GDP and employment and that hydropower and thermally generated electricity are the key sources for the sector (Table 4-29). While employment in construction activities in Mombasa County has been on the increase between 2011 and 2014, there has been no change in employment in real estate activities (Fig. 4-11).

Table 4-29: City planners' views on the cities and built environment sector contribution to the County GDP and employment.

Indicators	Remarks
Economic significance - GDP	Quite important to very important
Economic significance - Employment	Important to very important
Key sources of energy for the sector	Hydropower; Thermally generated electricity



Kenya National Bureau of Standards: Economic Survey, 2016

Figure 4-11: Growth in employment in construction and real estate activities (Cities and Built Environment Sector), Mombasa County, 2011 - 2015

II. Environmental issues and related indicators

Congestion of buildings and structures on the Island, as well as the adjacent mainly parts of the country are some of the issues that need to be dealt with, along with their associated noise, dust, and management of solid and liquid wastes. Along the coastal areas, coastal erosion has been exacerbated by human occupation and intensive development of infrastructure that is concentrated along the coastal strip (Omuombo et al., 2013a). In Diani–Bamburi area, shoreline recession of up to 10 m in 10 years is reported to have occurred over the period 1996–2006 (Ballot et al., 2006) due to modification to current flows caused by the construction of seawalls to protect the beachfront tourist hotels and resorts. They have also developed creation of beach gaming and relaxation areas that have increased the human traffic in the beach, creating loose sands on the beach that are easily washed into the sea during the high tides (Omuombo et al., 2013a), leading to the characterization of these areas as coastal erosion ‘hotspots’ (Government of Kenya, 2009, 2010).

The ecological conditions are evolving fast due to numerous developments in the County, including the recent dredging to deepen the Kilindini Channel of the port of Mombasa (Mombasa County Annual Development Plan, 2014). The tourist hotels and resorts keep their beaches clean through the clearing of seaweed. Frequent storms are expected off the East African coast due to climate change-related global sea level rise (Omuombo et al., 2013a), and it is estimated that by 2100, 17% of Mombasa city could be submerged by a 30 cm sea level rise (Awuor et al., 2008; Kabede et al., 2009). As a consequence of these changes, the low lying areas of Mombasa County, and the Kenya coast in general is at risk of related adverse impacts to biodiversity, land, properties, and the disruption of economic activities beyond the country’s borders since it has East Africa’s largest seaport (Kabede et al., 2009; Brown et al., 2011).

The Export Processing Zone, industries and mines within the County are known to contribute to pollution of the coastal waters and ecosystems as the industries discharge untreated solid and liquid effluents into the sea (Government of Kenya, 2009, 2010; Hashim et al., 2004). They also contribute to soil, vegetation, and groundwater pollution as they discharge solid and liquid effluents on land (cf. Owuor, 2011). High concentrations of Co, Cr, Cu, Ni, Pb, V and Zn have been measured in the surface samples of sediments from Makupa Creek, Mombasa City, and have been attributed to urban development in the industrial city (Williams et al., 1997; Muohi et al., 2003). Quality poles have been wiped out in most mangrove areas of Mombasa and this is associated with high human populations (Abuodha and Kairo, 2001).

Across the City, various types of renewable energy are relied upon, such as hydropower which is generated externally, and to a much lesser extent, solar and biogas, with the latter two being eco-labelled (Table 4-30). The respondents to the baseline survey ranked the energy use efficiency as ranging from not efficient to slightly efficient, and a wide range of waste water treatment and management options are employed, although the large scale ones such as Kizingo and Kiparu are not operating optimally as reported in section 4.2.1.

Table 4-30: County planners' views on issues relating to the energy, wastewater treatment, and eco-labelling indicators in the cities and built environment sector.

Indicators	Remarks
Renewable energy used	- National grid hydropower/geothermal (high); Solar, Biogas (low)
Resource efficiency - energy use	- Not efficient to slightly efficient - Old architecture - more cooling required, not very efficient.
Resource efficiency - products/materials recycled	- Plastic and paper - Iron sheets (hoarding), scaffolding, cement bags
Waste water treatment and management methods	- Kizingo & Kiparu treatment - Septic tanks, soak pits, municipal sewage. - Bio-digester/septic tank, soak pits
Ecolabelling of products from the sector	- Solar, eco-jiko, coconut briquettes - Paints

Three major environmental challenges are observed by the planners in relation to this sector, namely, land disturbance/deforestation (in the general sense of cutting down of trees or shrubs), waste management, and poor sewerage networks (Table 4-31), and these are largely a consequence of poverty, ignorance, inadequate infrastructure, and a general disregard for the environment. Hence, mitigation measures largely revolve around civic education and sensitisation.

Table 4-31: County planners' views on the environmental issues and impacts, mitigation measures in place and challenges faced in the County cities and built environment sector.

Environmental Issues	Related Societal Problems	Major Impacts on Sector	Mitigation Measures in Place	Challenges faced
Land disturbance	- Poverty	- Displacement	- Civic education	- Poverty

			- Backfilling	
<i>Deforestation</i>	- Due to ignorance	- Low water supply especially in water pans - Planting trees - Sensitisation programs - Compensation	- Encourage tree planting	- None stated
<i>Waste Management</i>	- Collapsing buildings - Industrial discharge	- Poor development - Hazards	- None stated	- Lack of capacity - Community not sensitised.
<i>Poor Sewerage</i>	- None stated	- None stated	- None stated	- Public awareness - Flooding in old town - Drainage

III. Wellbeing and equity indicators

Some major planning issues identified in the ISUDP and Zoning Plan that address wellbeing and equity issues include (i) expansion of the road network to meet the increasing human and vehicle demands (ii) the problem of absentee landlords and squatters under lease arrangements but with no security of tenure and (iii) lack of key plans and adequate support services like sewer and storm water drainage system.

In this sector the ratio of male to female employees is estimated by the baseline survey respondents to be 70% to 30% (see Annex 3), but the actual ratio at national level is 4:1 and this is likely the same at the County level (Table 4-32). The respondents indicated that the contribution of the sector to household budget is medium to high, and that the employees engage in other livelihood activities.

Table 4-32: : National wage employment in the cities and built environment sector (number in thousands)

Year	2012		2013		2014	
Employment By Industry (population count in '000s)	Male	Female	Male	Female	Male	Female
Construction	94.2	21.9	106.7	23.1	117.1	26.6
Real Estate Activities	2.0	1.7	2.8	1.0	2.9	1.0

(Source: Kenya National Bureau of Standards: Kenya Facts and Figures, 2015)

IV. Green economy policy interventions

(a) National Level

The GESIP has a provision for improving urban areas and cities. Under Thematic Area 1: Promoting Sustainable Infrastructure, urban infrastructure is fundamental in creating a healthy and thriving economic climate for transformation and expansion of economic activities and enhancing sustainability. The

Government is already implementing major projects aimed at upgrading Mombasa Town. The aim of this thematic area is to promote sustainable infrastructure which encompasses the designing, building and operating of energy and infrastructure elements so as to reduce negative impacts on social, economic and ecological processes necessary for human equity, diversity and integrity of natural resources. Infrastructure management also entails balancing natural and built infrastructure in optimal combinations for large and small-scale developments.

Objective 1.4 of GESIP seeks to promote sustainable design, construction and maintenance of buildings. It aims at ensuring 75 per cent of new and renovated public (national and county) and private large scale buildings are green by 2030; capacity building for architects, engineers and contractors and other stakeholders on integrated green technologies in design and construction; and developing and implementing certification standards for green buildings.

(b) County Level

The county GE initiatives as presented in Table 4-33 below are derived from the Counties development documents i.e. CIDP, ADP, ISUDP and Annual County Environmental Report. The challenges and opportunities as indicated are also extracted from these documents. In general, the table shows a government that is inclined to improve its urban and city infrastructure by embracing modern and tested strategies such as green urban spaces and solar powered street lighting programme.

Table 4-33: Existing Green Economy initiatives, challenges, and opportunities in the cities and built environment sector

GE Priorities	Challenges	Opportunities
- Formalization of informal settlements - County/ city master plan	- The problem of absentee landlords and squatters with no security of tenure. - Mombasa County lacks key plans and adequate support services like sewer and storm water drainage system. - Existing haphazard and unplanned development structures	- Facilitate land tenure/ownership - Provide development control framework - Initiate economic empowerment strategies - Provide an all inclusive framework for physical planning and development of the County/city

Source: CIDP, ADP, ISUDP

(c) County Baseline Survey, November 2015

The Green Economy initiatives cited by respondents to the baseline survey don't appear to be well fleshed out and coordinated with the sector specifically in mind (Table 4-34), so opportunity exists to mainstream such initiatives particularly in light of the proposed development of the City from now through to the year 2035. Other broader ranging initiatives such as the shoreline management strategy which is the basis for the formulation of the National Coastal Zone Management Plan can be used to address some of the issues relating to actual or proposed developments in this sector. In this approach, issues affecting predefined individual sediment cells are identified and prioritized as areas of concern with regard to

management (Government of Kenya, 2009, 2010; Muthiga, 2009). For example, the success in the implementation of the marine protected areas as a tool for protection and management of ecosystems in the coastal zone has influenced the shoreline management strategy that forms the basis of ICZM to use a site-specific approach (Omuombo et al., 2013b), and thereby informs to some extent on allowable developments along the coastal fringe. Green buildings and support infrastructure can be designed and can be supported by development of appropriate building codes and incentives that can be drawn into the framework of County policy and legislative formulation and review processes.

Table 4-34: Existing Green Economy initiatives, challenges, and opportunities in the cities and built environment sector

GE Initiatives	Challenges	Opportunities
- Planting trees - Green schools project - Climate change adaptation project - Coastal water security - Climate resilience project - Clean up events	- High cost of power - Funding - Conflict of interest in implementation - Incitement, politicisation of project implementation. - Late remittance of funds	- Public support and willingness to participate. - Resources available (funds).

The planners in the sector use strategic plans to guide the development planning, relying on documents such as: the Urban Areas and Cities Act of 2011, the Mombasa County Integrated Development Plan (CIDP) 2013-2017, and the Draft Mombasa Gate City Master Plan.

(d) Proposed Options, Approaches and Considerations

In addition to the key options proposed earlier, other options that can be considered include:

- Ensure that built infrastructure accounts for rising sea levels and increased occurrence of extreme weather events and flooding;
- Ensure that built infrastructure is designed to complement and work in synergy with natural infrastructure (e.g. wetlands) so that the latter's goods and services are not compromised;
- Promote energy efficient appliances and clean technologies (LPG, LED lamps, improved cook-stoves);
- Enhance enforcement of the building codes;
- Integrate green technologies in design and construction;
- Roll out pollution prevention programs across manufacturing and service industries;
- Roll out recycling and industrial symbiosis projects through private public partnerships;
- Roll out “trees for pay” program for youth;
- Promote rain water harvesting (at household and institutional) level through increased water collection and storage;
- Support and enhance existing Green Economy initiatives (see section 4.2.3d).

Factors that should be considered when integrating GE into the Cities and Build environment at the county level include:

- Prepare for the national government plan to roll out energy centers to counties;
- Provision of adequate legal/regulatory framework and funding; and
- Creation of locally appropriate incentives.

4.4.4 Transport and Infrastructure

I. General economic sector indicators

At the national level, the transport sector contributes about 16% to the GDP, and registered a growth rate of 6 per cent in 2013 (KNBS) compared to about 17% and 6.2 per cent respectively based on GDP by Activity data in the KNBS Economic Survey Report, 2016 (Table 4-35).

Table 4-35: National GDP by Transport and Infrastructure Sector Activity: 2011-2014

Percentage Contribution to GDP	2011	2012	2013	2014	2015
Transport and storage	7.1	8.0	7.9	8.6	8.4
Land transport	5.0	6.0	6.0	6.5	6.2
Air transport including support services	0.4	0.4	0.4	0.3	0.3
All other transport including postal and courier activities	1.7	1.6	1.6	1.8	1.8

(Source: KNBS Economic Survey Report, 2016)

The length of the road infrastructure is 475 km (Mombasa County Annual Development Plan, 2013), but there are no specific data on the population of cars/trucks within the County. Data was also not available at County level for road and rail freight volumes, total person kilometres travelled, and total public transport boarding per capita. In terms of investment in road infrastructure, the County is embarking on expanding the existing road network in order to make transport affordable and faster for economic efficiency. The Government of Mombasa has contracted firms to improve and construct 14.1 km of roads, and install traffic lights at six junctions at a total cost of KSh226 million. The works are at various stages of execution though lagging behind schedule (Hazel J Koitaba, County Government of Mombasa, Fiscal Strategic Plan 2015). The works to improve and construct 15.13 km of roads and one bridge at a total cost of KSh 292 million have recently commenced though they were originally scheduled to be completed by June 30 2015.

The transport and related infrastructure situation is as follows (Mombasa County Annual Development Plan, 2014):

Roads: There are a total of 257.17 km of bitumen surface roads, 127 km of gravel surface roads and 91.29 km of earth surface roads in the County (Mombasa County Annual Development Plan, 2014). While the major roads are in a fair condition, access roads within the residential and industrial areas are in deplorable state. The roads are maintained by the national government through Kenya Rural Roads Authority (KeRRA), Kenya Urban Roads Authority (KURA), the Kenya National Highways Authority (KeNHA) and Mombasa County.

Railway: The County has ten kilometres of railway line and three railway stations. The County Government of Mombasa envisions having light commuter rail within the city in the long run. Construction of the Standard Gauge Railway Line from Mombasa to Nairobi was started in 2014 and will be completed soon.

Water Transport: Mombasa's port is the largest in Kenya, with 19 deep water berths, two additional berths nearing completion and two oil terminals. Rail connects the port to the interior of the country. There is little or no scheduled passenger service. International cruise ships frequent the port. Container traffic at the sea port has also maintained an increasing trajectory except for a minor dip in 2013 (Table 4-36).

Table 4-36: Mombasa port activity indicators

Mombasa Port Activity Indicators	2011	2012	2013	2014
Freight handled by Kenya Ports	19,953	21,920	22,307	24,875
Container Traffic at the sea port TEUs	770,804	903,463	894,000	1,012,002
Ships Docking No.	1,684	1,763	1,768	1,832

(Source: KNBS, 2016)

Ferry: There is no bridge between Mombasa Island and the South Coast, instead the distance is served by ferries operated by the Kenya Ferry Service from Kilindini and Mtongwe to Likoni in the south coast of Mombasa. As a result of the major build-up of more luxurious hotels on the South Coast and a lack of a direct bridge linking the South Coast to the North Coast, visiting tourists have the option of flying directly into the South Coast airstrip using the domestic airline, Air Kenya.

Port: The port of Mombasa is also a key resource and the gateway to the East and Central African region, as it serves the entire region's export and import needs. In 2012, dredging was undertaken with the idea of deepening the Likoni channel to facilitate usage of the port by larger vessels. The ongoing construction of a second container terminal at Kipevu will increase the capacity of the containers being handled to three times the current capacity. Over recent years the amount of freight handled by Kenya Ports Authority, and the numbers of ships docking, has been rising steadily (Table 4.34).

Airports: The County has one international airport, the Moi International Airport within Changamwe sub-County. The airport is the second largest airport in Kenya and is used by both domestic and international flights. The airport is essential in the promotion of tourism and investment opportunities in the County and in the Coast region.

CO₂ emissions from transport (million metric tons) in Kenya were reported in 2011 to be 4.71. According to the World Bank, CO₂ emissions from transport include emissions from the combustion of fuel for all transport activity, regardless of the sector, except for international marine bunkers and international aviation. Data is not available at County level for emissions, consumption, NO₂ contributions, as well as costs of damages related to climate variability.

The baseline survey respondents ranked the economic significance of the sector with respect to GDP as quite important to very important, and with respect to employment as slightly important to quite important (Table 4-37).

Table 4-37: City planners' views on the transport and infrastructure sector contribution to the County GDP and employment.

Indicators	Remarks
Economic significance - GDP	Quite Important to very important
Economic significance - Employment	Slightly important to Quite Important
Key sources of energy for the sector	Fossil fuels, electricity

II. Environmental issues and related indicators

Noise, dust, poor garbage disposal, exhaust fumes, and fuel as well as other transported goods spills that lead to environmental pollution, degradation and erosion, are some of the environmental issues related to the transport industry. However, there is no data at County level as per the indicators specified in the data sheet for this sector. Flash flooding as a consequence of storm drainage systems being blocked by solid waste is a problem made worse by the fact that the system itself is largely dilapidated. Oil spill events have generated devastating effects on the ecosystems; in 1988, for example, about 10 ha of mangrove forests were lost, while in 2005, an oil spill from Kipevu oil terminal affected 234 ha of mangroves at Port Reitz (Abuodha and Kairo, 2001; Kairo et al., 2005). The full impacts of oil pollution are still unknown, although it is common to sight discharges from ships and tankers that are washed offshore (Government of Kenya, 2009).

Renewable energy is used for street lighting (hydro and solar power) and is deemed to be efficiently used (Table 4.36). A number of by-products generated by the sector activities tend to be recycled, but eco-labelling of products from the sector is not done (Table 4-38).

Table 4-38: County planners' views on issues relating to the energy, wastewater treatment, and ecolabelling indicators in the transport and infrastructure sector.

Indicators	Remarks
Renewable energy used	- Hydropower & Solar power for street lighting.
Resource efficiency - energy use	- Efficient
Resource efficiency - products/materials recycled	- Tarmac on destroyed roads is used to make pedestrian pathways - Used oil - lubrication - Milled Asphalt layers are blended with gravel to give a superior base quality material

	- Topsoil that is stripped is used again on the side slopes for grassing - Rock - Used paper - for making tissue paper
Waste water treatment and management methods	- Channelled to the drains - Waste water is treated through a plant where contaminants are removed - The semi-solid waste (sewage sludge) also undergoes further treatments before being disposed/applied to other uses like land application.
Ecolabelling of products from the sector	- None stated

The environmental issues related to the transport and infrastructure sector are noise, dust, and air pollution from motorised traffic, poor garbage disposal that blocks drains, soil erosion and environmental degradation (Table 4-39).

Table 4-39: County planners' views on the environmental issues and impacts, mitigation measures in place and challenges faced in the County transport and infrastructure sector.

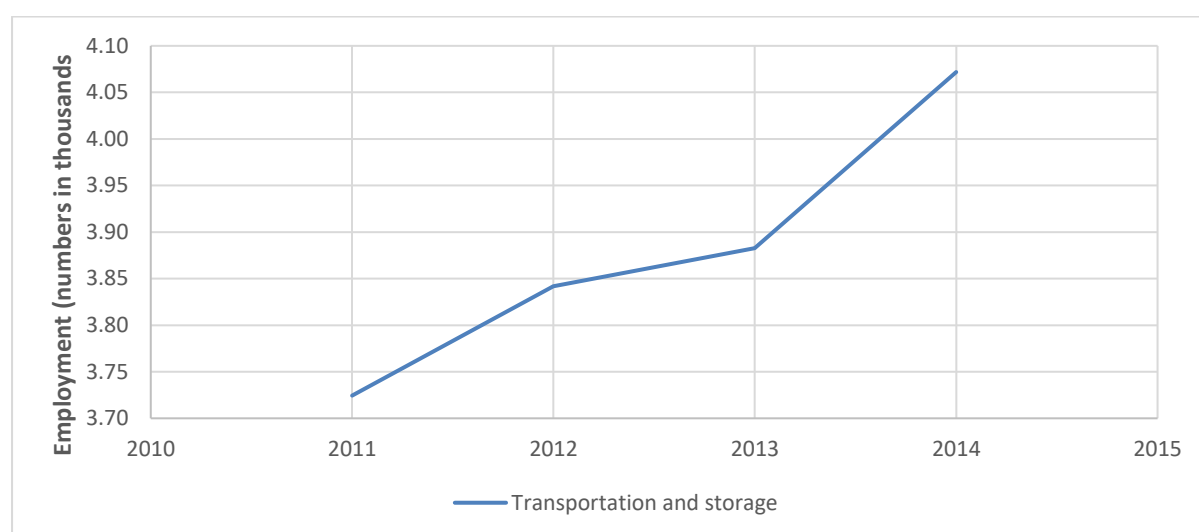
Environmental Issues	Related Societal Problems	Major Impacts on Sector	Mitigation Measures in Place	Challenges faced
Noise	- Community Complaints	- None stated	- None stated	- None stated
Dust / Air Pollution (exhaust fumes)	- Community Complaints	- Air Pollution	- Environmental measurement done for air, soil and water every year; - Eco-friendly fuel	- Alternative sources of fuels to run motorised traffic not researched/not in use.
Poor garbage disposal	- None stated	- None stated	- None stated	- None stated
Soil erosion emanating from change in land use;	- May lead to displacement of people - May lead to collapse of embankments - Sink holes	- Washout/undermining embankments	- Protection works to be done e.g. gabions	- Land use keeps changing and therefore erosion is almost inevitable.
Spills	- Water contamination	- None stated	- Immediate action in case of spillage - Monitoring systems in place - Treat the contaminated ground.	- Capital intensive - Delays due to limited capacity/service providers.

III. Wellbeing and equity indicators

The number of people employed in the sector at County level is not specified. The Dongo-Kundu by-pass is expected to ease congestion at the Central Business District, as traffic from Nairobi to the South coast shall be diverted at Miritini towards Likoni and Diani. The County Government of Mombasa is also

planning to have a second bridge fixed to the north mainland to ease traffic congestion to the north. The steady rise in numbers of people employed in transportation and storage indicates that this is a fast growing sector (Figure 4-12), and about one quarter of the work force is female (Table 4-40).

The County Government has procured road construction equipment for in-house road construction and maintenance. These include two motor graders, two wheel loaders, two drum rollers, one backhole excavator and five tippers this financial year. These are intended to increase coverage in opening of access roads while reducing the cost of works compared to contracting. This is aimed at a continued focus on road infrastructure improvement through maintaining the already existing road networks while also opening up new access roads so as to restore confidence of residents and making the cost of doing business favorable. Transport infrastructure development will also help revamp the tourism sector and foster social integration in the County.



(Source: Table 4.7a, data - KNBS Kenya Facts and Figures, 2015)

Figure 4-12: Mombasa wage employment (numbers in thousands), transportation and storage activities

Table 4-40: National wage employment in transportation and storage (number in thousands) by gender: 2012 -2014.

Year	2012		2013		2014	
Employment By Industry (population count in '000s)	Male	Female	Male	Female	Male	Female
Transportation and storage	62.2	13.0	55.7	20.4	58.3	21.4

(Source: KNBS Kenya Facts and Figures, 2015)

IV. Green economy policy interventions

(a) National Level

The GESIP has a provision for improving urban areas and cities. Under Thematic Area 1: Promoting Sustainable Infrastructure, urban infrastructure is fundamental in creating a healthy and thriving economic

climate for transformation and expansion of economic activities and enhancing sustainability. The Government is already implementing major projects aimed at upgrading Mombasa City. The aim of this thematic area is to promote sustainable infrastructure which encompasses the designing, building and operating of energy and infrastructure elements so as to reduce negative impacts on social, economic and ecological processes necessary for human equity, diversity and integrity of natural resources. Infrastructure management also entails balancing natural and built infrastructure in optimal combinations for large and small-scale developments.

Objective 1.1 of GESIP aims at enhancing sustainable mobility by establishing mass rapid transit in major urban areas especially Nairobi, Mombasa and Kisumu cities; Integrating non-motorized transport in the design and construction of roads in all county headquarters; reducing vehicular emissions through fiscal reform; and incorporating climate proofing into road and rail design, construction and maintenance.

(b) County Level

At the county level, GE based activities are guided by such documents as County Integrated Development Plan, Annual Development Plans and Integrated Sustainable Urban Development Project (ISUDP) and other local strategies. These are shown in Table 4-41.

Table 4-41: Existing Green Economy initiatives, challenges, and opportunities in the cities and built environment sector according to county documents

GE Initiatives	Challenges	Opportunities
- Pedestrian Facilities - Non-Motorized Transport Facilities (constitutes 2% to 6% of the total traffic in a day and 15% to 20% in some roads in north mainland) - Development Integrated with High Speed Corridor - Construction of Bypasses e.g. Dongo Kundu Bypass - Parking Management	- High establishment costs including land acquisition - Lack of sufficient policies - Limited space especially in town centres - Disjointed development between port and the city expansion plans - Congestion brought about by cargo and traffic	- Traffic lights installed powered by renewable energy e.g. solar - Greening of transport corridors - Improved maritime infrastructure - Alternative means of transport e.g. electric cars, tram/rail systems, bus lanes

(Source: CIDPs, ADPs, ISUDP and Mombasa Gate Master Plan)

Existence of traffic on major roads within Mombasa exceeds its acceptable limit; this is both under existing and projected traffic conditions. Moreover, transport system is becoming more increasingly oriented toward the use of private cars. This significantly increases the number of vehicles on the road, which increases traffic congestion, GHGs emission, air pollution, storm water pollution, noise pollution and road fatalities amongst other things. Despite this, there is no specific data on the population of cars/trucks within the county.

(c) County Baseline Survey, November 2015

The Green Economy initiatives in this sector revolve around tree and grass planting along roads (Table 4-42), so there is opportunity to expand the scope of such activities to entrain transportation and construction as well.

Table 4-42: Existing Green Economy initiatives, challenges, and opportunities in the transport and infrastructure sector according to the County planners.

GE Initiatives	Challenges	Opportunities
- Collaboration with KURA, KERRA and KENHA to plant trees along roads - Tree planting in project sites - Grass planting (scour checks) - on slopes to check soil erosion.	- Some trees can't grow well in town centre - Limited space especially in town centres - Costly - Tree planting after site clearance has to wait until the pavement is constructed. This may bring about disquiet among communities, conservation (environmental) lobby groups etc. - Lack of capacity - Lack of sufficient policies - Lack of awareness/resistance	- Community willing - Partner with other institutions.
- Target of zero waste on hydrocarbons	- monitoring system to ensure no leakage - Spillages are promptly attended to.	- Policy guidelines on use of alternative sources of energy.

Similar to the other sectors, strategic planning, stakeholder engagement, and learning from past experiences anchor the development planning, and the sector is funded to carry out a number of activities during the fiscal year 2015/2016.

(d) Proposed Options, Approaches and Considerations

In addition to the key options in the sections (a), (b), and (c) above, other options may include:

- Community car share: System that involves an organized group of participants (usually required to be members) with one or more self-accessing shared vehicles, the usage of which is booked in advance for short periods of time;
- Intermodality: Intermodality means combining several means of transport during the same journey, using different types of vehicles to get from one place to another; for example, train + car or bus + bicycle + walking; and
- Mass Rapid Transit Systems: The high speed corridor can initially have Bus Rapid Transit (BRT) with transit stops distributed at regular intervals or alternatively, in the future with enough population catchment and necessary funding, Light Rail Transit (LRT) system may be explored.
- Promote non-motorized transport and use of vehicles powered by non-fossil fuel energy;
- Incentives to facilitate the transition of the County's passenger and freight vehicle fleet to modern vehicles;

- Ensure that transportation infrastructure (including port facilities, roads, railways and bridges) are climate proofed e.g. accounts for rising sea levels and increased occurrence of extreme weather events and flooding;
- Construct, upgrade and/or rehabilitate sewerage and drainage infrastructure;
- Roll out pollution prevention and “trees for pay” program for youth;
- Adopt clean technologies;
- Support and enhance existing Green Economy initiatives (see section 4.2.4d).

Factors that should be considered when integrating the green economy options county transport sector develop plans include:

- Land: current land ownership and high cost of land acquisition has made it difficult for physically planning of the County to be undertaken;
- Informal settlement (Bangladesh, Magongo, Likoni, Kisauni and Baburi) and rapid urbanisation should be addressed;
- ACreate conditions for lower private car usage;
- Reduce transport-based CO₂ emissions; and
- Minimise health repercussions (e.g. reduce premature deaths from traffic-related pollution)

4.4.5 Fisheries

Mombasa County boasts of fisheries as an important resource that consists of artisanal and pelagic fisheries. Traditionally, inhabitants of the County have relied on fisheries and mangrove exploitation. Unpublished statistics by Fisheries Department, estimate that over 2000 fishermen are directly engaged in artisanal fishing with most fishing activity undertaken along the reef (NEMA, 2013). The fishermen’s main fishing gears include hook and line, gillnets, shark nets, beach seine and basket traps (*malema*). They also use simple fishing vessels (mainly dugout canoes) and gears. Types of fisheries include marine and coastal, dam and reverie, aquaculture (tilapia and mud fish). Fish farming practices in Mombasa County include pond culture, raceways culture, and tank culture.

I. General economic sector indicators

The marine fisheries resources are both inshore and offshore: the inshore fish species are overexploited by local fishers, while the offshore resources outside the territorial waters are exploited mainly by Distant Waters Fishing Nations (DWFN) (FAO, 2007). The marine fisheries contribute on average only 4 per cent of the total national fish landings (FAO, 2007). There are no County level data for the suite of general economic indicators as specified in the data sheets (Annex 4). However, extraction of fish and other resources from the narrow continental shelf, coral reef and mangrove ecosystems increases each year with inadequate monitoring and management structures to protect the resource bases. Mombasa has 19% of the fish landing sites, with the other towns along the coast having the following: Kwale (29%), Kilifi

(18%), Malindi (19%), Lamu (12%), and Tana Delta, (3%) (Ministry of Fisheries Development, 2012). Fishing statistics reflect a steady trend in the rise of both quantity and earnings over the period 2010 – 2014 (Tables 4-43 and 4-44).

Table 4-43: Quantity of Fish Landed, 2010 - 2014

Mombasa Fishing Statistics	Quantity (Metric Ton)				
Year	2010	2011	2012	2013	2014
Fresh Water Fish	-	-	-	-	-
Marine Fish	715	700	893	893	990
Crustaceans	132	145	230	223	165
Other Marine Fish	66	63	62	62	59
Mombasa Totals	913	908	1,185	1,178	1,214
National Totals	140,751	149,046	154,015	163,389	167,859

(Derived from KNBS, 2015).

Table 4-44: Value of Fish Landed, 2010 - 2014

Mombasa Fishing Statistics	Value to Fishermen (Kshs '000)				
Year	2010	2011	2012	2013	2014
Fresh Water Fish	-	-	-	-	-
Marine Fish	80,476	102,355	155,090	147,539	183,604
Crustaceans	20,100	35,871	42,703	53,115	49,772
Other Marine Fish	7,452	5,835	7,097	6,726	8,711
Mombasa Totals	108,028	144,061	204,890	207,380	242,087
National Totals	13,002,253	16,677,221	18,073,933	21,282,503	21,856,661

(Derived from KNBS, 2015).

The respondents ranked the economic significance of fisheries in terms of GDP as quite important to very important, and in terms of employment as being slightly to quite important (Table 4-45). Various energy types are used in the sector as indicated below for transportation, freezing and processing of fisheries products.

Table 4-45; City planners' views on the fisheries sector contribution to the County GDP and employment.

Indicators	Remarks
Economic significance - GDP	Quite important to very important
Economic significance - Employment	Slightly important to quite important
Key sources of energy for the sector	Electricity, fossil fuel; for informal sector fish processing - charcoal, wood fuel

II. Environmental issues and related indicators

Serious threats to fisheries in the area come from the ever increasing human pressure through tourism, industrial pollution, overfishing, destructive fishing, mangrove logging and other unsustainable uses (cf. Kairu et al., 2001). Experience over the years has proved the value of gazettement, such as that of the Mombasa Marine Reserve, as a primary conservation tool even if resources often lack to follow through on monitoring and active management (Githitho 2004). Fisheries production systems do impact the environment directly and indirectly (Table 4-46). Anthropogenic activities, particularly crop production alongside the discharge of raw domestic and industrial effluents coupled up with agrochemical wastes have negatively impacted upon the floral and faunal biodiversity within the Tudor Creek ecosystem, particularly impacting the fishing industry that supports a large population of the ecosystem's human population (Government of Kenya, 2009). In 2006, Beach Management Units (BMUs) were implemented through the Ministry of Fisheries in a co-management approach and to date they are still doing so. The BMUs empower the local communities as decision-makers in the management of the resources and thus work with the government to enforce legislation on fishing and conservation (McClanahan, 1997; Cinner et al., 2012).

Table 4-46: Fisheries production systems and their impacts to the environment (Source, Draft ISUDP)

Production system	Impacts	Mitigation measures
Marine and Coastal fisheries	- Introduction of biodegradable substances e.g. oils & detergents	- Enactment and enforcement of existing laws - Limiting of access to the fishery - Awareness creation to resource users
Aquaculture	- Introduction of effluents from agriculture inputs into the water system - Soil erosion and silting	- Regulation of pond construction - Regulation of types of inputs - Soil erosion control practices

In the informal sector, solar energy is used in fish processing, and is ranked as being not efficient to efficient, and products associated with the processing are waste paper and manure (Table 4-47). Environmental issues relate to mangrove degradation impacts on fish breeding, and the impacts of oil spillages and discharge of effluents into the ocean that heighten fish mortality (Table 4-48).

Table 4-47: County planners' views on issues relating to the energy, wastewater treatment, and ecolabelling indicators in the fisheries sector.

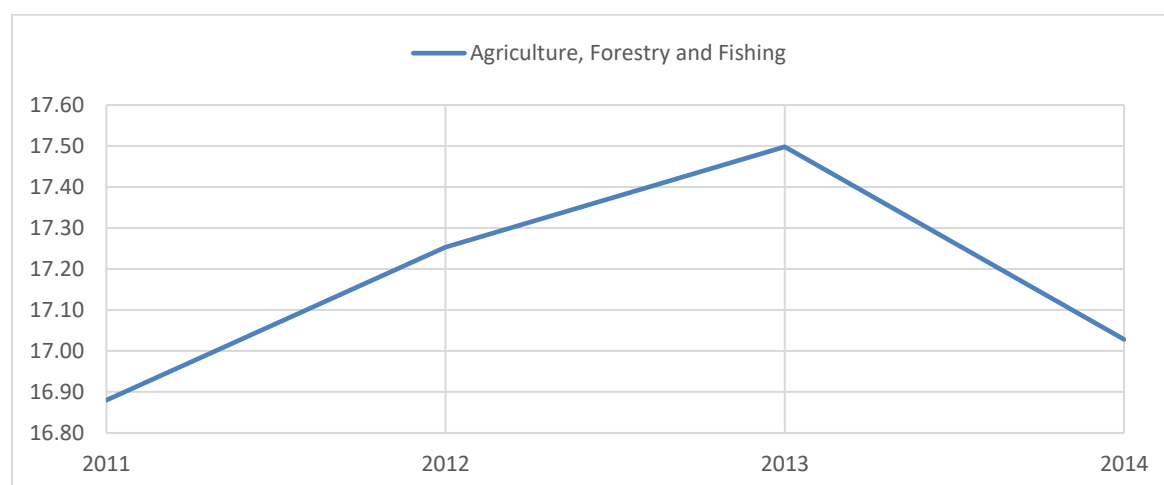
Indicators	Remarks
Renewable energy used	- National grid power; Solar used in fish processing (informal sector)
Resource efficiency - energy use	- Not efficient to efficient
Resource efficiency - products/materials recycled	- Waste paper; Farm produce; Manure/fertiliser/compost
Waste water treatment and management methods	- County sewage infrastructure
Ecolabelling of products from the sector	- N/A

Table 4-48: County planners' views on the environmental issues and impacts, mitigation measures in place and challenges faced in the County fisheries sector.

Environmental Issues	Related Societal Problems	Major Impacts on Sector	Mitigation Measures in Place	Challenges faced
<i>Mangrove degradation under harsh climatic conditions</i>	- Poverty	- Mangroves act as a breeding place for fish	- Sensitise the communities on the importance of mangroves	- None stated
<i>Oil Spillages</i>	- Heavy pollution impact	- Death of marine life e.g. fish	- None stated	- None stated
<i>Waste Management</i>	- Illegal dumping	- Death of marine life e.g. fish - Bureaucracy	- More deep sea fishing boats - Compensation of fishermen	- None stated

III. Wellbeing and equity indicators

The number of people employed in the sector is not specified at the County level, but the number of fishers in Mombasa is about 1500 (Ministry of Fisheries Development, 2012). From the estimates made in Table 4.7, and considering that agriculture and forestry contributions to the workforce are quite low, it is evident that the estimates of the number of fishers in Mombasa are somewhat accurate (Figure 4-13).



(Source: KNBS Facts and Figures 2015)

Figure 4-13: Mombasa wage employment (numbers in thousands) in agriculture, forestry and fishing, 2011 - 2014

The respondents to the baseline survey indicated that the ratio of male to female employees in the sector is 9:1, and that for some, work income is the sole livelihood for the employee household. They also ranked the contribution of the sector to household budgets as medium to very high. The male to female ratio is 2:1 (Table 4-49), but direct fishing at sea is strongly dominated by men, and women support in receiving and processing the fish at the landing sites.

Table 4-49: National wage employment in agriculture, fishing and forestry (number in thousands)

Year	2012		2013		2014	
Employment By Industry (population count in '000s)	Male	Female	Male	Female	Male	Female
Agriculture, Forestry and Fishing	209.9	127.8	217.8	124.7	220.6	112.7

(Source: Table 4.7, data - KNBS Facts and Figures 2015)

NB: In Mombasa, agriculture and forestry is very low compared to fishing.

IV. Green economy policy interventions

(a) National Level

The Ministry of Fisheries Development in recognition of depleted wild fish stocks in both fresh and marine water, and to meet the objective of promoting aquaculture, begun the Economic Stimulus Programme (ESP) in order to highlight fish farming as a viable economic activity in the country by raising the income of farmers and other stakeholders in the fishing industry (Ogada and Nyingi, 2013).

It was noted that up to 2013, the nation was still awaiting the decision of parliament in regard to the fisheries policy (Ogada and Nyingi, 2013). Today, the Agriculture, Fisheries and Food Authority (Amendment) Bill is still pending and in August 2015 the President called for its speedy passage. Ogada and Nyingi (2013) note that when the Fisheries 2008 Policy is enacted into law, the government will make institutional arrangement to improve coordination of research and management of fisheries including the following: create the Kenya Oceans and Fisheries Council chaired by the minister to advise and review the oceans and fisheries sector; create the Kenya Oceans and Fisheries Service (KOFS) to harmonize fisheries extension, development, management and marketing; and restructure the Kenya Marine and Fisheries Research Institute to offer effective and timely research to guide KOFS. As can be seen from Table 4.44, there isn't much in terms of GE for the fisheries sector.

According to the Agriculture Sector Development Strategy (2010-2020), the aquaculture subsector in Kenya has the potential of significantly contributing to the national economy by creating employment, earning foreign exchange, reducing poverty and supporting food security. Under the GESIP, Thematic Area 3 on Sustainable Natural Resource Management, the focus is on the economy-environment nexus to optimize the contribution of Kenya's natural resources to the economy, industrialization and livelihoods. It encompasses agriculture, forestry, water, fisheries, wildlife, land use, and extractive industries. Through this thematic area, GESIP seeks to address the drivers of change in both the biological and physical aspects of natural resources emphasizing the need for decoupling development from sustainable natural resources management and conservation of Kenya's natural capital. Objective 3.2 aims at promoting sustainable natural resources restoration by investing in conservation of marine and aquatic resources.

(b) County Level

The County level GE initiatives are guided by a numbers of documents such as CIDP, ADPs and ISUDP. The following table is populated with data on GE priority areas and strategies for mainstreaming GE into Fisheries sector. This is shown in table 4-50.

Table 4-50: Existing Green Economy initiatives, challenges, and opportunities according to County documents

GE Priorities	Challenges	GE Opportunities
- Commercial fishing vessels for landing, transshipment and access to services - Ease access of landing sites - Reduce the level of post-harvest losses; boost the level of income to fisher folks - Fish and fishery products handling - Promotion of aquaculture	- Inadequate budgetary allocation - Reduced effectiveness of extension services - Low absorption of modern technology - High cost and increased adulteration of key inputs - Inadequate legal, regulatory and disaster response framework - Multiple taxes - Weak surveillance on offshore fishing. - Inadequate infrastructure for e.g. storage and processing facilities. - Inadequate markets and marketing infrastructure. - Water-borne and zoonotic diseases.	- Acquisition of gazetted and ungazetted fish landing sites - Construction of infrastructure and provision of electricity, potable water and sanitary facilities - Routine visits and inspection of fish landing beaches, markets, factories and vessels - Laboratory testing of product and water samples - Ensure compliance with the fisheries law and regulations by the fisherfolk

(Sources: CIDPs, ADPs, ISUDP and Mombasa Gate Master Plan)

(c) County Baseline Survey, November 2015

There were few baseline survey respondents who were able to comfortably address the broad fisheries issues in the questionnaire. Table 4-51 shows data obtained from respondents interviewed during the county baseline survey on the GE initiatives, challenges and opportunities.

Table 4-51: Existing Green Economy initiatives, challenges, and opportunities in the fisheries sector according to the County planners

GE Initiatives	Challenges	Opportunities
- Energy saving jikos for fish processing	- Fireless cookers were viewed as an expensive undertaking. - Metrics for developing those targets and maintaining	- Alternative jikos

- Solar energy for fish processing	- High cost of renewable energy	- None stated
- Protection of mangroves	- None stated	- None stated

(d) Proposed Options, Approaches and Considerations

In addition to the options proposed in the previous sections above, other options that may be considered to address the challenges and constraints, in the fisheries subsector include

- Develop marine capture fisheries in partnership with the national government;
- Develop inland capture fisheries, targeting community fisherfolk;
- Develop aquaculture in large scale to target the growing market; and
- Promote fish safety, quality assurance, value addition and marketing.

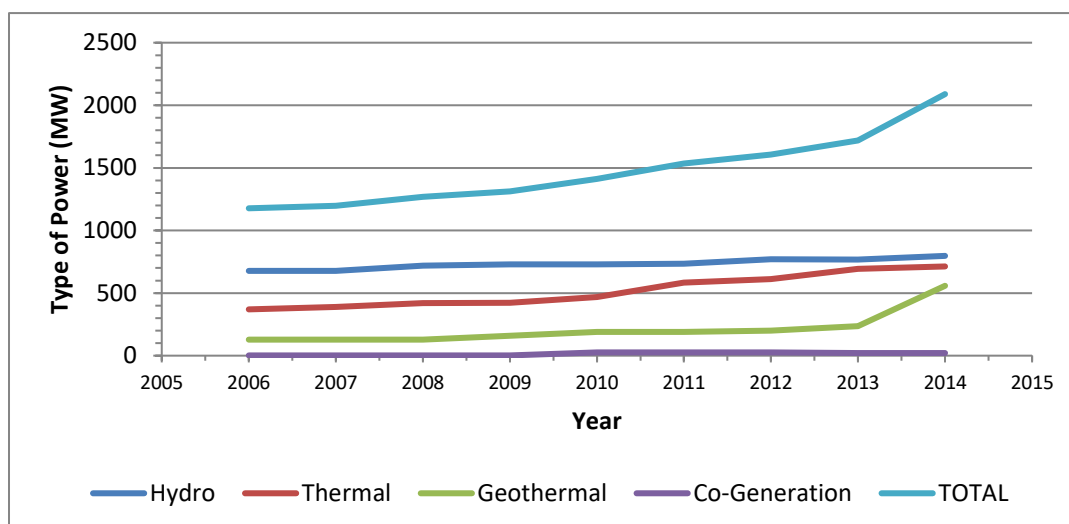
The performance of the fisheries sector depends on several factors that are closely linked, but are external to the sector. These include:

- Macroeconomic stability: **Ensure that macroeconomic stability is achieved and maintained through continued implementation of prudent fiscal, monetary systems;**
- Taxation system: **Review all taxation laws and regulations to rationalize taxes, cesses, fees and levies charged on fishery products;**
- Education, training, and technologies: **Education on sustainable management of the fisheries resource, as well as adoption of eco-friendly state-of-the-art technologies will enhance the value of the fisheries products;**

4.4.6 Energy

I. General economic sector indicators

The national installed capacity has been rising steadily, with the geothermal sector injecting most of the new energy from 2013 to 2014 (Fig. 4-14). Indeed, while hydroelectric power remains an important power source, geothermal energy recently overtook it by end of June 2015 as the main source of renewable energy supplies in the country (Kenya Investment Network, 2016). Fossil thermal energy accounted for 19.2% of Kenya's total production in June 2015; wind power accounts for just 0.4% while solar generation still remains limited to small-scale, off-grid projects (Kenya Investment Network, 2016).

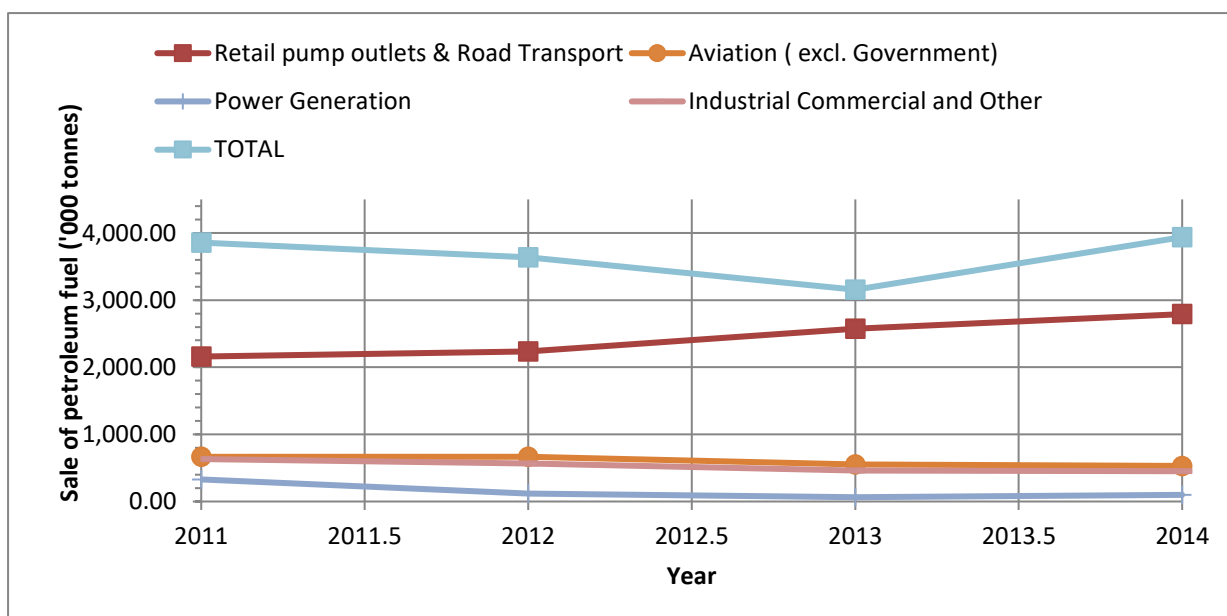


Source: KNBS Statistical Abstract 2015, from Kenya Power and Lighting Co. Ltd.

Figure 4-14: Nationally installed capacity by type of power (MW)

The KNBS (2016) report notes that Kenya has made remarkable steps towards achieving a green economy, with renewable energy development and restoration of dilapidated forestry land being some of the initiatives the country undertook towards transformation into a green economy. This was supported, in 2015, by Government enhancement of Public-Private Partnerships (PPPs) in order to reinforce the Intended Nationally Determined Contributions (INDC's) towards reducing the country's' greenhouse gas emissions.

At the national level, the total installed electricity capacity increased by 6.3% to 2,333.6 MW, and total electricity generation expanded by 4.1% to 9,514.6 GWh in 2015, while demand for electricity increased from 7,415.4 GWh to 7,826.4 GWh during the same period (KNBS, 2016). The volume of petroleum product imports increased marginally from 4,409.0 thousand tonnes in 2014 to 4,431.7 thousand tonnes in 2015. Total sales of petroleum fuel remained just below 4,000,000 tonnes between 2011 and 2014, having been most affected by the drop in power generation and aviation fuel demand in 2013, but sale for retail pump outlets and transportation steadily increased over this period (Fig. 4-15).

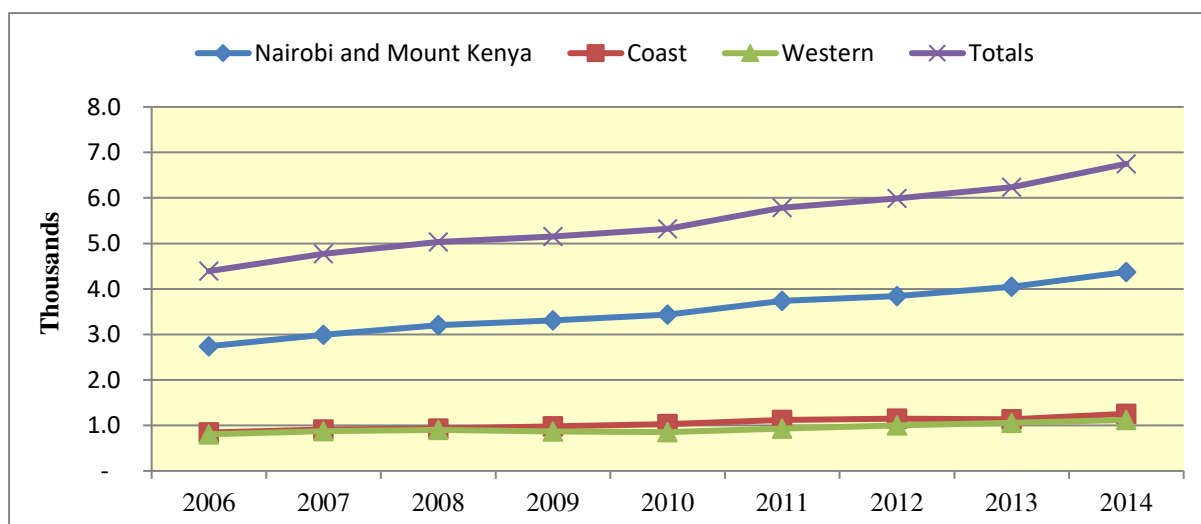


Source: KNBS, 2015

Figure 4-15: Domestic sale of petroleum fuels ('000 tonnes) by consumer category

At the County level, there is a total dependence on power from the national grid. The County is well endowed with a good electricity supply network. However, the urban areas have better coverage than the peri-urban areas with some areas, such as Mwakirunge, Maunguja II, Dongo Kundu and Shika Adabu, lack connections (Mombasa County Annual Development Plan, 2015). There is also a disparity among the sub-counties in the level of access to electricity with the gap between the ward with the highest and least access to electricity being about 95 percentage points (Ngugi, 2013).

The County has some thermal generating power stations, Kipevu I, Kipevu II and Kipevu III, installed feeding power to the national grid. Kipevu I generate 73.5 MW, Kipevu II 74.5 and Kipevu III, the largest thermal power generating station in East Africa, delivering 120MW to the national grid. Additionally, there are plans to construct an 800MW LNG power plant in Dongo Kundu area (Mombasa County Annual Development Plan, 2013). However these plans have faced several challenges and construction has not started yet. The County has a high potential for generation of solar and wind energy, which remains unexploited currently. Other sources of energy are wood, charcoal, kerosene, LPG and biogas. Below is an indication of the contribution of the Coast (Mombasa area) sales of electricity against the backdrop of two other major consumer zones in Kenya (Fig. 4-16). The respondents of the baseline survey noted that energy sector is very important with respect to the county GDP and employment (Table 4-52).



(Derived from KNBS, Statistical Abstract, 2015)

Figure 4-16: Electricity Sales By Area in Millions of KWh (June 2006 -2014)

Table 4-52 City planners' views on the energy sector contribution to the County GDP and employment.

Indicators	Remarks
Economic significance - GDP	Very Important
Economic significance - Employment	Very Important
Key sources of energy for the sector	Fossil fuel, hydropower

II. Environmental issues and related indicators

Ngugi (2013) notes the following (Fig. 4-17): (i) Only nine per cent of residents in Mombasa County use liquefied petroleum gas (LPG), and 39 per cent use paraffin; (ii) Six per cent use firewood and 41 per cent use charcoal; (iii) the most common cooking fuels among male headed households are paraffin and charcoal at 40 per cent each while the most common cooking fuels for female headed households is charcoal 45 per cent, (iv) Mvita constituency has the highest level of LPG use in Mombasa County at 23%, almost eight times Likoni constituency, which has the lowest share at 3%, (v) Likoni constituency has the highest level of charcoal use in Mombasa County at 46%, while Nyali constituency has the lowest at 37%, (vi) Changanwe constituency has the highest level of use of paraffin at 49%, and Mvita constituency the lowest at 20%, (vii) Mji wa Kale/Makadara ward has the highest level of electricity use at 97%, which is 95 percentage points above Mwakirunge ward which has the lowest level of electricity use. This sector is ranked by the baseline survey respondents as very important for GDP and employment (Table 4-53).

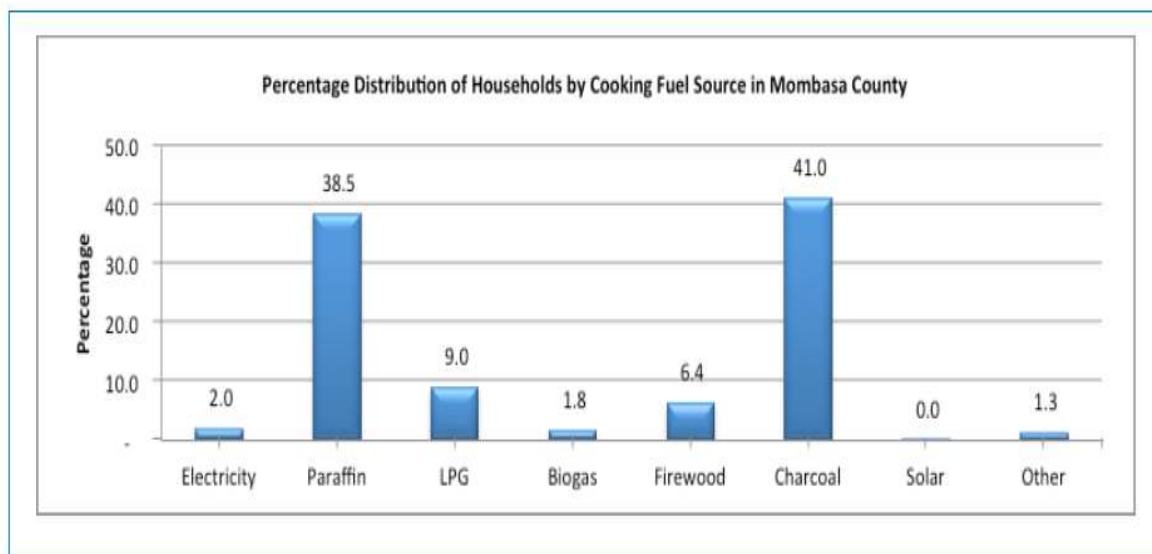


Figure 4-17: Mombasa County percentage distribution of households by cooking fuel source

Mombasa County has an electricity supply gap with current demand being around 350MW and is projected to be 1470MW by the year 2030 (Fig. 4-18).

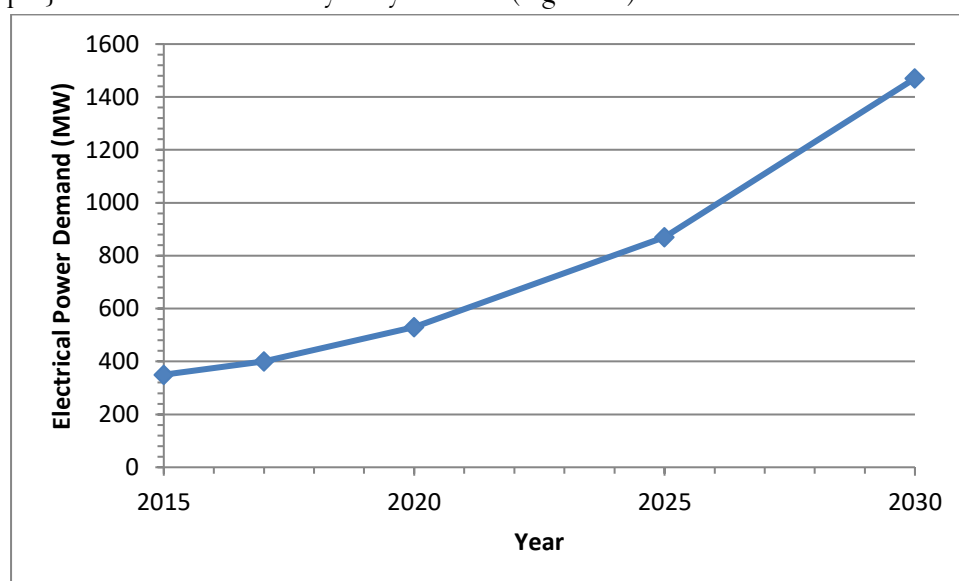


Figure 4-18: Current and Projected Power Demand (source - ISUDP).

Geothermal and hydropower are sourced from the national grid and the County does not have any production sources. Locally derived renewable energy is wind and solar, and these have low environmental impacts, and, along with geothermal and hydropower, are viewed as being quite efficiently utilised. It was not, however, clear to what extent the water used in the energy sector is treated and managed.

The use of fossil fuel in the transportation sector on land and at sea poses the largest environmental threat in the area. On land, fuel leakages from fuel storage depots such as at Port Reitz have contaminated shallow groundwater (Owuor, 2011) and coupled with leaks from vehicles, garages, etc., have contaminated soils as well. In the offshore harbour area, oil spills have occurred with devastating effect on marine and coastal zone flora and fauna (see Section 4.2.4). Consequently, it is no surprise that respondents to the baseline survey rank poor disaster management and exposure to exhaust fumes as being the two most significant environmental issues associated with the energy sector (Table 4-54).

Table 4-53: County planners' views on issues relating to the energy, wastewater treatment, and ecolabelling indicators in the energy sector.

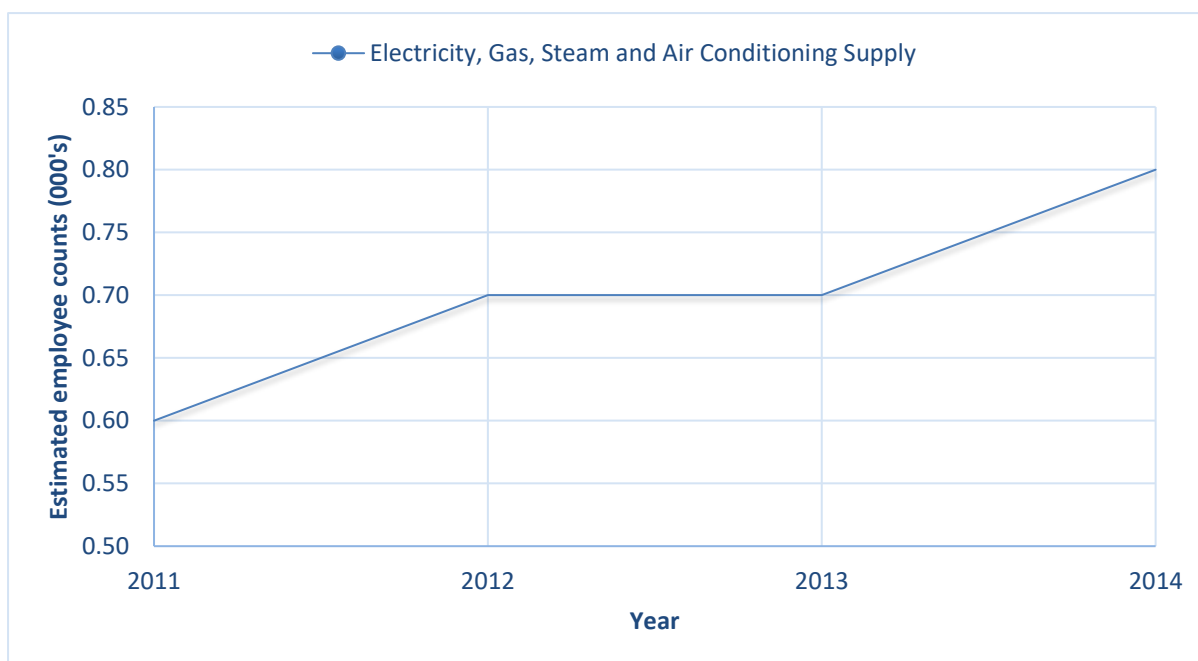
Indicators	Remarks
Renewable energy used	- Geothermal, hydropower, wind, solar
Resource efficiency - energy use	- Quite efficient
Resource efficiency - products/materials recycled	- Oil to heat energy (sludge)
Waste water treatment and management methods	- Sold to vendors, segregated and waste disposed.
Ecolabelling of products from the sector	- Energy

Table 4-54: County planners' views on the environmental issues and impacts, mitigation measures in place and challenges faced in the County energy sector

Environmental Issues	Related Societal Problems	Major Impacts on Sector	Mitigation Measures in Place	Challenges faced
Poor disaster management system	- None stated	- None stated	- Maintain the system	- None stated
Pollution and related hazards	- Smoke emissions	- None stated	- Compliance to World Bank standards	- None stated

III. Wellbeing and equity indicators

Mombasa County is well endowed with an electricity supply network: the urban area has more coverage than the peri-urban while areas such as Mwakirunge, Maunguja II, DongoKundu and ShikaAdabu lack connections (Mombasa County Annual Development Plan, 2013). It has been noted, however, that the gaps within counties in access to electricity for lighting are also enormous: the most severe disparities in the gap between the ward with the most access to electricity and the least access, more than 40 percentage points, are in Mombasa (95 percentage point gap between highest and lowest ward), Garissa (92 percentage points), and Nakuru (89 percentage points) (Ngugi, 2013). The Tariff arrangement in Mombasa County is similar to everywhere in Kenya. While individual customers pay their bills directly to Kenya Power, the County Government pays for street lighting. The Tariff arrangement has an array of options such as mobile phone, Postal Corporation of Kenya, ATMs among others for making payments.



(Source: Table 4.7, data - KNBS 2016)

Figure 4-19: Estimated Mombasa County wage employment (numbers in thousands)

The number of people employed in the energy sector has been steadily rising over the past five years (Fig. 4-19), but the ratio of male to female workers has remained more or less the same at about 5 to 2 (Table 4-55).

Table 4-55: Mombasa Wage Employment (in numbers of thousands) in Energy Sector and Gender.

Year	2012		2013		2014	
Employment By Industry (population count in '000s)	Male	Female	Male	Female	Male	Female
Electricity, Gas, Steam and Air Conditioning Supply	10.3	4.0	10.4	4.2	11.1	4.2

(Source: KNBS Kenya Facts and Figures 2015)

IV. Green economy policy interventions

There is a dearth of green economy policy interventions, (Table 4-56) while the scope to initiate such interventions is vast, given the large scale and cross-cutting character of the sector.

Table 4-56: Existing Green Economy initiatives, challenges, and opportunities in the energy sector according to the County planners.

GE Initiatives	Challenges	Opportunities
- Maintenance of mangroves - Tree planting - Support community - Tree Nurseries	- Finances	- Sustainability

(a) National Level

Under Thematic Area 1 on Promoting Sustainable Infrastructure (Kenya GESIP Report 2015), energy is underscored as fundamental in creating a healthy and thriving economic climate for transformation and expansion of economic activities and enhancing sustainability, and its development is a key foundation of Vision 2030. The aim of Thematic Area 1 of the Kenya GESIP report is to promote sustainable energy infrastructure which encompasses the designing, building and operating of energy elements so as to reduce negative impacts on social, economic and ecological processes necessary for human equity, diversity and integrity of natural resources.

At the national level, Kenya GESIP Report (2015) has two objectives covering the energy sector specifically. Objective 1.3 proposes to increase the share of renewable energy in the energy mix to 75% by reviewing the Feed-in-Tariff (FiT) policy to include more off-grid generation and net-metering, and establishing large-scale community biogas /solar PV/ wind-farm generating systems in off-grid areas, while objective 1.4 calls for the embracing of energy efficiency and conservation through the rolling out of energy centres to counties and promotion of energy efficiency appliances and technologies (LPG, LED lamps, improved cook-stoves). It proposes: developing sector-specific energy efficiency indicators and benchmarks for all sectors; rolling out demand-side energy efficiency programmes in urban residential, commercial and industrial establishments; rolling out supply side energy efficiency programmes through system reinforcement, efficient transformers, grid extension projects; adopting minimum energy efficiency performance standards for lighting and industrial products; developing technical and infrastructural capacity for energy efficient audits, equipment testing and certification; continually reviewing national and county policies and legal frameworks to respond to new technology and innovation; and enhancing the application of voluntary management approaches to energy efficiency, clean and renewable energy.

(b) County Level

The County GE plans for the energy sector given here are extracted from the county development plans (Table 4-57). The County Integrated Development Plan (CIDP) 2013-2017 proposes to develop green power plant generation through recycling of waste, and use of solar and wind sources. On the other hand, the Integrated Strategic Urban Development Plan (ISUDP) Mombasa Vision 2035 aims at installing solar lighting and heating in public buildings, learning institutions and health facilities, and also oversee the construction of the 700 MW gas-fired plant at Dongo Kundu proposed by the National Government.

Table 4-57: Existing Green Economy initiatives, challenges, and opportunities

GE Initiatives	Challenges	Opportunities
- Affordable and accessible energy - Development of infrastructure capacities - Provision of clean, sustainable, affordable, reliable and secure energy services according to national energy policies - Increase the share of renewable energy in the energy mix - Better grid access	- Lack of funds to implement projects	- Expansion of energy transmission network; more households, businesses, factories, get connected to the national grid; - High potential for generation of solar and wind energy

(Sources: CIDPs, ADPs, ISUDP and Mombasa Gate Master Plan)

(c) County Baseline Survey, November 2015

The respondents to the county baseline survey identified just a few GE initiatives that relate to the energy sector, and outlined the challenges and opportunities as summarised in Table 4-58, and most of these initiatives are focused on the management and conservation of the forest environment in response to the high demand for wood fuel as shown in Figure 4-17.

Table 4-58 Existing Green Economy initiatives, challenges, and opportunities in the energy sector according to the County planners.

GE Initiatives	Challenges	Opportunities
- Maintenance of mangroves - Tree planting - Support community - Tree Nurseries	- Inadequate finances - The use of fossil fuel in the transportation sector on land and at sea poses the largest environmental threat in the area. - In the offshore harbour area, oil spills have occurred with devastating effect on marine and coastal zone flora and fauna	- Sustainability

(d) Proposed Options, Approaches and Considerations

In order to enhance the uptake of GE initiatives in the County, several other options can also be considered. The County is dependent on electricity from the national grid whose capacity is controlled by the national needs. One approach that the County can adopt to increase energy availability and access over and above that provided by the national grid, and reduce costs to the users, is by encouraging the use of distributed systems. These are generally small-scale electric power generators that produce electricity at a site close to customers or that are tied to an electric distribution system (American Public Power Association, 2013). They may be developed from fossil and renewable energy technologies (e.g., photovoltaic arrays, wind turbines, micro-turbines, reciprocating engines, fuel cells, combustion turbines, and steam turbines); energy storage devices (e.g., batteries and flywheels); and combined heat and power systems. Another

approach that the County can take is by investing in alternative sources of energy, including solar, wind and solid waste. Finally, the County can also promote GE by embracing the concept of energy use efficiency within offices, homes and industries.

In addition to the options outlined in sections (a), (b), and (c) above, other options may include:

- Use of solid waste: Solid waste offers a good alternative opportunity for energy production. However, its use is inhibited due to socio-cultural issues among communities. For example, many communities have a negative attitude towards the use of human waste. Sensitization efforts must accompany the successful utilization of this option;
- Biogas: The initial cost of installation is high; there is limited capacity to construct and control quality of installation; private ownership of land could be a hindrance particularly in cases where the aim is to have a system that supplies a small community;
- Solar energy options:
 - Solar Home Systems: High initial cost of installation hinders uptake; installation standards not maintained; there is lack of awareness on potential for solar energy and lack of quality control on equipment (use of solar photovoltaic (PV) technologies).
 - Solar dryer: A special structure that uses the sun's energy to dry agricultural produce (fruits, vegetables, meat).
 - Solar water heaters: Devices that use solar energy to heat water for domestic, institutional, commercial and industrial use.
- Energy use efficiency
 - Energy saving bulbs: Compared to conventional bulbs, energy-efficient light bulbs such as halogen incandescent, compact fluorescent lamps (CFLs), and light emitting diodes (LEDs) typically use about 25%-80% less energy. They can also last 3-25 times longer.
 - Efficient cooling systems: due to the high temperatures, most buildings in the County are fitted with cooling systems. The buildings should be designed appropriately to minimize wastage of energy.
 - Improved Cook Stoves: Use of improved cook-stoves can improve the quality of lives of the people in the County, particularly that of women and children in rural and urban areas. This is achieved through reduction of the time to collect wood fuel, and reduction of indoor air pollution and thus potentially introducing cost savings to households. These stoves also enhance the efficiency in energy use. However, there are low levels of public awareness amongst households and institutions, limited distribution across the County, and quality control of locally manufactured ICS remains a key concern.
- Wind energy options
 - Wind turbines: Devices used to generate electricity using kinetic energy from wind
 - Wind pumps: Devices that use wind energy to lift water from underground sources
- Other options

-
- Biofuels: Liquid fuels and blending components produced from biomass (plant and animal) feedstocks, used primarily for transportation.
 - Independent Power Producer (IPP): Privately owned power company that produces electricity and sells it for a profit to the national grid or to a distribution utility.

4.4.7 Prioritisation of Analysed Sectors

The sectors that were prioritised during the county stakeholder meetings, and which have been analysed in this report, are: Waste, Tourism, Cities and Built Environment, Transport and Infrastructure, Fisheries, and Energy. Those which were not analysed are: Agriculture, Forestry, Manufacturing, and Water.

This section presents an analysis of the availability and adequacy of indicators developed by UNEP and which are important for monitoring and evaluation purposes for sub-national planning and green economy performance tracking (see Annex 1), using a ten point criteria (Tables 4.56 to 4.62). This analysis outlines the gaps in availability and adequacy of indicator datasets for monitoring and evaluation in each of the analysed sectors. It points to where effort is needed to build the requisite county specific datasets in order to support an informative and dynamic monitoring and evaluation system for each sector to measure and track performance in green economy growth.

The sectors, ranked in terms of lowest to highest effort needed to build the requisite datasets based on the scores in Tables 4-59 to 4-64, are as follows:

- | | | |
|-----|------------------------------|---------------|
| 7) | Transport and Infrastructure | (Score 1.80); |
| 8) | Waste; Energy | (Score 1.70) |
| 9) | Cities and Built Environment | (Score 1.50) |
| 10) | Tourism; Fisheries | (Score 1.40) |

Table 4-59: Assessment of the waste sector

(1) WASTE SECTOR			
	CRITERIA	REMARKS	MARKS
1.	Importance of the sector according to the respondents	- Considered by respondents to be <i>important</i> for GDP and employment	3
2.	Adequacy of GDP related data (Annex 4)	- No sector disaggregated data	1
3.	Adequacy of other General Sector data (Annex 4)	- Snapshot data available, no longitudinal data	2
4.	Adequacy of data on GHG emissions (Annex 4)	- No data available	1
5.	Adequacy of other Environmental & related data (Annex 4)	- Snapshot data available, no longitudinal data	2
6.	Adequacy of Green Economy policy interventions and other related data (Annex 4; Section 4.2.1b)	- GE initiatives are few and uncoordinated, but there are policy reviews ongoing where GE principles can be embedded	2
7.	Adequacy of well-being and equity data (Annex 4)	- No data available	1
8.	Level of renewable energy use excluding on-grid geothermal and hydro power, based on (1) existing documents and (2) respondent analysis	- Low (see Section 4.2.1b)	1
9.	Level of waste treatment and recycling	- Waste treatment available but inadequate - Recycling is undertaken but in an uncoordinated fashion and to varying degrees depending on waste product type (see Section 4.2.1b)	2
10.	Level of funding available to the sector <i>vis à vis</i> (1) sector funding needs and (2) proportion of budgetary allocations to the sector compared to all sectors combined	- Funding available to rehabilitate existing facilities but not for expansion to meet demand. - Sector ranks low in budgetary allocation under Water, Environment and Natural Resources (2015/2016 fiscal year)	2
	TOTAL SCORE		17
	AVERAGE SCORE		1.7

The 10 point criteria for sector prioritisation (Availability/Adequacy: 1- low; 2- medium; 3 - High).

Table 4-60 Assessment of the waste sector

(2) TOURISM			
	CRITERIA	REMARKS	MARKS
1.	Importance of the sector according to the respondents	- Considered by respondents to be important for GDP and employment	3
2.	Adequacy of GDP related data (Annex 4)	- None available	1
3.	Adequacy of other General Sector data (Annex 4)	- Snapshot data available but mostly at coast region level, no longitudinal data	2
4.	Adequacy of data on GHG emissions (Annex 4)	- No data available	1
5.	Adequacy of other Environmental & related data (Annex 4)	- No data available	1
6.	Adequacy of Green Economy policy interventions and other related data (Annex 4; Section 4.2.2b)	- There are some structured GE-related initiatives but they are not directly linked to the sector	2
7.	Adequacy of well-being and equity data (Annex 4)	- No data available but some indicators can be estimated from national data	1
8.	Level of renewable energy use excluding on-grid geothermal and hydro power, based on (1) existing documents and (2) respondent analysis	- Low (see Section 4.2.2b)	1
9.	Level of waste treatment and recycling	- The sector generally does not treat waste products in the desired fashion (see Section 4.2.2b) - Very few actors have taken up recycling but those which have set a good example for replication (see Section 4.2.1b)	1
10.	Level of funding available to the sector <i>vis à vis</i> (1) sector funding needs and (2) proportion of budgetary allocations to the sector compared to all sectors combined	- Tourism sector is generally facilitated by public private partnerships rather than directly by County. - Sector ranks low in budgetary allocation, under Tourism and Culture Development (2015/2016 fiscal year)	1
	TOTAL SCORE		14
	AVERAGE SCORE		1.4

The 10 point criteria for sector prioritisation (Availability/Adequacy: 1- low; 2- medium; 3 - High).

Table 4-61: Assessment of the Cities and Built Environment sector

(3) CITIES & BUILT ENVIRONMENT			
	CRITERIA	REMARKS	MARKS
1.	Importance of the sector according to the respondents	- Considered by respondents to be important for GDP and employment	3
2.	Adequacy of GDP related data (Annex 4)	- No data available, only at national level	1
3.	Adequacy of other General Sector data (Annex 4)	- No systematic collection of such data	1
4.	Adequacy of data on GHG emissions (Annex 4)	- No data available	1
5.	Adequacy of other Environmental & related data (Annex 4)	- Some snapshot data, some can be derived from national datasets	2
6.	Adequacy of Green Economy policy interventions and other related data (Annex 4; Section 4.2.3b)	- There are some structured GE-related initiatives but they are not directly linked to the sector	1
7.	Adequacy of well-being and equity data (Annex 4)	- No data available but some indicators can be estimated from national data	1
8.	Level of renewable energy use excluding on-grid geothermal and hydropower, based on (1) existing documents and (2) respondent analysis	- Low (see Section 4.2.3b)	1
9.	Level of waste treatment and recycling	- The sector is generally a major land and water polluter (particularly from industrial activities) (see Section 4.2.3b) - Poor sewerage works within the City and low level of recycling of wastes generally (see Section 4.2.3b)	1
10.	Level of funding available to the sector <i>vis à vis</i> (1) sector funding needs and (2) proportion of budgetary allocations to the sector compared to all sectors combined	- Funding available for mega projects nested within Vision 2030 Flagship projects. - Sector ranks high in budgetary allocation, under Transport and Infrastructure & Lands, Planning and Housing (2015/2016 fiscal year)	3
	TOTAL SCORE		15
	AVERAGE SCORE		1.5

The 10 point criteria for sector prioritisation (Availability/Adequacy: 1- low; 2- medium; 3 - High).

Table 4-62: Assessment of the Transport and Infrastructure sector

(4) TRANSPORT & INFRASTRUCTURE			
	CRITERIA	REMARKS	MARKS
1.	Importance of the sector according to the respondents	- Considered by respondents to be <i>important</i> for GDP and employment	3
2.	Adequacy of GDP related data (Annex 4)	- Some data available but mostly at national level	2
3.	Adequacy of other General Sector data (Annex 4)	- Some data available but mostly at national level	2
4.	Adequacy of data on GHG emissions (Annex 4)	- No data available but some indicators can be estimated from national data	1
5.	Adequacy of other Environmental & related data (Annex 4)	- Some snapshot data, some can be derived from national datasets	2
6.	Adequacy of Green Economy policy interventions and other related data (Annex 4; Section 4.2.4b)	- Sector is low on GE initiatives which are not well structured and lack diversity	1
7.	Adequacy of well-being and equity data (Annex 4)	- No data available but some indicators can be estimated from national data	1
8.	Level of renewable energy use excluding on-grid geothermal and hydro power, based on (1) existing documents and (2) respondent analysis	- Used for street lighting (see Section 4.2.4b)	2
9.	Level of waste treatment and recycling	- The sector is generally a land and water polluter (see Section 4.2.4b) - There is low level recycling of waste materials and oils but is done in an <i>ad hoc</i> manner (see Section 4.2.3b)	1
10.	Level of funding available to the sector <i>vis à vis</i> (1) sector funding needs and (2) proportion of budgetary allocations to the sector compared to all sectors combined	- Funding available for mega projects (roads/railway) nested within Vision 2030 Flagship projects, County undertaking major infrastructure works. - Sector ranks high in budgetary allocation (2015/2016 fiscal year)	3
	TOTAL SCORE		18
	AVERAGE SCORE		1.8

Table 4-63: Assessment of the Fishery sector

(5) FISHERIES			
	CRITERIA	REMARKS	MARKS
1.	Importance of the sector according to the respondents	- Considered by respondents to be important for GDP and employment	3
2.	Adequacy of GDP related data (Annex 4)	- No data available but some indicators can be estimated from national data	1
3.	Adequacy of other General Sector data (Annex 4)	- Some data available but mostly at national level	2
4.	Adequacy of data on GHG emissions (Annex 4)	- No data available	1
5.	Adequacy of other Environmental & related data (Annex 4)	- Some data available but scattered, few trend data	2
6.	Adequacy of Green Economy policy interventions and other related data (Annex 4; Section 4.2.5b)	- Sector is low on GE initiatives which are not well structured and lack diversity	1
7.	Adequacy of well-being and equity data (Annex 4)	- No data available but some indicators can be estimated from national data	1
8.	Level of renewable energy use excluding on-grid geothermal and hydro power, based on (1) existing documents and (2) respondent analysis	- Low level of renewable energy use attributed to high cost (see Section 4.2.5b) -	1
9.	Level of waste treatment and recycling	- Low level of mostly <i>ad hoc</i> waste recycling (see Section 4.2.5b)	1
10.	Level of funding available to the sector <i>vis à vis</i> (1) sector funding needs and (2) proportion of budgetary allocations to the sector compared to all sectors combined	- Low funding - Sector ranks average in budgetary allocation, under Agriculture, Livestock, Veterinary and Fisheries (2015/2016 fiscal year)	1
	TOTAL SCORE		14
	AVERAGE SCORE		1.4

Table 4.62: Assessment of the waste sector against the 10 point criteria for sector prioritisation (Availability/Adequacy: 1- low; 2- medium; 3 - High).

(6) ENERGY			
	CRITERIA	REMARKS	MARKS
1.	Importance of the sector according to the respondents	- Considered by respondents to be <i>very important</i> for GDP and employment	3
2.	Adequacy of GDP related data (Annex 4)	- No data available but maybe can be estimated from national data	1
3.	Adequacy of other General Sector data (Annex 4)	- Some data available but mostly at national level	2
4.	Adequacy of data on GHG emissions (Annex 4)	- No data available but maybe can be estimated from national data	1
5.	Adequacy of other Environmental & related data (Annex 4)	- Some data for most indicators at national level, not at county level	2
6.	Adequacy of Green Economy policy interventions and other related data (Annex 4; Section 4.2.6b)	- Sector is low on GE initiatives which are not well structured and lack diversity	1
7.	Adequacy of well-being and equity data (Annex 4)	- No data available but indicators can be estimated from available data at national level	2
8.	Level of renewable energy use excluding on-grid geothermal and hydro power, based on (1) existing documents and (2) respondent analysis	- Energy sector is generally moving towards a higher mix of renewable vs non-renewable energy sources but at national level (see Section 4.2.6b)	2
9.	Level of waste treatment and recycling	- There is little information on waste treatment and recycling from energy generation (see Section 4.2.6b)	1
10.	Level of funding available to the sector <i>vis à vis</i> (1) sector funding needs and (2) proportion of budgetary allocations to the sector compared to all sectors combined	- Energy sector is generally a service sector so not many projects under County government. - Sector ranks medium in budgetary allocation, under Trade, Energy and Industry (2015/2016 fiscal year)	2
TOTAL SCORE			17
AVERAGE SCORE			1.7

4.5 County Planning Cycle

4.5.1 Institutional arrangements for annual development planning

County governments are structures with almost similar institutional frameworks with autonomous and differentiated roles and responsibilities. The various institutions and their mandates are discussed here below.

a) Governor

Some of the relevant functions of a Governor include: Representing the county in national and international fora and events; appointing, with the approval of the county assembly, the county executive committee; submitting the county plans and policies to the county assembly for approval; considering, approve and assent to bills passed by the county assembly; chairing meetings of the county executive committee; submitting to the county assembly an annual report on the implementation status of the county policies and plans; providing leadership, promoting citizen participation, democracy, good governance, unity and cohesion within the county; and promoting the competitiveness of the county and be accountable for the management and use of the county resources.

b) Council of County Governors

The Council provides a forum for: Consultation; sharing of information; considering matters of common interest; dispute resolution; capacity building for governors; receiving reports and monitoring the implementation of projects.

c) Intergovernmental Relations Technical Committee

Its functions include: taking responsibility for the day to day administration of the Summit and of the Council and in particular facilitate the activities of the Summit and of the Council; implementing the decisions of the Summit and of the Council of governors.

d) Senate

The main role of Senators is to: represent the counties, while also serving to protect the interests of the counties and their governments; participating in the law-making function of Parliament by considering, debating and approving Bills concerning counties; determining the allocation of national revenue among counties, as provided in Art. 217, and exercises oversight over national revenue allocated to the county governments.

e) County Assembly

The County assembly is mandated to make any laws that are necessary for, or incidental to, the effective performance of the functions and exercise of the powers of the county government; exercise oversight over the county executive committee and any other county executive organs; and receive and approve plans and policies for the management and exploitation of the county's resources; and the development and management of its infrastructure and institutions.

f) County Executives

There functions include: Implementing county and national legislations; managing and coordinating the functions of the county administration and its departments; preparing proposed legislation for consideration by the county assembly; providing the county assembly with full and regular reports on matters relating to the county.

g) County Public Service Board

Their functions are: Establishing and abolish offices in the county public service; appointing persons to hold or act in offices of the county public service; exercising disciplinary controls; preparing regular reports for submission to the county assembly; facilitate the development of human resource planning and budgeting for personnel emoluments; and advising the County Government on human resource management and development.

h) City and Municipality Boards

The City and Municipality boards are mandated to: oversee the affairs of the city or municipality; develop and adopt policies, plans, strategies and programmes, and may set targets for delivery of services; formulate and implement an integrated development plan; control land use, land sub-division, land development and zoning by public and private sectors for any purpose; develop and manage schemes; maintain a comprehensive database and information system; implement applicable national and county legislation; monitor and, regulate city and municipal; settle and implement tariff, rates and tax and debt collection policies as delegated by the county government.

i) Strategic Documents guiding development and decision making

There are a number of key documents, both at National and County levels that influence development planning at the County level.

At the **national level**, some of these documents include:

- The Public Finance Management Act 2012 which provides for the establishment of a County Budget and Economic Forum (CBEF). The CBEFs provide a platform for the County and public to consult on preparation of a County Fiscal Strategy Paper and County Budget Review and Outlook Paper (BROP).
- The Urban Areas and Cities Act of 2011 - makes provisions for public participation.
- The National Climate Change Response Strategy (NCCRS) 2010.
- The National Integrated Monitoring and Evaluation system (NIMES).
- EMCA 1999 that exercise general supervision and coordination of lead agencies that execute environmental functions.

At the County level, the documents include:

- The Mombasa County Integrated Development Plan (CIDP) 2013-2017 where stakeholders gave their views in six stakeholder forums
- Mombasa County Annual Development Plan 2015-2016 Fiscal Year. Budget Office of the County Government of Mombasa, Finance and Economic Department.

- The Mombasa County Monitoring and Evaluation system, aligned to the National Integrated Monitoring and Evaluation system (NIMES), to serve the needs of the County. The system takes cognizance of the projects and programmes included in the County Integrated Development Plan as well as indicators facilitating the MTEF process and development of the country.
- Draft Mombasa County Environment Action Plan: Sustainable Development towards Achievement of Vision 2030 (CEAP) 2013-2018.
- Mombasa County State of Environment Reports and County Environment Action Plans
- Draft Mombasa Gate City Master Plan: The proposed Master Plan will integrate all the existing Master Plans of various infrastructures within the city of Mombasa and its surrounding. The purpose of the master plan is to provide a guiding framework to manage urban development in Mombasa City County from 2014-2030, integrate all urban development sectors and realize the goals of Kenya Vision 2030 for the city county of Mombasa.
- Draft Integrated Strategic Urban Development Plan (ISUDP) for Cluster Towns is part of the Kenya Municipal Program (KMP). The KMP aims to strengthen local governance and improve urban service delivery in selected urban municipalities by reforming frameworks for urban governance, municipal restructuring, strengthening of planning mechanism, financing and capacity building, and investment in infrastructure and service delivery improvements in the towns.
- Draft Mombasa City Zoning Plan: This is a spatial framework guiding development of land by defining the allowed standards and regulations for development of various land uses in different places (zones) within the 5 years from the date of inception.
- County Zoning Plan: This is a spatial framework guiding development of land by defining the allowed standards and regulations for development of various land uses in different places (zones) within the 5 years from the date of inception.

j) Aspects of the Development Planning Cycle

This table below is consolidated from the sectoral information that was obtained from respondents (see Annex 5(2)). It captures the generic and common issues derived from the sectoral perspectives.

Table 4.63: Aspects of the development planning cycle undertaken by the Mombasa County planners.

Development Planning		Status/Remarks
County Government (role(s) in county development planning process)		- Development of plans to improve and maintain services and related facilities - Plan and implement new development projects - Enacting policies/regulations and by-laws - Sharing of Plans Civic education - Working with developers - Consultations and engagement with stakeholders - Budgetary planning - Development oversight

Key steps in planning process	- County board and top management conceive and draft development plans - Employees provide information that are relevant to County planning - Involve consultants - Stakeholder consultations	
Timelines for planning process	- Dependent on project magnitude - 1 year (Annual Cycle) to 25 years (SEA) span - Adhere to the Master Plan where there are short-term, medium and long term schedules. -	
Key objectives of institutions development plan.	- Provision and maintenance of services and improvement of support infrastructure - Create a clean conducive environment for investment growth and habitation by the county residents. - Achieve the standards expected internationally and locally. - Enact and implement regulations - Risk management - Enhance revenue and growth - Improve institutional capacity - Support development of industry output and capacity - Sustainability - Succession planning	
Key (a) Methodologies (b) Tools (c) Platforms used to implement development plans.	a) methodologies	- Collection and archiving of data from self, other institutions and other reliable sources - Develop Master plans, Strategic plans, Zoning plans, Land use plans - Informal procedures - Planning on a need basis
	b) tools	- SWOT - Cost/ benefit analysis - Rapid results initiatives - Performance contracts - Monitoring committees (sectoral) - EIA and SEAs - Internal/self-Audits - Risk management systems e.g. Enterprise risk management system - IFMIS

	c) platforms	- Monitoring committees (sectoral) - Sectoral and departmental consultations - Public-Private partnerships - Public / Community / Stakeholder involvement; field days. - Umbrella authorities (sectoral or cross-sectoral) - Seminars/Workshops - Mass media (print, radio, TV, website) - Policy briefs - ICT (e.g. e-citizen, websites, fora with toll free number)
Monitoring and evaluation for implementation of the development plans		- Activity tracking - QA/QC e.g. Inspections of buildings, testing of drinking water quality - Monitoring and evaluation committees - A balanced scorecard - ISO certification.
Expectations for a toolkit and knowledge management platform		- Makes it easier to interrogate issues and convince decision makers/strategic organisations in the planning process - Mainstream environmental issues identified in the CIDP (County Integrated Development Plan). - Be able to analyse current status and project goals using the toolkit and management platform - The platform should have GIS capabilities - Capacity building to utilise the platform

4.5.2 Relevant actors

	Sector	Actors
1.	Waste Management	County Government, Ministry of Youth and Sports, Community Development Trust Fund, Destiny Africa Ecotact, NEMA, JICA, Kenya Forestry Service, Ministry of Health, Kenya Handicraft Association, KUZA/British High Commission, UNEP, USAID, Society for Protection of Environment in Kenya.
2.	Tourism	County Government, Kenya Tourism Board, Kenya Wildlife Service, Kenya Handicraft Association, Tourism Regulatory Authority, Kenya Tourism Federation, Mombasa and Coast Tourist Association, Kenya Tourist Development Corporation, Tourism Finance Corporation, Kenya Association of Hotel Keepers and Caterers.
3.	Transport	County Government, Kenya Urban Roads Authority, Kenya Rural Roads Authority, Kenya Highways Authority, Kenya Ports Authority, Northern Corridor Stakeholders Consultative Forum, Northern Corridor Transit and Transport Coordination Authority, Kenya Airports Authority, Kenya Ferry Services, Dock Workers Union, National Transport Safety Authority
4.	Cities	County Government, Ministry of Lands, Urban and Housing, JICA, World Bank
5.	Fisheries	Mombasa County Government, Kenya Marine and Fisheries Research Institute, Coast-Rural Development Organisation, Tuna Fishery Alliance of Kenya, UNEP, IUCN, UNESCO, FAO, KWS, National Museums of Kenya, Kenya Forestry Research Institute, East African Wildlife Society (EAWLS).
6.	Energy	County Government, KPLC, KenGen, Kenya Pipeline Corporation, Kenya Refineries Limited.
7.	Water	County Government, WRMA, Kenya Maritime Authority, Kenya Marine Research Institute, MOWASCO, Maji na Ufanisi,
8.	Forestry	County Government, Kenya Forests Service, NEMA, Bamburi/LARFARGE, Kenya Forestry Research Institute, Kenya Forests Working Group, Society for Protection of Environment in Kenya, Nature Kenya
9.	Buildings	County Government; Department of Planning; Ministry of Lands, Urban and Housing; Ministry of Devolution and Planning.
10.	Manufacturing	County Government, Kenya National Chamber of Commerce and Industry, KEPSA, Kenya Ships Agents Association, Kenya Transporters Association, The Mombasa Port Community Charter, The Mombasa Tea Auction
11.	Agriculture	Ministry of Agriculture, Livestock and Fisheries, NEMA, County Government, Meteorology Department

4.5.3 Best entry point for the project in Mombasa County

The Government of Kenya, in its Medium Term Plan 2013-2017, committed to developing a green economy strategy to support development efforts towards addressing key challenges such as poverty, unemployment, inequality, environmental degradation, climate change and variability, infrastructure gaps and food insecurity (Republic of Kenya, 2015). This commitment offers the Mombasa County government the opportunity to adopt and align its development plans with this strategy. A smooth transition to a Green Economy depends upon a number of related conditions that include: political commitment, a legal and regulatory foundation, implementation of financial instruments, technological viability, human capital, and effective institutions. The development pathway in a green economy will deliver resilient, equitable and pro-poor development through investments that reduce carbon dependency, promote resource and energy efficiency, and lessen environmental degradation (UNEP, 2015). It is noted that the Government of Kenya, development partners and other non-state actors are already supporting and implementing green economy-related policies and initiatives (UNEP, 2015). This is also evident from the results of the field survey in this study. Kenya's key programmes that relate to the green economy transition include (UNEP, 2015): investments in renewable energy, promotion of resource-efficient and clean production, pollution control as well as waste management, environmental planning plus governance, and restoration of forest ecosystems.

The analysed sectors for prioritisation of green economy activities in Mombasa County are: (1) Transport and Infrastructure, (2) Energy, (3) Cities and Built Environment, (4) Waste, (5) Tourism, and (6) Fisheries (see Section 4.3). Of these, the energy sector has pervasive and significant cross-cutting elements that underlie all of the other analysed (and non-analysed) sectors as energy is the driving force without which transportation cannot take place, infrastructure cannot be built, cities and the built environment cannot be developed, waste cannot be processed into inert products, and so on. Further, it is noted that all the sectors will produce waste of one type or another, so the waste sector can also be viewed as having pervasive and cross-cutting elements. Thus, some aspects of the energy and waste sectors will of essence need to be addressed even when implementing green economy plans, strategies, and actions that are focused on a particular sector. The tourism sector is important for the national and County GDP as well as economy and employment, and, although green economy data was less for this sector as compared to some of the others analysed, it also embraces the national pride and gives the country a strong visibility in the international community. Green economy activities undertaken in the other sectors would also, both directly and indirectly, contribute to green economy entrainment efforts that specifically target the sector. The fisheries sector contribution is small but ranks high in terms of well-being and equity as it supports the livelihoods of poor coastal zone inhabitants.

For each of the analysed sectors, green economy options have already been outlined. Here, the focus is on cross-cutting elements that can help underscore the success of green economy interventions if incorporated in the planning and implementation of the activities. UNEP (2015), Republic of Kenya (2015), and Mombasa County respondents to the baseline survey during this study, outline some of the actions that Mombasa County can take to implement the transition to the green economy in the sectors named above. These are customised or added to, to focus on the specific attributes of Mombasa County as outlined below.

Crosscutting, generic actions to undertake to support the green economy in the various sectors described in this report, and other sectors as well, would include the following:

- Establish high-level support and buy-in from the County governance structure and key public and private sector institutions and actors;
- Develop a well-coordinated and strategic overall approach to transition to a green economy with clear measurable aims to channel achievements;
- Develop, through engagement with local communities/stakeholders (including the private sector), supporting and robust County policies, regulations and standards that embrace eco-labelling, technical standards related to product standards and labelling, and environmental standards for green technologies, goods and services;
- Strengthen relevant institutions to improve on regulatory enforcement and compliance;
- Create a system of targeted subsidies and taxes to support nascent green sectors and overall development;
- Promote the trade and purchase of green products and services;
- Encourage transparency, accountability, and solid fiscal management as this will ensure buy-in and attract funding for projects from local, national and international donors;
- Establish, and where existing, improve green economy indicator data² collection, management, and the dissemination of statistical findings which is crucial for performance assessment and in measuring progress on the green economy evolution;
- Develop the capacity to evaluate the costs and performance characteristics of available green technologies as many studies have shown that the costs of turning over the current fossil-fuel based technology stock in transport and power supply to green alternatives are low relative to its benefits;
- Work with other County governments and the national government to engage nationally and internationally, respectively, on the green economy;
- Institute programmes to develop skills, capacity and awareness of green economy issues; and finally
- Address the cross-sectoral barriers to a green transition which have been identified as including:
 - Inaccessibility to local/international markets;
 - The growing concerns over youth employment;
 - Gender inequalities;
 - Training and skills necessary for new green opportunities;
 - Insufficient of awareness about green economy best practices;
 - Obsolete and slow adoption of technology;
 - **Devolution transitional challenges regarding capacity and policy coordination;**
 - **Capacity to leverage private sector investment;**
 - **Insufficient incentives, low rate of return on green investment; and**
 - **Inadequate access to information on climate and weather data, which can impact resilience.**

²The Green Economy indicators are as specified in Annex 1 of this Report.

CHAPTER 5 - CONCLUSIONS

5.1. Priority sectors for each study county

A stakeholder identification and prioritization of the sectors for GE were done and are summarized as;

(1) Nakuru

Agriculture, Water, Forestry, Energy, Tourism, and Waste Management,

(2) Mombasa

Waste, Tourism, Cities and Built environment, Transport and Infrastructure, Fisheries, and Energy.

5.2. The predefined indicators

- The predefined indicators for green economy in the sectors proved to be difficult to measure, due in part to the following reasons; county functions are in transition from the national government; lack of collection and archiving of green economy indicator data
- The counties have no database on performance and the historic data and information are still held at national office.
- Most of the information acquired during the baseline are thus basic as derived from the documents done at county level in the last three years. In some cases national indicators were inferred.

5.3. Analysis and main findings of Green Economy

- All the sectors analysed are in the process of implementing performance based service delivery and are thus good candidates for the GE initiative.
- The resource base and utilization have been given priority in the strategic and planning documents, but are currently not verifiable.
- There is, however, a challenge in adequacy of data and information for implementation of Green Economy Plans within the Counties.

5.4. Areas that need to be addressed for an effective Green Economy implementation

- Sensitisation on what Green Economy would improve in comparison to Business-as-Usual:
 - Job creation,
 - CO₂ reduction,
 - Resource use efficiency
 - Effective and efficient service delivery
 - And other benefits
- Piloting case study on GE projects and showing reliability.
- Mainstream Green Economy in the county annual/development plans for the key sectors.

-
- Political commitment to the green growth path ensuring that fiscal and monetary tools are used to steer economies away from business-as-usual.
 - Investing in research and development (R&D), demonstration, deployment and the commercialization of different renewable technologies in production and consumption activities.
 - Legal reforms requiring revisiting existing laws and aligning these with GE principles.
 - Investments into human capital, a crucial expenditure that should not be compromised and, as part of public funding, **ought to enhance the know-what, know-how and the know-why.**

The Counties may need to embrace protocols for local green initiatives with modest goals to attract new investment and to bring in new revenue.

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ANNEXES

ANNEX 1 - THE PERFORMANCE (SECTOR) INDICATORS

Baseline Survey for ‘Operationalizing Green Economy Transition in Africa: Status of green economy initiatives in Ethiopia, Ghana, Kenya and Rwanda’

The UNEP approach on Green Economy categorizes 10 industry sectors. We further split the sector Agriculture & Fishery into two sectors. For M&E purposes meaningful indicators on sub-national planning and Green Economy have been developed. These indicators are comprehensive but not exclusive and can be reduced or enhanced in the further process of the project.

After a chapter on general cross-sectorial indicators for all of the eleven industry sectors the identified indicators were grouped into four categories:

- I. General sector indicators**
- II. Environmental issues and related indicators**
- III. Green economy policy interventions and related indicators**
- IV. Well-being and equity indicator**

See below the list of the industry sectors covered:

- 1. General (cross-sectorial, complex indicators)**
- 2. Agriculture**
- 3. Buildings**
- 4. Cities**
- 5. Energy**
- 6. Fishery**
- 7. Forestry**
- 8. Manufacturing**
- 9. Tourism**
- 10. Transport**
- 11. Waste**
- 12. Water**

1. General (cross-sectorial, complex indicators)

I. Economic indicators	- GDP [USD] - GDP growth [%, average 5 years] - Trade balance [USD] - Population [people] - Gender female/male [%] - Unemployment rate [%]
II. Environmental issues and related indicators	- CO₂ emission [tonne/year] - CO₂ emissions per unit GDP [kg] - CO₂ emissions per capita [kg] - Energy consumption per capita [KJ] - Energy productivity [tbd] - Resource productivity (materials, water, nutrients) [tbd] - Waste generation per capita [kg] - Waste recycling and reuse [%] - Water intensity or productivity [m³/USD] - Coal consumption intensity ([tonnnes/GDP] - Storm-related damages [USD/year]
III. Green economy policy interventions and related indicators	- Extent of land and marine conservation areas [ha] - Total green investments [USD/year] - Green Economy initiatives [number] - Extent of land and marine conservation areas [ha] - Share of renewable electricity production [%] - Subsidies green economy [USD] - Downloads toolkit [downloads] - Share of plans on sub-national level with toolkit [%] - Number of private sector partners in GE [partners] - Replication of cases/best practices [replications]
IV. Well-being and equity indicator	- Average income [USD] - Primary sector employment (people) - Dietary energy supply [Kcal/day per person] - Share of new-borns with low birth weight (<2500g) [%] - Life expectancy [years]

V. Complex indicators	- Genuine Progress Indicator (GPI) - Multidimensional Poverty Index (MPI), UNDP - Your Better Life Index, OECD - Indices of Social Development (ISD), ISS - Human Development Index (HDI), UNDP - Sustainable Society Index (SSI), SSF - Social Institutions and Gender Index (SIGI), OECD - Gender Equity Index (GEI), Social Watch - EPI Index
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2. Agriculture

I. General sector indicators	- GDP agriculture sector [USD] - GDP growth agriculture sector [%, average 5 years] - Contribution of agriculture sector to GDP [%] - Agricultural output [tonnes] - Net exports of agricultural commodities [USD] - Agricultural area used [ha] - Number of farms [number]
II. Environmental issues and related indicators	- CO2 emission agriculture sector [tonne/year] - CO2 emissions agriculture sector per tonne agricultural commodities [kg] - CO2 emissions agriculture sector per tonne of livestock [kg] - Energy consumption agriculture sector [MJ/year] - Share of renewable energy agriculture sector [%] - Degree of top soil losses on agricultural land [%] - Fertilizer consumption (kilograms per hectare of arable land) [kg/ha] - Cereal yield [kg/ha] - Agricultural irrigated land (% of total agricultural land) [%]
III. Green economy policy interventions and related indicators	- Green investments in agriculture sector [USD] - Green Economy initiatives in agriculture sector [number] - Areas of agricultural ecosystems under sustainable management [ha] - Share of organic farming (based on output) [%] - Agriculture GE subsidies [USD] - Downloads agriculture sector toolkit [downloads] - Share of sector plans on sub-national level with toolkit [%] - Number of private sector partners in GE agriculture sector [partners] - Replication of cases/best practices agriculture sector [replications]
IV. Well-being and equity indicator	- Employment in agriculture sector [people] - Employment growth in agriculture sector [%, average 5 years] - Agriculture employment multiplier [factor]

	- Average farm income [USD]
V. Data sources	- data.worldbank.org/indicator#topic-1

3. Buildings

I. General sector indicators	- GDP buildings sector [USD] - GDP growth buildings sector [%, average 5 years] - Contribution of buildings sector to GDP [%] - New built living space [m²] - New built office space [m²] - New built industrial space [m²] - Number of construction firms [number]
II. Environmental issues and related indicators	- CO₂ emission buildings sector [tonne/year] - CO₂ emissions buildings sector per m² [kg] - Energy consumption buildings sector [MJ/year] - Energy-use efficiency buildings sector [tbd] - Share of renewable energy buildings sector [%] - Buildings retrofitting [% of portfolio] - Penetration of high efficiency light bulbs [%] for residential, commercial, and institutional buildings
III. Green economy policy interventions and related indicators	- Green investments in buildings sector [USD] - Green Economy initiatives in buildings sector [number] - Share of new buildings to meet the energy efficiency criteria set out in South African National Standard 204 [%] - Buildings GE subsidies [USD] - Downloads buildings sector toolkit [downloads] - Share of sector plans on sub-national level with toolkit [%] - Number of private sector partners in GE buildings sector [partners] - Replication of cases/best practices buildings sector [replications]
IV. Well-being and equity indicator	- Employment in buildings sector [people] - Employment growth in buildings sector [%, average 5 years] - Buildings employment multiplier [factor] - Share of people with decent housing [tbd]
V. Data sources	-

4. Cities

I. General sector indicators	- GDP cities sector [USD] - GDP growth cities sector [%, average 5 years] - Contribution of cities sector to GDP [%]
II. Environmental issues and related indicators	- Air pollution [ppm] - Water pollution [BOD mg/L] - Urbanised area growth [%, average 5 years] - Percentage of trees in the city in relation to city area [%] - Percentage of each mode of transportation, i.e. private, public, bicycle, pedestrian [%] - Generation and disposal of municipal solid waste [tonnes] - Recycling rates [%] - Energy-use efficiency cities sector [tbd] - Percentage of city population with access to water-born or alternative (and effective) sanitary [%] - Percentage of green space (public parks) coverage in relation to city area and/or population size [%] - Population connected to urban wastewater treatment plants [%]
III. Green economy policy interventions and related indicators	- Green investments in urbanisation [USD] - Green Economy initiatives in urbanisation [number] - Percentage of green jobs in the municipal economy [%] - Subsidies green cities [USD] - Downloads cities sector toolkit [downloads] - Share of sector plans on sub-national level with toolkit [%] - Number of private sector partners in GE cities sector [partners] - Replication of cases/best practices cities sector [replications]
IV. Well-being and equity indicator	- Number of people reported to have been affected by diseases arising from drinking/bathing water over a defined period per capita [number/capita] - Population exposure to harmful levels of air pollution [%] - Access to basic services in urban areas [%]

	- Share of women in city councils [%]
V. Data sources	-

5. Energy

VI. General sector indicators	- GDP energy sector [USD] - GDP growth energy sector [%, average 5 years] - Contribution of energy sector to GDP [%] - Total Primary Energy Supply [PJ] - Energy self-sufficiency [%] - Electricity production [GWh/year] - Electricity capacity [MW] - Petroleum consumption [tonne/year]
VII. Environmental issues and related indicators	- CO₂ emission energy sector [tonne/year] - CO₂ emissions energy sector per KWh [kg] - Electricity production from renewable sources [GWh/year] - Share of renewable electricity production [%] - Electricity capacity from renewable sources [MW] - Share of renewable electricity capacity [%] - CO₂ intensity of energy production [CO₂/KWh] - Share of population with access to electricity [%] - Share of population using solid fuels [%] - Electricity use per capita [KWh]
VIII. Green economy policy interventions and related indicators	- Green investments in energy sector [USD] - Green Economy initiatives in energy sector [number] - Energy subsidies in renewable energies [USD] - Downloads energy sector toolkit [downloads] - Share of sector plans on sub-national level with toolkit [%] - Number of private sector partners in GE energy sector [partners] - Replication of cases/best practices energy sector [replications]
IX. Well-being and equity indicator	- Employment in energy sector [people] - Employment growth in energy sector [%, average 5 years] - Energy employment multiplier [factor]
X. Data sources	- www.irena.org/REmaps/africamap.aspx - International Energy Agency (IEA), CO₂ Emissions from Fuel Combustion - data.worldbank.org/indicator/EG.ELC.ACCS.ZS

6. Fishery

I. General sector indicators	- GDP fishery sector [USD] - GDP growth fishery sector [%, average 5 years] - Contribution of fishery sector to GDP [%] - Fish catch landed [tonnes] - Net exports of fish [USD] - Number of fishers/vessels [number] - Number of fish processing enterprises [number]
II. Environmental issues and related indicators	- CO₂ emission fishery sector [tonne/year] - CO₂ emissions fishery sector per tonne of fish catch [kg] - Energy consumption fishery sector [MJ/year] - Energy consumption fishery sector per tonne of fish catch [KJ/tonne] - Fraction of fish stocks overexploited and collapsed by exclusive economic zone [fraction] - Trawling catch per exclusive economic zone [tonne/km²]
III. Green economy policy interventions and related indicators	- Green investments in fishery sector [USD] - Green Economy initiatives in fishery sector [number] - Coverage of marine conservation areas [km²] - Areas of aquaculture ecosystems under sustainable management [km²] - Fishery GE subsidies [USD] - Downloads fishery sector toolkit [downloads] - Share of sector plans on sub-national level with toolkit [%] - Number of private sector partners in GE fishery sector [partners] - Replication of cases/best practices fishery sector [replications]
IV. Well-being and equity indicator	- Employment in fishery sector [people] - Employment growth in fishery sector [%, average 5 years] - Fishery employment multiplier [factor] - Number of small-scale fishers [number]
V. Data sources	- Sea Around Us Project, University of British Columbia Fisheries Centre

7. Forestry

I. General sector indicators	- GDP forestry sector [USD] - GDP growth forestry sector [%, average 5 years] - Contribution of forestry sector to GDP [%] - Value of timber products [USD/year] - Net exports of timber [USD/year]
II. Environmental issues and related indicators	- CO₂ emission forestry sector [tonne/year] - Deforestation [ha/year] - Energy consumption forestry sector [MJ/year] - Forest land cover [ha] - Trend in extend of forest [% in 5 years] - Degraded forest land [ha] - Protected forest areas [%] - Fuelwood consumption [tonnes/year] - Forest fragmentation [factor Biofrac]
III. Green economy policy interventions and related indicators	- Green investments in forestry sector [USD] - Green Economy initiatives in forestry sector [number] - Area of forest under sustainable management [% certified] - Forestry GE subsidies [USD] - Reforestation [ha/year] - Share of eco labelled timber enterprises [%] - Ecotourism [USD/year] - Downloads forestry sector toolkit [downloads] - Share of sector plans on sub-national level with toolkit [%] - Number of private sector partners in GE forestry sector [partners] - Replication of cases/best practices forestry sector [replications]
IV. Well-being and equity indicator	- Employment in forestry sector [people] - Employment growth in forestry sector [%, average 5 years] - Forestry employment multiplier [factor] - Income of forest communities [USD/Capita/year] - Reduced household fuelwood demand [fuel alternatives]

V. Data sources	- www.bipindicators.net/forestfragmentation
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8. Manufacturing

I. General sector indicators	- GDP manufacturing sector [USD] - GDP growth manufacturing sector [%, average 5 years] - Contribution of manufacturing sector to GDP [%] - Net exports of manufactured products [USD/year] - Number of manufacturing enterprises [number]
II. Environmental issues and related indicators	- CO₂ emission manufacturing sector [tonne/year] - Energy consumption manufacturing sector [MJ/year] - Energy productivity manufacturing sector [tbd] - Water productivity manufacturing sector [tbd] - Resource productivity (materials, water, nutrients) [tbd] - Recycled/reused content of material inputs [%; recycled + reused material/total material input]
III. Green economy policy interventions and related indicators	- Green investments in manufacturing sector [USD] - Green Economy initiatives in manufacturing sector [number] - Manufacturing GE subsidies [USD] - Share of certified Green manufacturing enterprises [%] - Downloads manufacturing sector toolkit [downloads] - Share of sector plans on sub-national level with toolkit [%] - Number of private sector partners in GE manufacturing sector [partners] - Replication of cases/best practices manufacturing sector [replications]
IV. Well-being and equity indicator	- Employment in manufacturing sector [people] - Employment growth in manufacturing sector [%, average 5 years] - Manufacturing employment multiplier [factor]
V. Data sources	-

9. Tourism

I. General sector indicators	- GDP tourism sector [USD] - GDP growth tourism sector [%, average 5 years] - Contribution of tourism sector to GDP [%] - Guest nights [nights] - Occupancy rate hotel beds [%] - Number of tourism enterprises [number]
II. Environmental issues and related indicators	- CO₂ emission tourism sector [tonne/year] - CO₂ emissions tourism sector per guest night [kg] - Energy consumption tourism sector [MJ/year] - Energy consumption tourism sector per guest night [KJ] - Share of renewable energy tourism sector [%] - Water consumption tourism sector [m³/year] - Water consumption tourism sector per guest night [m³] - Share of wastewater treatment tourism sector [%] - Waste generation tourism sector [tonnes/year] - Waste recycling and reuse tourism sector [%] - Effective management of built and cultural heritage tourism sector [%] - Consideration loss of biological diversity tourism sector [tbd]
III. Green economy policy interventions and related indicators	- Green investments in tourism sector [USD] - Green Economy initiatives in tourism sector [number] - Share of eco labelled tourism enterprises [% certified] - Tourism GE subsidies [USD] - Destination governance, planning and development [%] - Downloads tourism sector toolkit [downloads] - Share of sector plans on sub-national level with toolkit [%] - Number of private sector partners in GE tourism sector [partners] - Replication of cases/best practices tourism sector [replications]
IV. Well-being and equity indicator	- Employment in tourism sector [people]

	- Employment growth in tourism sector [%, average 5 years] - Tourism employment multiplier [factor]
V. Data sources	-

10. Transport

I. General sector indicators	- GDP transport sector [USD] - GDP growth transport sector [%, average 5 years] - Contribution of transport sector to GDP [%] - Population of cars/trucks [vehicles] - Length of road infrastructure [km] - Investment in road infrastructure [USD] - Road freight volume [tonnes] - Rail freight volume [tonnes] - Total person kilometres travelled [km] - Total public transport boarding per capita [factor]
II. Environmental issues and related indicators	- CO₂ emission transport sector [tonne/year] - CO₂ emissions transport sector per capita [kg] - Energy consumption transport sector [MJ/year] - Energy consumption transport sector per capita [KJ] - Annual NO₂ concentrations in urban areas [ppm] - Cost of destruction of roads, bridges, railway lines due to temperature and rainfall variability [USD]
III. Green economy policy interventions and related indicators	- Green investments in transport sector [USD] - Green Economy initiatives in transport sector [number] - Transport GE subsidies [USD] - Share of renewable energy use in transport sector [%] - Efficiency of transport sector [tbd] - Downloads transport sector toolkit [downloads] - Share of sector plans on sub-national level with toolkit [%] - Number of private sector partners in GE transport sector [partners] - Replication of cases/best practices transport sector [replications]
IV. Well-being and equity indicator	- Employment in transport sector [people] - Employment growth in transport sector [%, average 5 years] - Transport employment multiplier [factor] - Road traffic fatalities per 100 000 inhabitants (transport related) [fatalities]
V. Data sources	-

11. Waste

I. General sector indicators	- GDP waste sector [USD] - GDP growth waste sector [%, average 5 years] - Contribution of waste sector to GDP [%] - Total waste generation [tonnes] - Average waste per capita [kg]
II. Environmental issues and related indicators	- CO₂ emission waste sector [tonne/year] - Methane production from liquid waste [tonnes] - Waste recycling rate [%] - Resource efficiency or productivity (GDP/natural resources) [factor] - Waste landfilled [tonnes] - Energy generation from waste [TJ]
III. Green economy policy interventions and related indicators	- Green investments in waste sector [USD] - Green Economy initiatives in waste sector [number] - Waste GE subsidies [USD] - Share of renewable energy use in waste sector [%] - Downloads waste sector toolkit [downloads] - Share of sector plans on sub-national level with toolkit [%] - Number of private sector partners in GE waste sector [partners] - Replication of cases/best practices waste sector [replications]
IV. Well-being and equity indicator	- Employment in waste sector [people] - Employment growth in waste sector [%, average 5 years] - Waste employment multiplier [factor]
V. Data sources	-

12. Water

I. General sector indicators	- GDP water sector [USD] - GDP growth water sector [%, average 5 years] - Contribution of water sector to GDP [%] - Water consumption [m³] - Average water use rate per capita per day [l] - Number of dams [number] - Number of water supply systems [number] - Ground water boreholes [number]
II. Environmental issues and related indicators	- Proportion of total freshwater resources used [%] - Share of wastewater that receives treatment [%] - Share of population with access to sanitation [%] - Share of population with access to improved drinking water source [%] - Water productivity (average gross value of product per unit of water) [factor]
III. Green economy policy interventions and related indicators	- Green investments in water sector [USD] - Green Economy initiatives in water sector [number] - Water GE subsidies [USD] - Share of renewable energy use in water sector [%] - Downloads water sector toolkit [downloads] - Share of sector plans on sub-national level with toolkit [%] - Number of private sector partners in GE water sector [partners] - Replication of cases/best practices water sector [replications]
IV. Well-being and equity indicator	- Employment in water sector [people] - Employment growth in water sector [%, average 5 years] - Water employment multiplier [factor]
V. Data sources	- UNDP: www.data360.org/dsg.aspx?Data_Set_Group_Id=757 - Global database of National Wastewater Treatment - WHO/UNICEF Joint Monitoring Programme (JMP) for Water Supply and Sanitation

ANNEX 2 – STAKEHOLDER MEETING MINUTES AND REPORTS

Baseline Survey for ‘Operationalizing Green Economy Transition in Africa: Status of green economy initiatives in Kenya’

k) Nakuru Fieldwork Report November 2015

■ 5th November 2015

The team left for Nakuru on the 5th November 2015. Initial contacts were made to identify the key informants to be met on arrival. The summary was drawn from the stakeholder prioritised sectors. The calls made on the first day identified a potential 11 institutions/organisations to be targeted the second day of the fieldwork.

■ 6th November 2015

The team held a joint stakeholder briefing which took place at the county livestock boardroom at the former PCs offices. Dr. Dulo took participants through the questionnaire and explained the process, types of information and data required. He emphasized the need to refer to other relevant private stakeholders in sub-counties within relevant sectors

Some observations raised and clarified include:

1. Timeline-explained that the timeline of operation doesn’t necessarily stick to 10 years but leave it open to cover the years the institution/company been in operation.
2. GDP-most contribute to it but exact figures don’t exist so just give estimate contributions
3. Competitive margins and profits-remains a contentious issue!
4. Enterprise targets and sub-sector segmentation- if many list all
5. Turnover and the role of eco-friendly labelling and appreciating farmers/producers

After lengthy discussions, the meeting ended and participants were urged to fill the forms and hand over to Purity by Tue 10 am including contacts of other private stakeholders to contact in their relevant sub-sectors.

Table 1. Attendees List

Name	Affiliation
John Kagunda	Water and Irrigation
David Ndegwa	GO Green CBO
Purity Kaburu	KAPAP/KALRO
Grace Kirui	Agriculture
Michael Wachira	Livestock
Jacksn Raini	Flamingonet
Mary Nyamu	Forestry
John Maina	SCODE
Grace Kibuku	NBA/GDB
Nelson Maina	
Grace Karanja	Environment

Name	Affiliation
Simeon Dulo Christopher Oludhe William Ogara Gideon Muchiri Eunice Boruru Fred Atieno	ICCA

■ 9th – 13th November 2015

The team split into two with one embarking on the Naivasha area for two consecutive days.

The second team members targeted areas around Nakuru, Njoro, Menengai and Nyahururu

Table 2. Summary of Respondents

Questionnaires administered					
	Name of respondent	Organization	Sector	Planning (31)	Sector (21)
1	Immaculate Maina	KALRO Njoro	Agriculture	YES	YES
2	Collins Psinen	Agric And Food Security	Agriculture	YES	
3	Purity Kaburu	KAPAP/KALRO	Agriculture	YES	YES
4	Grace Kirui	Agriculture-County	Agriculture	YES	YES
5	Michael Wachira	Livestock-County	Agriculture	YES	YES
6	Centre Director-Mbuku	KALRO-Beef Research Institute	Agriculture	YES	YES
7	Charles Welimo	ADC Lanet	Agriculture	YES	YES
8	John Kagunda	Water&Irrigation-County	Agriculture	YES	YES
9	Richard Mulewa	Egerton University Farm	Agriculture and Energy	YES	YES
10	Zedekiah Orioki	GDC Naivasha	Energy	YES	
11	Geoffrey Mibey	GDC-Nakuru	Energy	YES	
12	Baraza Philip	Ken-Gen Naivasha	Energy	YES	
13	David Ndegwa	GO Green CBO	Environment	YES	YES
14	Jackson Raini	Flamingonet	Environment	YES	YES
15	John Maina	SCODE	Environment	YES	YES
16	Grace Karanja	CEC Environment	Environment	YES	YES
17	Daniel Koro	WWF-Naivasha	Environment	YES	
18	Japheth Kirui	GIZ-IWASP-Naivasha	Environment	YES	
19	Peter Njuguna	Meteorology Dept	Environment	YES	YES
20	Nelson Maina	Fisheries-County	Fisheries	YES	YES
21	Mary Nyamu	Forestry-County	Forestry	YES	YES
22	Alfred Abongo	Mau Conservancy	Forestry	YES	
23	John Karenju	KWS-Naivasha	Tourism	YES	
24	James Mushori	Merica Hotel Nakuru	Tourism	YES	YES
25	Alfred Kitur	Cathay Hotel Nakuru	Tourism	YES	YES

	Questionnaires administered				
	Name of respondent	Organization	Sector	Planning (31)	Sector (21)
26	Maurine Musimbi	Nakuru National Park	Tourism	YES	
27	Rose Nyamori	WARMA-Naivasha	Water	YES	
28	Peter Ngubu	WARMA-Nakuru	Water	YES	
29	John Kagunda	Water&Irrigation-County	Water	YES	YES
30	Kaman Mbogo	Imaarisha Naivasha	Water & Environment	YES	YES
31	Isaac Makori/Chege	NAWASCO	Water & Sewerage		YES
32	Vincent-Planning Officer	Rift Valley Water Board Services	Water & Sewerage	YES	YES

(2) MOMBASA FIELD TRIP REPORT

The Mombasa County field survey was undertaken from 2nd to 6th November 2015. A total of 31 questionnaires were filled in over the five days, comprising of 24 planning and 7 private sector questionnaires filled in. The questionnaires had been tested in Nairobi, but some further issues were detected during the first day in Mombasa and these were corrected and the slightly modified questionnaires were used from day two to five. These modifications were shared with the Nakuru team to implement as their field trip was undertaken a bit later than that of the Mombasa team. The team was split into two groups, and support to the team in terms of scheduling and networking was provided by Mr. Cristom Mwalimu and Ms. Beatrice Obunga of the Mombasa County Department of Environment.

Challenges during the administration of the questionnaires included: (1) Time taken to answer the questionnaire was on average about 2.5 hrs and this was reduced to about 1.5 with the modifications made to the questionnaire. (2) Moving around from one place to another plus waiting for the interviewee to host the team added on average some 30 to 60 minutes, hence administration of each questionnaire including travel and waiting time was taking about 2.5-3 hrs - on this basis, it was only possible for each team to handle 4 questionnaires per day, hence a total of 8 for the full Mombasa team (3) Rescheduling of appointments was common (4) Some institutions/companies (private sector in particular) would request further letters from the County Department of Environment, which were furnished by Mr. Mwalimu, or for clearance letters from their head offices in Nairobi, which we were unable to procure (5) Some interviewees requested that they be left with the questionnaires to fill in and the team to follow up with them - the return rate under this scenario was extremely low (5) Some interviewees declined to fill in the forms either because they felt they did not have enough knowledge to fill in the forms, or unless sanctioned by top management.

A summary of the questionnaires and stakeholders are provided below.

Mombasa Questionnaires Summary

	Tourism	Fisheries	Cities	Transport	Waste	Total
Planning	6	5	4	4	5	24
Private	3	Nil	1	Nil	3	7
Total	9	5	5	4	8	31

Mombasa Stakeholders

	Name	Institution	Contact
1	M Otieno	NCA-Regional Off	m.otieno@nca.go.ke
2	Twahir Fatuma	NMK-Old Town	0722722345
3	Ronda Josephine	CDA-Research	rondajosphine@yahoo.com 0722 301086
4	Antony Nyaramba	Fisheries	Antonynyaraba@gmail.com 0721 453146

	Name	Institution	Contact
5	Enock Wakwaba	KMFRI D-Director	ewakwabi@kmfri.co.ke
6	Harman Muga	KPC-Depot Mnger	Harman.muga@kpc.co.ke
7	Wamera Auto	Min of Infra & Transport-Regional Eng.	wameraauto@yahoo.com 0722 645183
8	S. Murage	KeNHA	s.murage@kenha.co.ke 0728147933
9	B.G. Abdilla	KURA-Regional Mnger	gedibash@yahoo.com 0720879082
10	Ndeke Julie	KCTA-Project Mnger	ndekejully@gmail.com 0724221400
11	Mwanasiti M. B	CDA-Environ Off	sitideflag@gmail.com 0724 793887
12	Tsofa M.Mweni	WCK	wckmsa@yahoo.com 0710770103
13	David Apili	Travellers-F&B	travhtls@africaonline.com
14	J Mwamburi	MW S& S-Oper. Mnger	0722583546,j.mwamburi@Mombasa water.ac.ke
15	C. K Mwalimu	Min Environ County Govt	ckmwalimu@gmail.com
16	David Kimutai	Severin Hotel	Maintainance.manager@severinsealodge.com 0725 334742
17	Patrick Riungu	Tsavo Plant	Patrict.riungu@wartsila.com 0733477011
18	K. Kirui	Keen Kleener	0721 786856 keenkleener@africaonline.co.ke
19	Gilbert Muariega	Pwani	0725410936,gilbert.mariake@pwani.net

ANNEX 3 - SAMPLE QUESTIONNAIRES

(1) Planning for Development at County Level

University of Nairobi
Institute for Climate Change and Adaptation

Questionnaire for the Green Economy Baseline Study - October 2015
Planning for Development at County Level

Baseline Survey for ‘Operationalizing Green Economy Transition in Africa: Status of green economy initiatives in Nakuru and Mombasa County

This study seeks to assess the status of green economy (GE), including priority development sectors, environmental, social and economic issues, and assess the planning cycle at the county level

Your participation in this survey will be highly appreciated. Please rest assured that whatever information you provide will be treated in strict confidence.

1) Interviewee Information

Sector	Institution/Department	Title of the Contact person	Contacts (Email, Mobile No.)	Responsibilities of the contact person

Relevant experience		
Description of your experience	Yes /No	Number of years
GE and climate change mitigation		
Strategies and planning		
Implementation plans		
Monitoring and evaluation		

2) Economic structure

County	Economic significance									
	GDP					Employment				
Currently (rank)	1	2	3	4	5	1	2	3	4	5

Key 1-not important, 2-slightly important, 3- important, 4-quite important, 5-very important

GDP Precise figure if available_____

Please provide the number of Employees (Precise figure if available)

County	How has the sector developed in the last ten years?				
	Very slow (1)	Slow (2)	Moderate (3)	Fast (4)	Very Fast (5)

3) Sector profile (If applicable)

	Remarks
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Indicators and trends	Ask questions based on indicators relevant for the sector annex 1				
Energy consumption	What are the key sources of energy in your sector (name 3 max):				
	What sources of renewable energy are used in your sector?				
	How efficient is the sector in energy use?				
	1-not efficient	2-slightly efficient	3- efficient	4-quite efficient	5-highly efficient
Resource efficiency	What products/materials are recycled?				
Waste water	How is waste water treated and managed?				
Eco labelling	Identify products which are eco labeled in your sector?				

GE initiatives	Identify GE initiatives/investments in your sector (recently completed, ongoing and planned):

4) Social structure

NB/Population (Age structure and Gender) and Income structure from census report 2009

What is the ratio of male to female employees in your sector?	Male (%)			Female (%)	
Is the income from this sector the sole livelihood source for the employees household? (YES/NO)					
What is your opinion on the contribution of the sector to the household budget? (Very low if the household budget is supported by diverse income sources, and very high if it is the sole source of household budget support)	1	2	3	4	5
What is the percentage of people employed in the sector at (a) country level or (b) county level relative to the total workforce at (a) country level or (b) county level. (Answer either (a) or (b), not both).	0-20%	21-40%	41-60%	61-80%	81-100%

Key 1-very low, 2-low, 3-medium, 4-high, 5-very high

5) Environmental issues

Issues/challenges			
Which are the THREE major environmental issues in this sector? (List in order of priority)			
What are the most pressing societal problems related to each of the above environmental issues?			
What are the major impacts of these issues on your sector?			
What are the mitigation measures put in place to check the issues?			
What are the challenges faced in dealing with these environmental issues?			

6) Development plans [Questions are for Planner in Public and Private Sector]

What role(s) do you play in the County development planning process?		
What are the key steps in the planning process? [jump to next question if you are not a county planner]		
What are the timelines for the planning process? Jump to the next question if you are not a county planner]		
What is your experience in the County development planning process?		
What are the key objectives of your own institutions development plan?		
What are the key (a) methodologies (b) tools (c) platforms used to implement development plans?	a) methodologies	
	b) tools	
	c) platforms	
List the activities with a budget allocation in the 2015/2016 development plan for your sector?		

How many of the above activities that were budgeted for have been implemented?	
From your experience how is monitoring and evaluation done for implementation of the development plans	
A tool kit is a practical planning tool that can be used to facilitate the operationalisation of existing strategies at national or county scales. What would be your expectations for a tool kit and knowledge management platform?	

7) Implementation of GE initiatives

How do the GE initiatives that you listed in Section 3 above relate with your institutional mandate?		
For an on-going initiative [name] describe the challenges and opportunities	Challenges	Opportunities

8) Green cross-sectorial themes

NB: Explanatory statement can accompany the ranking.

Green Cross Sectorial Themes	Climate Change	Ecosystem management	Resource Efficiency	Chemicals and waste
How relevant are these issues to your planning unit (5 scale) [1 - not relevant; 2 - slightly relevant; 3 - relevant; 4 - quite relevant; 5 - Highly relevant]				
To what extent have they been incorporated in your planning process. [1 - not at all; 2 - a little; 3 - occasionally/sometimes; 4 - often; 5 - always]				
Are these themes incorporated in your measurement and evaluation system (YES/NO) If yes, how relevant are these themes to your measurement and evaluation system? [1 - not at all; 2 - a little; 3 - occasionally/sometimes; 4 - often; 5 - always]				

THANK YOU FOR YOUR COOPERATION

(2) State of Green Economy in Private Sector

University of Nairobi
Institute for Climate Change and Adaptation

Questionnaire for State of Green Economy in Private Sector – October 2015

Baseline Survey for ‘Operationalizing Green Economy Transition in Africa: Status of green economy initiatives in Mombasa and Nakuru Counties

This study seeks to assess the status of green economy (GE), including priority development sectors, environmental, social and economic issues, and assess the planning cycle at the county level

Your participation in this survey will be highly appreciated. Please rest assured that whatever information you provide will be treated in strict confidence.

I. Background Information

Name				
Acronym				
Address				
Tel				
e-mail				
Ownership e.g. limited co.				
Company size	SE	ME	LE	Others
a) Number of employees				
b) Annual revenue (USD)				
c) Company structure	To request for company organogram.			
Business Activities				
a) Activity				
b) Market				
c) Geographical coverage				
d) Exports				
Strategy and background information	To request for strategic plan and any other relevant information.			

II. Sector information

1. What is the level of importance of the enterprise for the sector in terms of (tick appropriately)

a) Market share as annual turnover/output in the county

V/high	High	Moderate	Low	V/ low
--------	------	----------	-----	--------

*Precise figure if available*_____

b) People employed.

Less than 100	101 - 1000	More than 1000
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2. Describe of the sector in terms of segmentation

a. Subsectors (e.g. food processing, packaging, value addition etc)

b. What are the enterprise consumer targets (e.g. households, institutions, industries etc)

c. Regions (spatial coverage)

d. Others, specify

e. What is the growth trend of the sector over the past ten years?

Positive	No change	Negative
----------	-----------	----------

*Precise figure if available*_____

3. What is the effect of the following factors on competitive dynamics?

<i>Please assess the following factors according to your view and experiences on competitive dynamics</i>	1	2	3	4	5
i) Margins (profit)					
ii) Technologies, specify up to three technologies that gives competitive advantage._____					
iii) Rivalry, specify upto three key rivals_____					

(1 = very low; 2 = low; 3 = moderate; 4 = high; 5 = very high)

4(a) What are your organization/business's strengths and weaknesses

<i>Please assess the following sentences according to your view and experiences</i>	1	2	3	4	5
i) Successful in achieving strategic goals.					
ii) Goals have been met or exceeded.					
iii) The current initiatives meet market requirements.					
iv) The links with other key organizations/agencies are effective.					
v) The financial position is secure.					
vi) There has been tremendous achievements over the last five years.					
vii) There is adequate capacity and expertise necessary to meet sectoral objectives.					
viii) The organization/business is regarded successful externally.					
ix) There is positive relationship with funders/supporters					
x) There is effective communication with external agencies.					
xi)The top regulatory agencies and government have the capacity/expertise to effectively monitor the sector					
xii) Policies are sufficient to drive growth for the next 20-30 years					
xiii) Availability of clear frameworks for environmental protection					
xiv) Corporate Social Responsibility is adequately addressed					
xv) Environmental friendly technologies have been adopted					

(1 = Strongly disagree; 2 = Disagree; 3 = Neither disagree nor agree; 4 = Agree; 5 = Strongly agree)

b) What are the opportunities and threats?

<i>Please assess the following sentences according to your view and experiences</i>	1	2	3	4	5
i) Policies introduced in past 5 years have brought opportunities					
ii) Information on current resource use efficiency practices is available					
iii) Technological trends in the sector have a positive influence					
iv) Are there incentives to promote green economy initiatives					
v) Natural factors such as weather and climatic changes gives us competitive edge					
vi) Our products have high environmental impact					
vii) New technology will result to new skills and competences to do the work					

(1 = Strongly disagree; 2 = Disagree; 3 = Neither disagree nor agree; 4 = Agree; 5 = Strongly agree)

5) Impact of the sector's activities on the environment.

Please highlight the impact appropriately

<i>Please select the number that best represents the impact on</i>	1	2	3	4	5
i) Air quality/ Air pollution					
ii) Water pollution					
iii) Extraction of water from river, lake and ocean					
iv) Hazardous substances					
v) Deforestation					
vi) Loss of biodiversity (damage to habitats)					
vii) Waste management challenges					
viii) Land degradation					
ix) Soil pollution					
x) Climate change (greenhouse effect)					
xi) Other impacts, specify _____					

(1=No impact; 2=Low impact; 3=Moderate impact; 4=Slight high; 5=high impact)

6. What is your primary source of energy?

6.1 How much do you spend on energy annually? _____

6.2 Which factors do you consider important when it comes to energy source.

<i>Please select the number that best represents the factor in terms of importance</i>	1	2	3	4	5

i)Reliability					
ii)Cost					
iii)Eco-friendly (low carbon footprint)					
iv)Availability					
v)Other factors, specify					

(1=not important; 2=slightly important; 3= important; 4=Quite important; 5=Very important)

7. Does your enterprise have environmental and Green Energy regulations? Yes () No ()

7.1. If yes, give at least three key regulations

7.2. What has been the experience in implementation of the regulations?

a) Success

b) Challenges

III. Strategy on Green Economy

Definition of Green Economy (GE) is one that results in improved human well-being and social equity, while significantly reducing environmental risks and ecological scarcities (UNEP).

8. What opportunities do you see in Green Economy for your organisation/business?

9. What do you see as the potential risks of GE in your organisation/business?

10. What strategies if any exist for embedding GE in the private sector?

11. What needs to be done to attract GE investments in terms of?

a) Policies and legislation

b) Technologies

c) Capacity building

d) Funding

e) Incentives

IV. What is your general comments on National and County planning

a. Experiences (Do they involve you)

b. List any agency who has contacted or collaborated with you in planning

c. What are your expectations and recommendations

THANK YOU FOR YOUR COOPERATION

ANNEX 4 - FILLED INDICATOR DATA SHEETS FOR NAKURU AND MOMBASA

Notes to the compilation of County data

1. The data has been completed to the extent possible; taking into consideration the suggested data sources and others also gleaned from the internet and external sources e.g. library texts. Any non-recommended sources used are from websites of authoritative county, national or global governance/oversight organisations or bodies, hence the data should be reliable barring any errors or inadvertent omissions.
2. However there are clear gaps that have been noted. It would appear that a significant amount of data relating to both Nakuru and Mombasa County are lacking from the internet. Most of the data available is at the national/country level, which in most cases is not analysed at lower county/city levels. It was therefore necessary to refine searches to target the counties, but clearly not much county detail has been uploaded online that specifically matches the indicators/metrics being researched. Sometimes the data is found, but are not aligned to the keywords and search phrases that are relevant, making them harder to find. In other online searches, Kenya national data was available but analysis by sector was lacking.
3. Data searches were in most cases run through Google search, including advanced searches. Exact phrase matches of the indicators/metrics being researched were first used, and details then searched for within the top ranked websites for each indicator. In most cases the initial phrase matches would provide no results, prompting further search variations and combinations to yield better results. Further detailed searches also needed to be made within PDF documents for key words like 'Nakuru' 'Mombasa' and 'Kenya Counties' in order to narrow down to Mombasa related data. All in all high search volumes were required to arrive at more recent and hence more reliable findings.
4. Green economy data, being a relatively new concept in Kenya, had the least available information. Documents available had a lot of valuable data for planning and implementation, but not quite adequate to match the required indicator data, or to allow derivation of the same. Data available did not provide much insight as to "where we are now", which would have probably provided the details sought after.
5. Data values, URLs, Document references and explanatory details, as well as dates of publication have all been included for ease of reference/validation. In some cases data needed to be derived, calculated or extrapolated from the data sources. In these cases, notes have been made in the "Details" column. Data calculations can be viewed by clicking on each data cell that qualifies. Wherever required, the current Central Bank of Kenya (CBK) exchange rate used is Kshs 102 to 1 USD.
6. Where County details are unavailable, national data has been indicated as a placeholder so that where possible and applicable, extrapolation can be considered to obtain an estimate of what Mombasa details should look like. The absence of county data and the references to national data have been marked in red text in the 'Details' column. Update: Further local websites provided will be reviewed to search for the data.
7. Indicators where data is missing have had the entire row highlighted in orange colour.

8. Several indicators appeared inconclusive, such as 'Energy productivity [tbd]'. Such were also not researched as they did not appear to be conclusively defined. These were also highlighted in orange colour.
9. Data related to Cities, Fisheries, Tourism, Transport and Waste is pending research and completion due to time constraints. Update: Further local websites provided will be reviewed to search for the data.
10. As far as possible, the only data located was the most recent that could be found. Data from official government bodies such as the Kenya National Bureau of Statistics (KNBS), was included in preference to other international research bodies whose data tended to be more outdated than locally available records.

Waste	Metrics/Indicators	Data	Data Sources	Details	Date of Data	Data Availability	Additional Suggested References
I. General sector indicators	GDP waste sector [USD]						None
	GDP growth waste sector [%, average 5 years]						None
	Contribution of waste sector to GDP [%]						None
	Total waste generation [tonnes]	850	http://www.slideshare.net/JoshuaPalfreman/mapping-out-waste-characteristics-in-mombasa-kenya	MAPPING OUT WASTE CHARACTERISTICS IN MOMBASA, KENYA J. PALFREMAN & M. CLARK	October-15		None
		875	Unpublished data from Mombasa County Government		January-16		

	Average waste per capita [kg]	0.93	http://www.slideshare.net/JoshuaPalfreman/mapping-out-waste-characteristics-in-mombasa-kenya	MAPPING OUT WASTE CHARACTERISTICS IN MOMBASA, KENYA J. PALFREMAN & M. CLARK	October-15		None
II. Environmental issues and related indicators	CO ₂ emission waste sector [tonne/year]						None
	Methane production from liquid waste [tonnes]						None
	Waste recycling rate [%]	23%	http://www.slideshare.net/JoshuaPalfreman/mapping-out-waste-characteristics-in-mombasa-kenya	MAPPING OUT WASTE CHARACTERISTICS IN MOMBASA, KENYA J. PALFREMAN & M. CLARK	October-15		None
	Resource efficiency or productivity (GDP/natural resources) [factor]						None
	Waste landfilled [tonnes]	460	http://www.slideshare.net/JoshuaPalfreman/mapping-out-waste-characteristics-in-mombasa-kenya	MAPPING OUT WASTE CHARACTERISTICS IN MOMBASA, KENYA J. PALFREMAN & M. CLARK	October-15		None
	Energy generation from waste [TJ]	16	http://www.slideshare.net/JoshuaPalfreman/mapping-out-waste-characteristics-in-mombasa-kenya	MAPPING OUT WASTE CHARACTERISTICS IN MOMBASA, KENYA J. PALFREMAN & M. CLARK	October-15		None
III. Green economy	Green investments in waste sector [USD]						None

policy interventions and related indicators	Green Economy initiatives in waste sector [number]						None
	Waste GE subsidies [USD]						None
	Proportion of renewable energy use in waste sector [%]						None
	Number of private sector partners in GE waste sector [partners]						None
	Replication of cases/best practices waste sector [replications]						None
IV. Well-being and equity indicator	Employment in waste sector [people]						None
	Employment growth in waste sector [%, average 5 years]						None
	Waste employment multiplier [factor]						None
V. Data sources							

Tourism	Metrics/Indicators	Data	Data Sources	Details	Date of Data	Data Availability	Additional Suggested References
I. General sector indicators	GDP tourism sector [USD]						None
	GDP growth tourism sector [%], average 5 years]						None
	Contribution of tourism sector to GDP [%]	11.60%	GIZ PROJECT,OCT 2015-HHS Mombasa 1	GIZ PROJECT,OCT 2015-HHS Mombasa 1 Page 2	October 2015		None
	Guest nights [nights]	2,756,800	http://www.knbs.or.ke/index.php?option=com_content&view=article&id=335:statistical-abstract-2015&catid=82:news&Itemid=593	Coastal data.STATISTICAL ABSTRACT 2015 from the KENYA NATIONAL BUREAU OF STATISTICS. Mombasa County data not specified. Table 19(d), Page 43	2014		None
	Occupancy rate hotel beds [%]	64%	http://www.knbs.or.ke/index.php?option=com_content&view=article&id=335:statistical-abstract-2015&catid=82:news&Itemid=593	MOMBASA COUNTY INTEGRATED DEVELOPMENT PLAN 2013 Page x	2013		None

	Number of tourism enterprises [number]	201	http://www.knbs.or.ke/index.php?option=com_content&view=article&id=335:statistical-abstract-2015&catid=82:news&Itemid=593	MOMBASA COUNTY INTEGRATED DEVELOPMENT PLAN 2013 Page x	2013		None
II. Environmental issues and related indicators	CO ₂ emission tourism sector [tonne/year]						None
	CO ₂ emissions tourism sector per guest night [kg]						None
	Energy consumption tourism sector [MJ/year]						None
	Energy consumption tourism sector per guest night [KJ]						None
	Proportion of renewable energy tourism sector [%]						None
	Water consumption tourism sector [m ³ /year]						None

	Water consumption tourism sector per guest night [m ³]						None
	Proportion of wastewater treatment tourism sector [%]						None
	Waste generation tourism sector [tonnes/year]						None
	Waste recycling and reuse tourism sector [%]						None
	Effective management of built and cultural heritage tourism sector [%]						None
	Consideration loss of biological diversity tourism sector [tbd]						None
III. Green economy	Green investments in tourism sector [USD]						None

policy interventions and related indicators	Green Economy initiatives in tourism sector [number]						None
	Proportion of eco labelled tourism enterprises [% certified]						None
	Tourism GE subsidies [USD]						None
	Destination governance, planning and development [%]						None
	Number of private sector partners in GE tourism sector [partners]						None
	Replication of cases/best practices tourism sector [replications]						None
IV. Well-being and equity indicator	Employment in tourism sector [people]	37,575	GIZ PROJECT,OCT 2015-HHS Mombasa 1	GIZ PROJECT,OCT 2015-HHS Mombasa 1 Page 2 Derived from 4% of population total from 2009 Census of 939,370	October 2015		None

	Employment growth in tourism sector [%, average 5 years]						None
	Tourism employment multiplier [factor]						None
V. Data sources							

Cities	Metrics/Indicators	Data	Data Sources	Details	Date of Data	Data Availability	Additional Suggested References
I. General sector indicators	GDP cities sector [USD]	52,526,193,137.25	http://www.knbs.or.ke/index.php?option=com_phocadownload&view=category&id=20&Itemid=595	National data.FACTS AND FIGURES 2015 from the KENYA NATIONAL BUREAU OF STATISTICS. Mombasa County data not specified (Available data derived from Page 1, Table 1)	2014		None
	GDP growth cities sector [%, average 5 years]	5.50	http://www.knbs.or.ke/index.php?option=com_phocadownload&view=category&id=20&Itemid=595	National data.FACTS AND FIGURES 2015 from the KENYA NATIONAL BUREAU OF STATISTICS. Mombasa County data not specified (Available data derived from Page 5, Table 5)	2014		None
	Contribution of cities sector to GDP [%]						None
II. Environmental issues and	Air pollution [ppm]						None
	Water pollution [BOD mg/L]						None

related indicators	Urbanised area growth [%, average 5 years]						None
	Percentage of trees in the city in relation to city area [%]						None
	Percentage of each mode of transportation , i.e. private, public, bicycle, pedestrian [%]						None
	Generation and disposal of municipal solid waste [tonnes]	460	http://www.slideshare.net/JoshuaPalfreman/mapping-out-waste-characteristics-in-mombasa-kenya	Mapping Out Waste Characteristics in Mombasa Kenya	Oct-15		None
	Recycling rates [%]	5%	http://www.slideshare.net/JoshuaPalfreman/mapping-out-waste-characteristics-in-mombasa-kenya	MAPPING OUT WASTE CHARACTERISTICS IN MOMBASA, KENYA J. PALFREMAN & M. CLARK Section 4 Paragraph 1	Oct-15		None

Energy-use efficiency cities sector [tbd]						None
Percentage of city population with access to water-born or alternative (and effective) sanitary [%]						None
Percentage of green space (public parks) coverage in relation to city area and/or population size [%]						None
Population connected to urban wastewater treatment plants [%]						None

III. Green economy policy interventions and related indicators	Green investments in urbanisation [USD]						None
	Green Economy initiatives in urbanisation [number]						None
	Percentage of green jobs in the municipal economy [%]						None
	Subsidies green cities [USD]						None
	Number of private sector partners in GE cities sector [partners]						None
	Replication of cases/best practices cities sector [replications]						None

IV. Well-being and equity indicator	Number of people reported to have been affected by diseases arising from drinking/bathing water over a defined period per capita [number/capita]						None
	Population exposure to harmful levels of air pollution [%]						None
	Access to basic services in urban areas [%]						None
	Share of women in city councils [%]	<20%	http://www.mombasa.go.ke/downloads/1st%20CIDP%202013-2017%20Mombasa%20County.pdf	MOMBASA COUNTY INTEGRATED DEVELOPMENT PLAN 2013 Page ix	2013		None

V. Data sources							
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Transport	Metrics/ Indicators	Data	Data Sources	Details	Date of Data	Data Availabilit y	Additional Suggested Reference s
I. General sector indicators	GDP transport sector [USD]	4,356,196,078	http://www.knbs.or.ke/index.php?option=com_content&view=article&id=335:statistical-abstract-2015&catid=82:news&Itemid=593	National data. STATISTICAL ABSTRACT 2015 from the KENYA NATIONAL BUREAU OF STATISTICS. Mombasa County data not specified. Table 26, Page 53	2014		None
	GDP growth transport sector [%, average 5 years]	6.40%	www.opendata.go.ke/GDP-inflation/Growth-Rates-GDP-of-various-industries	(Data from 2009 to 2013 GOK, 2015) First Annual Progress Report On Implementation of the Second Medium Term Plan (2013-2017) of the Kenya Vision 2030 Mombasa County data not specified.	2013		None
	Contribution of transport sector to GDP [%]	8.3% to 10% (See notes showing different sources)	http://www.knbs.or.ke/index.php?option=com_content&view=article&id=335:statistical-abstract-2015&catid=82:news&Itemid=593	1. National data. STATISTICAL ABSTRACT 2015 from the KENYA NATIONAL BUREAU OF STATISTICS. Mombasa County data not specified. Table 27, Page 54 2. 10 per cent of GDP (UNEP, 2014) Green Economy Assessment	2014		None

Population of cars/trucks [vehicles]	2,210,907	http://www.knbs.or.ke/index.php?option=com_content&view=article&id=335:statistical-abstract-2015&catid=82:news&Itemid=593	National data.STATISTICAL ABSTRACT 2015 from the KENYA NATIONAL BUREAU OF STATISTICS. Mombasa County data not specified.	2014		None
Length of road infrastructure [km]	475	http://www.knbs.or.ke/index.php?option=com_content&view=article&id=335:statistical-abstract-2015&catid=82:news&Itemid=593	MOMBASA COUNTY INTEGRATED DEVELOPMENT PLAN 2013 Derived from Section 2.4.2, Page 9 (257.17 + 127+ 91.29) NB: National Data shows 160, 886 with only 7% paved(ADB (2014); 178,000 km	2013		None
Investment in road infrastructure [USD]	2.89 Trillion	Link not provided	The cost of infrastructure projects under the MTP II is estimated at Ksh 245.63 trillion (USD 2.89 trillion) (ADB, 2014).ADB (2014). Kenya Country Strategic Paper 2014-2018, East Africa Resource Centre. Mombasa County data not specified.	2014		None
Road freight volume [tonnes]						None

	Rail freight volume [tonnes]	TBC	www.opendata.go.ke	National Data. The MTP II target is to increase the railways capacity to handle 50% of freight cargo (25 Million Tonnes) from the Mombasa Port Government of Kenya (GOK, 2015). See Also First Annual Progress Report On Implementation of the Second Medium Term Plan (2013-2017) of the Kenya Vision 2030 Mombasa County data not specified.			None
	Total person kilometres travelled [km]	3,845,000	http://www.knbs.or.ke/index.php?option=com_content&view=article&id=335:statistical-abstract-2015&catid=82:news&Itemid=593	National data.STATISTICAL ABSTRACT 2015 from the KENYA NATIONAL BUREAU OF STATISTICS. Mombasa County data not specified. Table 123, Page 165	2014		None
	Total public transport boarding per capita [factor]						None
II. Environmental issues and related indicators	CO ₂ emission transport sector [tonne/year]						None
	CO ₂ emissions transport						None

sector per capita [kg]						
Energy consumption transport sector [MJ/year]	TBC	Link not provided	National Data. GHG emissions from transport are projected to grow significantly from 6 MtCO ₂ e in 2010 to almost 18 MtCO ₂ e in 2030 UNEP (2014). Green Economy Assessment Report – Kenya Mombasa County data not specified.	2014		None
Energy consumption transport sector per capita [KJ]						None
Annual NO ₂ concentrations in urban areas [ppm]						None

	Cost of destruction of roads, bridges, railway lines due to temperature and rainfall variability [USD]	777,000,000	Link not provided	National Data. The 1997-98 El Niño floods are estimated to have caused damage at least equivalent to 11 per cent of GDP, including KSh62 billion (USD 777 million) in damage to transport infrastructure. Government of Kenya (GOK). (2013). National Climate Change Action Plan 2013-2017 Mombasa County data not specified.	2013		None
III. Green economy policy interventions and related indicators	Green investments in transport sector [USD]	TBC	http://www.greengrowthknowledge.org/sites/default/files/downloads/resource/ILO_Green_Jobs_Reality.pdf	National Data. ILO (2012) Assessing Green Jobs Potential in Developing Countries: A Practitioner's Guide. Mombasa County data not specified.	2012		None
	Green Economy initiatives in transport sector [number]	3 (proposed initiatives)	Link not provided	National Data. UNEP (2014). Green Economy Assessment Report – Kenya Mombasa County data not specified.	2014		None
	Transport GE subsidies [USD]						None

Proportion of renewable energy use in transport sector [%]							None
Efficiency of transport sector [tbd]							None
Number of private sector partners in GE transport sector [partners]							None

	Replication of cases/best practices transport sector [replications]	TBC	Link not provided	National Data. Employment of more effective traffic management technologies, especially traffic control lights to reduce vehicle travel time mainly in urban areas -A total of 3,405 streetlights and floodlights were installed in the Metropolitan area; Up-scaling mass transport systems to reduce the use of personal vehicles - Construction of the Syokimau train station was completed and commuter services commenced, The government has taken steps to reduce emissions of certain harmful pollutants. For example, in 2006, Kenya eliminated lead in petrol and reduced sulphur in imported refined diesel, Recent regulations have stipulated that vehicles imported into Kenya are required to be no older than eight years from the year of manufacture. Government of Kenya (GOK). (2015). First Annual Progress Report On Implementation of the Second Medium Term Plan (2013-2017) of the Kenya Vision 2030	2015		None
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				Mombasa County data not specified.		
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IV. Well-being and equity indicator	Employment in transport sector [people]	79,721	http://www.knbs.or.ke/index.php?option=com_content&view=article&id=335:statistical-abstract-2015&catid=82:news&Itemid=593	National data. STATISTICAL ABSTRACT 2015 from the KENYA NATIONAL BUREAU OF STATISTICS. Mombasa County data not specified. Table 56, Page 97	2014		None
	Employment growth in transport sector [%, average 5 years]						None
	Transport employment multiplier [factor]						None
	Road traffic fatalities per 100 000 inhabitants (transport related) [fatalities]	7.53	http://www.knbs.or.ke/index.php?option=com_content&view=article&id=335:statistical-abstract-2015&catid=82:news&Itemid=593	National data. STATISTICAL ABSTRACT 2015 from the KENYA NATIONAL BUREAU OF STATISTICS. Mombasa County data not specified. Table 129, Page 168	2014		None
V. Data sources							

Fishery	Metrics/ Indicators	Data	Data Sources	Details	Date of Data	Data Availability	Additional Suggested References
I. General sector indicators	GDP fishery sector [USD]	395,950,980	http://www.knbs.or.ke/index.php?option=com_content&view=article&id=335:statistical-abstract-2015&catid=82:news&Itemid=593	National data.STATISTICAL ABSTRACT 2015 from the KENYA NATIONAL BUREAU OF STATISTICS. Mombasa County data not specified Table 26, Page 53	2014		None
	GDP growth fishery sector [%, average 5 years]						None
	Contribution of fishery sector to GDP [%]	0.5%	GIZ PROJECT,OCT 2015-HHS Mombasa 1	GIZ PROJECT,OCT 2015-HHS Mombasa 1	October 2015		None
	Fish catch landed [tonnes]	1,214	http://www.knbs.or.ke/index.php?option=com_content&view=article&id=335:statistical-abstract-2015&catid=82:news&Itemid=593	National data.STATISTICAL ABSTRACT 2015 from the KENYA NATIONAL BUREAU OF STATISTICS. Mombasa County data not specified Table 74, Page 113	2014		None

Net exports of fish [USD]	25,185.16	http://www.kilimo.go.ke/fisheries/wp-content/uploads/2015/05/Annual-Fisheries-Statistical-Bulletin-2013.pdf	FISHERIES ANNUAL STATISTICAL BULLETIN 2013 Table 20 Page 54	2013		None
Number of fishers/vessels [number]	X/356	http://www.kilimo.go.ke/fisheries/wp-content/uploads/2015/05/Marine-Artisanal-Fisheries-Framework-Survey-2012-Report.pdf	MINISTRY AGRICULTURE LIVESTOCK AND FISHERIES STATE DEPARTMENT OF FISHERIES MARINE ARTISANAL FISHERIES FRAME SURVEY 2014 REPORT Number of fishers to be determined	2014		None
Number of fish processing enterprises [number]	9	http://www.google.com/url?sa=t&rct=j&q=&esrc=s&source=web&cd=1&ved=0ahUKEwi9od_W957JAhVE1hQKHbhLCFsQEgghMAA&url=http%3A%2F%2Fwww.fao.org%2Ffileadmin%2Fuser_upload%2Ffisheries%2Fdocs%2FBackground_Report_Kenya.doc&usg=AFQjCNGluToDdQ2b0Y6D-mGwq4VhOcDC-g	National data. STATISTICAL ABSTRACT 2015 from the KENYA NATIONAL BUREAU OF STATISTICS. Mombasa County data not specified (Page 123)	2009		None

II. Environmental issues and related indicators	CO ₂ emission fishery sector [tonne/year]		-				None
	CO ₂ emissions fishery sector per tonne of fish catch [kg]						None
	Energy consumption fishery sector [MJ/year]						None
	Energy consumption fishery sector per tonne of fish catch [KJ/tonne]						None
	Fraction of fish stocks overexploited and collapsed by exclusive economic zone [fraction]						None
	Trawling catch per exclusive economic zone [tonne/km ²]						None

III. Green economy policy interventions and related indicators	Green investments in fishery sector [USD]						None
	Green Economy initiatives in fishery sector [number]						None
	Coverage of marine conservation areas [km ²]						None
	Areas of aquaculture ecosystems under sustainable management [km ²]						None
	Fishery GE subsidies [USD]						None
	Number of private sector partners in GE fishery sector [partners]						None
	Replication of cases/best practices fishery sector [replications]						None

IV. Well-being and equity indicator	Employment in fishery sector [people]	583	http://www.knbs.or.ke/index.php?option=com_content&view=article&id=335:statistical-abstract-2015&catid=82:news&Itemid=593	National data.STATISTICAL ABSTRACT 2015 from the KENYA NATIONAL BUREAU OF STATISTICS. Mombasa County data not specified Table 56, Page 95	2014		None
	Employment growth in fishery sector [%, average 5 years]						None
	Fishery employment multiplier [factor]						None
	Number of small-scale fishers [number]						None
V. Data sources	Sea Around Us Project, University of British Columbia Fisheries Centre						

Energy	Metrics/ Indicators	Data	Data Sources	Details	Date of Data	Data Availabilit y	Additional Suggested Reference s
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I. General sector indicators	GDP energy sector [USD]	539,303,92 2	http://www.google.com/url?sa=t&rct=j&q=&esrc=s&source=web&cd=1&sqi=2&ved=0CCUQFjAAahUKEwilqbf2sZTJAhVJDxoKHU36AwI&url=http%3A%2F%2Fwww.knbs.or.ke%2Findex.php%3Foption%3Dcom_phocadownload%26view%3Dcategory%26download%3D720%3Aeconomic-survey-2015%26id%3D107%3Aeconomic-survey-publications%26Itemid%3D1181&usg=AFQjCNEuEaxKBNugwOguLZarE3YBZE7Q9w&bv m=bv.107467506,d.d24	National data.ECONOMIC SURVEY 2015 from the KENYA NATIONAL BUREAU OF STATISTICS. Mombasa County data not specified (Available data derived from Page 22)	2014		None
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	GDP growth energy sector [%, average 5 years]	9.40%	http://www.google.com/url?sa=t&rct=j&q=&esrc=s&source=web&cd=1&sqi=2&ved=0CCUQFjAAahUKEwilqbf2sZTJA hVJDxoKHU36AwI&url=http%3A%2F%2Fwww.knbs.or.ke%2Findex.php%3Foption%3Dcom_phocadownload%26view%3Dcategory%26download%3D720%3Aeconomic-survey-2015%26id%3D107%3Aeconomic-survey-publications%26Itemid%3D1181&usg=AFQjCNEuEaxKBNugwOguLZarE3YBZE7Q9w&bv m=bv.107467506,d.d24	National data.ECONOMIC SURVEY 2015 from the KENYA NATIONAL BUREAU OF STATISTICS. Mombasa County data not specified (Derived from Page 25)	2014		None
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	Contribution of energy sector to GDP [%]	0.1%	http://www.google.com/url?sa=t&rct=j&q=&esrc=s&source=web&cd=1&sqi=2&ved=0C CUQFjAAahUKEwilqbf2sZTJA hVJDxoKHU36AwI&url=http%3A%2F%2Fwww.knbs.or.ke%2Findex.php%3Foption%3Dcom_phocadownload%26view%3Dcategory%26download%3D720%3Aeconomic-survey-2015%26id%3D107%3Aeconomic-survey-publications%26Itemid%3D1181&usg=AFQjCNEuEaxKBNugwOguLZarE3YBZE7Q9w&bv m=bv.107467506,d.d24	National data.ECONOMIC SURVEY 2015 from the KENYA NATIONAL BUREAU OF STATISTICS. Mombasa County data not specified (Available data derived from Page 23)	2014		FROM COUNTY
	Total Primary Energy Supply [PJ]	18,215.99	http://www.cofek.co.ke/ERCStudy_ExecSummary_02082010.pdf	A COMPREHENSIVE STUDY AND ANALYSIS ON ENERGY CONSUMPTION PATTERNS IN KENYA A SYNOPSIS OF THE DRAFT FINAL REPORT Mombasa County data not specified (Available data derived from Section 4.2, Paragraph 1)	2009		None

	Energy self-sufficiency [%]						None
	Electricity production [GWh/year]	0.01	http://www.knbs.or.ke/index.php?option=com_content&view=article&id=335:statistical-abstract-2015&catid=82:news&Itemid=593	National data.STATISTICAL ABSTRACT 2015 from the KENYA NATIONAL BUREAU OF STATISTICS. Mombasa County data not specified (Derived from Page 138)	2014		None
	Electricity capacity [MW]	2,089.10	http://www.knbs.or.ke/index.php?option=com_content&view=article&id=335:statistical-abstract-2015&catid=82:news&Itemid=593	National data.STATISTICAL ABSTRACT 2015 from the KENYA NATIONAL BUREAU OF STATISTICS. Mombasa County data not specified (Page 123)	2014		None
	Petroleum consumption [tonne/year]	2,631,000	https://www.iea.org/statistics/statisticssearch/report/?year=2013&country=KENYA&product=Oil	National data.INTERNATIONAL ENERGY AGENCY. Mombasa County data not specified. Data derived from website	2013		None
II. Environmental issues and	CO ₂ emission energy sector [tonne/year]	11,700,000	https://www.iea.org/statistics/statisticssearch/report/?country=KENYA&product=indicators&year=2013	National data.INTERNATIONAL ENERGY AGENCY. Mombasa County data not specified. Data derived from website	2013		None

related indicators	Energy consumption buildings sector [MJ/year]	41,147,174	https://www.iea.org/statistics/statisticssearch/report/?country=KENYA&product=indicators&year=2013	National data.INTERNATIONAL ENERGY AGENCY. Mombasa County data not specified. Data derived from website	2013		None
	Electricity production from renewable sources [GWh/year]	9,596	http://erc.go.ke/images/docs/Electricity_Subsector_Medium_Term_Plan_2012-2016.pdf	Electricity Sub-Sector Medium Term Plan (2012-2016) National data. Mombasa County data not specified. Table 10, Page 38	2014		None
	Share of renewable energy buildings sector [%]	63%	http://erc.go.ke/images/docs/Electricity_Subsector_Medium_Term_Plan_2012-2016.pdf	Electricity Sub-Sector Medium Term Plan (2012-2016) National data. Mombasa County data not specified. Table 10, Page 38	2014		None
	Buildings retrofitting [% of portfolio]	6,080	http://erc.go.ke/images/docs/Electricity_Subsector_Medium_Term_Plan_2012-2016.pdf	Electricity Sub-Sector Medium Term Plan (2012-2016) National data. Mombasa County data not specified. Table 10, Page 38	2014		None
	Penetration of high efficiency light bulbs [%]	14%	http://climatestrategies.org/wp-content/uploads/2015/01/Climate-Technology-and-Development-Case-Study-	Climate Technology & Development Case study: Compact Fluorescent Lamps (CFLs) Section 3.2, Paragraph 2 National data. Mombasa County data not specified	2012		None

		Compact-Fluorescent-Lamps-CFLs-Rob-Byrne-final.pdf				
For residential, commercial, and institutional buildings						KENYA ENERGY AND ELECTRICITY BOARD REGULATION
Share of population with access to electricity [%]	4.44%	http://erc.go.ke/images/docs/Electricity_Subsector_Medium_Term_Plan_2012-2016.pdf	Electricity Sub-Sector Medium Term Plan (2012-2016) National data. Mombasa County data not specified. Page 12 See also Economic Survey 2015 for a listing of 2011 population figures.	2014		None
Share of population using solid fuels [%]						None

	Electricity use per capita [KWh]	170	https://www.iea.org/statistics/statisticssearch/report/?country=KENYA&product=indicators&year=2013	National data.INTERNATIONAL ENERGY AGENCY. Mombasa County data not specified. Data derived from website.	2013		None
III. Green economy policy interventions and related indicators	Green investments in energy sector [USD]						None
	Green Economy initiatives in energy sector [number]						None
	Energy subsidies in renewable energies [USD]						None
	Number of private sector partners in GE energy sector [partners]						None
	Replication of cases/best practices energy sector [replications]						None

IV. Well-being and equity indicator	Employment in energy sector [people]	15,300	http://www.google.com/url?sa=t&rct=j&q=&esrc=s&source=web&cd=1&sqi=2&ved=0C CUQFjAAahUKEwIlqbf2sZTJA hVJDxoKHU36AwI&url=http%3A%2F%2Fwww.knbs.or.ke%2Findex.php%3Foption%3Dcom_phocadownload%26view%3Dcategory%26download%3D720%3Aeconomic-survey-2015%26id%3D107%3Aeconomic-survey-publications%26Itemid%3D1181&usg=AFQjCNEuEaxKBNugwOguLZarE3YBZE7Q9w&bv m=bv.107467506,d.d24	National data.ECONOMIC SURVEY 2015 from the KENYA NATIONAL BUREAU OF STATISTICS. Mombasa County data not specified (Derived from Page 25)	2014		None
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	Employment growth in energy sector [% , average 5 years]	3.90%	http://www.google.com/url?sa=t&rct=j&q=&esrc=s&source=web&cd=1&sqi=2&ved=0C CUQFjAAahUKEwilqbf2sZTJA hVJDxoKHU36AwI&url=http%3A%2F%2Fwww.knbs.or.ke%2Findex.php%3Foption%3Dom_phocadownload%26view%3Dcategory%26download%3D720%3Aeconomic-survey-2015%26id%3D107%3Aeconomic-survey-publications%26Itemid%3D1181&usg=AFQjCNEuEaxKBNugwOguLZarE3YBZE7Q9w&bv m=bv.107467506,d.d24	National data.ECONOMIC SURVEY 2015 from the KENYA NATIONAL BUREAU OF STATISTICS. Mombasa County data not specified (Derived from Page 80)	2014		None
	Energy employment multiplier [factor]						None
V. Data sources	www.irena.org/REmaps/africamap.aspx						
	International Energy Agency (IEA), CO2 Emissions from						

	Fuel Combustion						
	data.worldbank .org/indicator/ EG.ELC.ACCS .ZS						

ANNEX 5 - THE DEVELOPMENT PLANNING CYCLE UNDERTAKEN BY MOMBASA COUNTY PLANNERS: RESPONDENT ANALYSIS BY SECTOR

MOMBASA COUNTY

Table A5.2.1: Aspects of the development planning cycle undertaken by the County planners with respect to the **waste sector**.

Development Planning	Status/Remarks
County Government (role(s) in county development planning process)	- There are plans to improve the sanitation facilities, given that only a small part of the population is served by conventional sewerage system. - Greening the County - tree and grass planting. - Civic education on good waste disposal methods - Working with developers to promote waste segregation - Enacting waste management and environmental policies/regulations for the County by-laws. - Consultations and active engagement - Budgetary planning - The environment department is central to decision making (all development projects have environmental considerations).
Key steps in planning process	- Through the top management (M.D). - Submission at start of financial skills - All projects are required to have an EIA - The city is in the process of developing a SEA.
Timelines for planning process	- Dependent on project magnitude - Adhere to the master plan where there are short term, medium and long term schedules. - 1 year (Annual Cycle) to 25 years (SEA) span
Experience in the County development planning process.	- Departmental Development Planning
Key objectives of your own institutions development plan.	- Making sure sewage is collected, transported and disposed of. - Short term: include improving infrastructure. - Long Term: development of the disposal site.

	- To create a clean conducive environment for investment growth and habitation by the country residents.	
Key (a) Methodologies (b) Tools (c) Platforms used to implement development plans.	a) methodologies	- Master plans - Strategic plans - Informal procedures
	b) tools	- SWOT - Cost/ benefit analysis - Rapid results initiatives - Performance contracts - Planning on a need basis - Monitoring committees (sectoral). - EIA - SEA
	c) platforms	- Monitoring committees (sectoral) - Public participation - Sectoral consultations (departmental) - Public Private partnerships - Community/Stakeholder involvement.
Activities with a budget allocation in the 2015/2016 development plan for your sector.	- Operations - Buying truck, bulldozer, supervision cars - Beautification (greening) - Reviving island water plant (Kizingo)	
Above activities budgeted for and implemented	- Implementation ongoing	
From your experience, how is monitoring and evaluation done for implementation of the development plans	- Poor implementation - This is an inspectorate department: does self –audits and NEMA audits (for discharge of sewage).	

Your expectations for a toolkit and knowledge management platform	- Marine environment to be considered - pollution as a result of effluents discharge into it - Makes it easier to interrogate issues and convince decision makers/strategic organisations in the planning process - Mainstream environmental issues identified in the CIDP (County Integrated Development Plan).
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Table A5.2.2: Aspects of the development planning cycle undertaken by the County planners with respect to the **tourism sector**.

Development Planning	Status/Remarks	
County Government (role(s) in county development planning process)	- Employees provide information that are relevant to county planning.	
Key steps in planning process	- Not stated	
Timelines for planning process	- Not stated	
Your experience in the County development planning process.	- Environment Department	
Key objectives of your own institutions development plan.	- To achieve the standards expected internationally and locally.	
Key (a) Methodologies (b) Tools (c) Platforms used to implement development plans.	a) methodologies	- Collect real data from institutions
	b) tools	- Audits and Issuing of permits to those who comply
	c) platforms	- Seminar forums and through umbrella authorities.
Activities with a budget allocation in the 2015/2016 development plan for your sector.	- Construction of Green Building - Staff Training and Development	
Above activities budgeted for and implemented	- Staff Training and Development has been implemented	
From your experience, how is monitoring and evaluation done for implementation of the development plans	- Putting up water and power meters to give data on real capacity consumption through inspections.	
Your expectations for a toolkit and knowledge management platform	- Expectation is to analyse current status and project goals to be achieved by a certain period using available and borrowed resources.	

Table A5.2.3: Aspects of the development planning cycle undertaken by the County planners with respect to the **cities and built environment sector**.

Development Planning	Status/Remarks	
County Government (role(s) in county development planning process)	- Consultative process - Representation in county board -	
Key steps in planning process	- Engage locals	
Timelines for planning process	- None - 12 months	
Your experience in the County development planning process.	- Positive	
Key objectives of your own institutions development plan.	- Regulation - Development: target to raise sum in a year - Maintenance and equipment - review equipment - Engage locals	
Key (a) Methodologies (b) Tools (c) Platforms used to implement development plans.	a) methodologies	- Strategic plans - Performance contracts - Collect and archive data -
	b) tools	- EIA - Equipment utilisation
	c) platforms	- Public-private partnerships - Workshops - All activities tracked (monthly) for all equipment and consumption - formula used to calculate cost effectiveness.
Activities with a budget allocation in the 2015/2016 development plan for your sector.	- Infrastructure increased in water service areas - Water project in marginalised areas - Rehabilitating Baricho treatment works - Expanding sewage works -	
Above activities budgeted for and implemented	- Ongoing implementation process - Maintenance of equipment	

From your experience, how is monitoring and evaluation done for implementation of the development plans	- Building inspections - County - Strategic plan - departmental objectives
Your expectations for a toolkit and knowledge management platform	- GIS based data base.

Table A5.2.4: Aspects of the development planning cycle undertaken by the County planners with respect to the **transport and infrastructure sector**.

Development Planning	Status/Remarks	
County Government (role(s) in county development planning process)	- Share plans	
Key steps in planning process	- Community/stakeholder sensitisation and engagement - Involve consultants - Drafting -	
Timelines for planning process	- 5 years	
Your experience in the County development planning process.	- Community Involvement - planning - Participative	
Key objectives of your own institutions development plan.	- Strategic planning - Takes into account all challenges experienced - Risk management -	
Key (a) Methodologies (b) Tools (c) Platforms used to implement development plans.	a) methodologies	- Stakeholders participation - Enterprise risk management system
	b) tools	- Laboratory testing of materials - Performance contracts - Monitoring and evaluation
	c) platforms	- Forums for different stakeholders
Activities with a budget allocation in the 2015/2016 development plan for your sector.	- Changamwe road, Roads planned for - concrete block payments/ access roads - in Changamwe, Jomvu, Nyali, Kisauni, Kisauni, and Likoni constituency. - Water supply system - a consultant engaged by World Bank	

	- Power supply and distribution consultant - to look on energy saving and green options - Pipeline from Mombasa to Nairobi - started August last year - Social responsibility - dispensary at Makindu
Above activities budgeted for and implemented	- Not yet - Some solar panels already installed in Lamu.
From your experience, how is monitoring and evaluation done for implementation of the development plans	- Monitoring and evaluation committee, community evaluation. - Not really taken into consideration for GE - Monitoring and evaluation reports -
Your expectations for a toolkit and knowledge management platform	- None stated

Table A5.2.5: Aspects of the development planning cycle undertaken by the County planners with respect to the **fisheries sector**.

Respondents Views (Planners)	
Development Planning	Status/Remarks
County Government (role(s) in county development planning process)	- Committees - Planning - Water - Energy - Tourism
Key steps in planning process (NB: county planners only)	- N/A - Stakeholder consultation - Bottom up approach
Timelines for planning process (NB: county planners only)	- N/A
Your experience in the County development planning process.	- N/A
Key objectives of your own institutions development plan.	- N/A - Increase tree cover - Enhance revenue through sustainable forestry - Improve institutional capacity - Growth of the sector - Develop industry output and capacity

Key (a) Methodologies (b) Tools (c) Platforms used to implement development plans.	a) methodologies	- Training and capacity building - Having yearly target - Public/private participation - Stakeholder involvement - Expert consultation -
	b) tools	- pamphlets - Visiting communities to give technical advise - Media /Radio for communication
	c) platforms	- Field days - Radio/media - Sensitisation through public fora
Activities with a budget allocation in the 2015/2016 development plan for your sector.	- Rehabilitation of catchment areas - Protection of bamboos seeding production - Restoration of degraded forests 38Ha. - Protection of tree and bamboo growing 47ha. -	
Above activities budgeted for and implemented	- Seeding production by farmers	
From your experience, how is monitoring and evaluation done for implementation of the development plans	- Field visits to assess success rates - Protection of seeds and replanting of dead seeds - Monitoring and evaluation for projects quarterly, half-yearly and annually, performance contract, ISO certified, global benchmarking.	
Your expectations for a toolkit and knowledge management platform	- Effective communication of knowledge. - Capacity Building at all levels	

Table A5.2.6: Aspects of the development planning cycle undertaken by the County planners with respect to the **energy sector**.

Development Planning	Status/Remarks	
County Government (role(s) in county development planning process)	- Not stated	
Key steps in planning process	- Not stated	
Timelines for planning process	- Not stated	
Your experience in the County development planning process.	- Not stated	
Key objectives of your own institutions development plan.	- Sustainability - Succession planning	
Key (a) Methodologies (b) Tools (c) Platforms used to implement development plans.	a) methodologies	- Strategic Planning
	b) tools	- Not stated
	c) platforms	- Energy sector forum
Activities with a budget allocation in the 2015/2016 development plan for your sector.	- Regulatory Framework	
Above activities budgeted for and implemented	- Not stated	
From your experience, how is monitoring and evaluation done for implementation of the development plans	- Not stated	
Your expectations for a toolkit and knowledge management platform	- A balanced scorecard - ISO certification.	